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**An Application of Rostering Algorithms to Task
Assignment in Restaurant Operations
A CASE STUDY OF Southern Airports Services JSC -
SASCO
CAPSTONE DESIGN COURSE**

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Ho Chi Minh City, Vietnam

June, 2026



ABSTRACT

Airport Buffet Lounge service levels are subject to large fluctuations in customer demand largely due to flight schedule arrivals, peak travel patterns, and random airport conditions. Consequently, fixed staffing numbers are inefficient because they create employee service gaps during peak periods and idle employee time during off-peak times. This capstone project creates an integrated workforce management system to support Le Saigonais International Lounge operated by SASCO with a focus on developing an equitable employee roster in response to passenger demand to support the delivery of quality service and improved utilization of labor resources.

The proposed solution for this problem is to implement three steps through Optimization-Simulation-Feedback. First, a passenger arrivals model is generated for fifteen minute intervals and translated into hourly role-based workloads across six general operational categories which include: Reception, Seating, Dining Area Support, Buffet Line Operations, Food Safety/Temperature Control, Dishwashing/Sanitation, and Service Logistics Support. The hourly workloads are converted to shift level employee staffing needs in alignment with three predetermined eight hour shifts for a 28 day planning period. The second step is a Mixed Integer Linear Program (MILP) to define employee assignment to shifts and service role by considering coverage requirements, skill eligibility and availability, maximum allowable work hours, rest break policies, and employee preference. Simulation Annealing is then utilized as a Post-Optimization improvement process to minimize soft constraint penalties and improve the overall quality of the employee work rostering.

The benefits of this workload-based work rostering methodology in comparison to existing Fixed Workload Models are significant. The use of simulation, optimization and validation of the employee roster will build a sounder basis for developing an airport lounge staffing process and aligning customer demand with operational workload and employee work assignments. This study provides airport lounge operators with a decision support system to enhance the determination of workforce planning, enhance the reliability of service delivery, and improve the utilization of labor resources.

Keywords: Airport lounge staffing; Workforce rostering; Task assignment; Mixed Integer Linear Programming (MILP); Simulated Annealing (SA); Discrete-Event Simulation (DES); Workload-driven staffing; Flight-wave demand; Service-level performance; Labor utilization.



ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our capstone project advisor Assoc. Prof. Phan Nguyen Ky Phuc for his guidance, constructive feedback, and support throughout the project development. His academic direction supported our improvement of our understanding of workforce rostering & optimization modeling, based on the principles of Simulation and Verification, as well as the application of our ideas as a formalized research framework.

Our appreciation also goes to the industrial engineering and management school, international university - Vietnam national university ho chi minh city, for providing us with the knowledge, learning environment, and educational foundation to complete the capstone design project. The education and training provided to us during our study helped us to understand how to apply the methods of industrial engineering to solve a real-world service operation challenge.

We would like to take this opportunity to thank the southern airports service joint-stock company (SASCO) for providing us with inspiration for this study through the context of the Le Saigonais International Lounge case. The characteristics of total quality cultures as applied to the operations of an airport buffet lounge provided an interesting context for the application of workforce planning techniques, multi-criteria integer linear programming (milp) for workforce rostering, simulated annealing procedures, and discrete event simulation modeling.

We would also like to thank our team members for their cooperation, dedication, and efforts on this project. Each member made significant contributions to the research, model development, report writing, and results dissemination. This project could not have been completed without the collaboration, dedication, and commitment of each member throughout the capstone project process.



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CHAPTER I: INTRODUCTION

1. Motivation:

The airport business lounge ecosystem is an integral aspect of the overall airport service system, as they are a primary contributor to the experience of premium travellers and a key factor in developing customer loyalty to both airline and service brands. In the airport service ecosystem, SASCO (Southern Airport Services JSC) manages business lounge services in Vietnam as part of its offering of non-aeronautical services. The manner in which operational staff are scheduled, assigned and utilised between the various service activities that comprise a business lounge service directly impacts upon both the performance of the lounges (in terms of quality of service and cost) and the passengers who use those services.

The operation of buffet lounges is unique in comparison to traditional restaurants due to their highly time-sensitive demand patterns. Specifically, demand by passengers arriving at an airport tends to be concentrated during periods of peak air traffic (e.g., flight-wave) rather than evenly distributed throughout the entire day. As passenger traffic peaks, workloads simultaneously rise for a number of service streams such as reception, dining-area support, buffet restocking, verifying food safety, dishwashing, sanitising and logistical services. A constraint in one service stream quickly affects how the remaining service streams operate. For example, when dishwashing accumulates a backlog and there are insufficient amounts of clean dishware to serve customers, the ability of servers to clear tables and reflect availability of buffets will be limited.

Therefore, the airport business lounge service presents an important dilemma between service quality and labour productivity. Specifically, insufficient staffing during peak periods can result in excessive passenger queue times, delays in replenishing buffets, inadequately prepared dining areas and low service quality perceptions; while excessive staffing during slack periods can create unnecessary idle time and labour costs. Consequently, the objective of this capstone project is to develop an improved and structured approach to workforce rostering that will more effectively align the supply of staff resources with time-varying, flight-dependent demands for business lounge services while allowing for feasible and equitable work schedules.

The specific application of this project is focused on the Le Saigon International Lounge located at Tan Son Nhat International Airport. The project's objective is to provide a more efficient, demand-responsive staffing process that will improve the consistency of lounge service levels at Le Saigon International Lounge by lowering the amount of inefficient labour utilized within that service model and also provide a data-driven basis for future roster decisions related to SASCO's operation of business lounge services.

1.1. Background:

The quality of service offered at an airport is generally managed by linking the experience of passengers (satisfaction), the operational performance of the airport's services (available capacity,...) and the costs associated with providing those services (expenses), using a service-level concept. The International Air Transport Association (IATA) describes a level of service (LoS) as an integrated planning and monitoring framework for service excellence, which links passenger experiences to system capacity and performance and promotes the economical provision of airport services (IATA, n.d.). There are best-practice guidelines published by IATA for implementing Airport Service Level Agreements (SLA), which consider how the service levels, measurement methods, and performance incentives that apply at the airport will be defined collectively and in such a way that are transparent to all parties involved (IATA, n.d.). IATA, at a more general governance level, has pointed out that service quality frameworks for airports provide both airlines and passengers with clear expectations of service quality and help ensure delivery of consistent, economical performance (IATA, 2022).

These frameworks are often discussed in relation to terminal subsystems (e.g. check-in and security); however, the inherent concepts of a service system define the target service outcomes (what you want to achieve), operationally define the target service outcomes to measurable indicators, and ultimately, results in the proper allocation of resources to achieve the target service outcomes as consistently as possible. The quality of service in buffet lounges is particularly influenced by “continuous availability” and visible indicators such as: the readiness of buffet lines (hot and cold), the availability of clean dishware on tables, timely turnover of tables, and cleanliness of customer touchpoints. If one of these service streams fails, for example due to a backlog in dishwashing that results in a shortage of utensils or delayed buffet replenishment, perceived service quality will decline even when staffing levels appear adequate.

The demand for service at Le Saigon International Lounge is inconsistent due to many factors including flight schedules, passenger eligibility, delays in flights and seasonal travel patterns. While a fixed staffing model may provide a baseline, it is difficult to adjust the staffing levels to respond quickly enough when the customers from flights arrive in short time frames. In a buffet environment, there are multiple processes that must occur simultaneously. The Front-of-House must manage the flow of guests and ensure tables are ready; the Food Service Line must maintain the buffet offerings and ensure that proper food safety is followed; and the Back-of-House must have enough staff to wash dishes, maintain sanitation and provide logistical support for the other areas. The uncertain passenger traffic creates significant variability in demand, making fixed and evenly distributed

staffing impractical. While any disruption in one service stream may create localized bottlenecks and contribute to crowding during peak periods.

Scheduling the workforce at a buffet environment is not only a headcount problem. There are many different task coverage and role allocation issues to be addressed. There are many employees who possess multiple skills, however cross-training does not imply that every employee can perform every role. When making assignment decisions, management must take into account role eligibility, labor rules, rest requirements, equitable allocation, vacation, preferences and operational coverage needs. These complexities make the situation similar to a real appropriateness problem, which is to effectively schedule employees to feasible shifts and roles and ensure service levels are maintained while minimizing avoidable cost and penalty violations.

This project is motivated by three concurrent pressures in airport buffet lounge operations: highly volatile flight-dependent demand, service outcomes that must be maintained through measurable performance standards, and the need to allocate sufficient staff since minimizing wasted labour capacity and unnecessary labour costs.

1.2. Ideal and Method:

Given the concurrent pressures mentioned previously, an optimal operational state for an airport buffet lounge will include a consistently stable service levels defined by continuous buffet availability, timely table turnover, sufficient clean wares, and high hygiene standards. However, in order to maintain this balance, lounge operators must also ensure that employee staffing patterns are service effective, fair and feasible. Buffet lounge environments can create particularly difficult challenges in maintaining this balance since service levels are driven by not only how many employees are scheduled to work but also the synchronization of numerous interdependent and continuously changing operational activities based on under highly variable and flight-dependent demand conditions.

From an operations management perspective, these challenges can be formulated as a workforce scheduling and task coverage problem, requiring structured analytical approaches. Rostering across service industries is an extensively researched problem within workforce scheduling, driven by its practical importance and combinatorial complexity, and has attracted a wide range of solution methodologies (Ngoo et al., 2022). Rostering is a complex assignment problem that is impacted by many considerations such as coverage requirements, constraints imposed by staff members having multiple skill sets, adherence to labour laws and regulations and fairness. Research indicates that using structured methods to create rosters is critical when quality of service outcomes are sensitive to

having the appropriate workforce available for duty at an appropriate time. This issue is particularly relevant in buffet lounge environments, where multiple service tasks must be synchronized in real time. In many service operations, task assignment is still often handled informally to assign employees to specific tasks. The lack of integration between task assignments and staffing decisions may lead to operational deficiencies which can create bottlenecks and reduce the delivery of consistent service.

The findings from these research streams lead us to consider how we approach the operation of buffet lounges in terms of an integrated placement of employees (rostering) and employee coverage of required tasks, as opposed to just an experiential basis of employee scheduling alone. In the case of buffet lounge services provided by Le Saigonais lounge at airports, the main operational goal is to have the right number of employees working at the right time to cover the various tasks required. By providing the appropriate number of employees at the appropriate times and for the appropriate tasks, Le Saigonais lounge will be able to consistently meet the key service level targets (e.g. minimum wait times for customers, timely replenishment of the buffet, timely cleaning of the buffet, and balanced workload distribution among employees). In addition to providing support for achieving labour savings, this alignment will also improve the overall reliability of their operation.

The OSF Study was conducted using an Optimization-Simulation-Feedback (OSF) Integrated Framework to find a balance between service quality and labour efficiency. First, customer arrivals were estimated at a detailed level of time intervals, and converted to operational workload in terms of role-specific work for the six lounge roles. The operational workload was then converted to staffing requirement levels on a shift basis. These requirements were then used as inputs to the MILP rostering model. The MILP model is a mathematical program that assigns employees to roles and shifts while preserving constraints, such as; labour, skill, availability and preference. The validity of the roster could then be confirmed using a Discrete-Event Simulation (DES) to verify that the service performance (under varying fixed customer arrival rates) could be achieved and that staffing updates can be made in order to achieve service level targets.

This framework creates a useful division in that the operational characteristics of customer arrivals and service processes can be represented by 30-minute time intervals within the simulation layer, whereas the roster (monthly) represents only shift level (8-hour) workflow, as this is the manner in which actual workforce planning is completed. This difference allows the framework to develop a construct of the short-term congestion, while allowing the rostering model to remain cutable (simple), which can be further used in a managerial capacity.

2. Problem Statement:

Tan Son Nhat Airport is home to the Le Saigonnais Business Lounge, which is one of the main amenities that offer a mix of fine dining and comfortable space for high-end passengers. Le Saigonnais International Lounge provides a premium buffet and resting environment for eligible airport passengers is open all night. Because the lounge operates in an airport setting, customer demand is shaped by flight activity and can vary sharply within the same day. The current staffing practice can be represented as a fixed shift-template approach based mainly on projected passenger volume. While this approach provides a simple baseline, it does not fully account for the timing of demand peaks or the role-specific workload generated by actual operational events.

The operation of buffet lounge can be assumed in three main functions are belows:

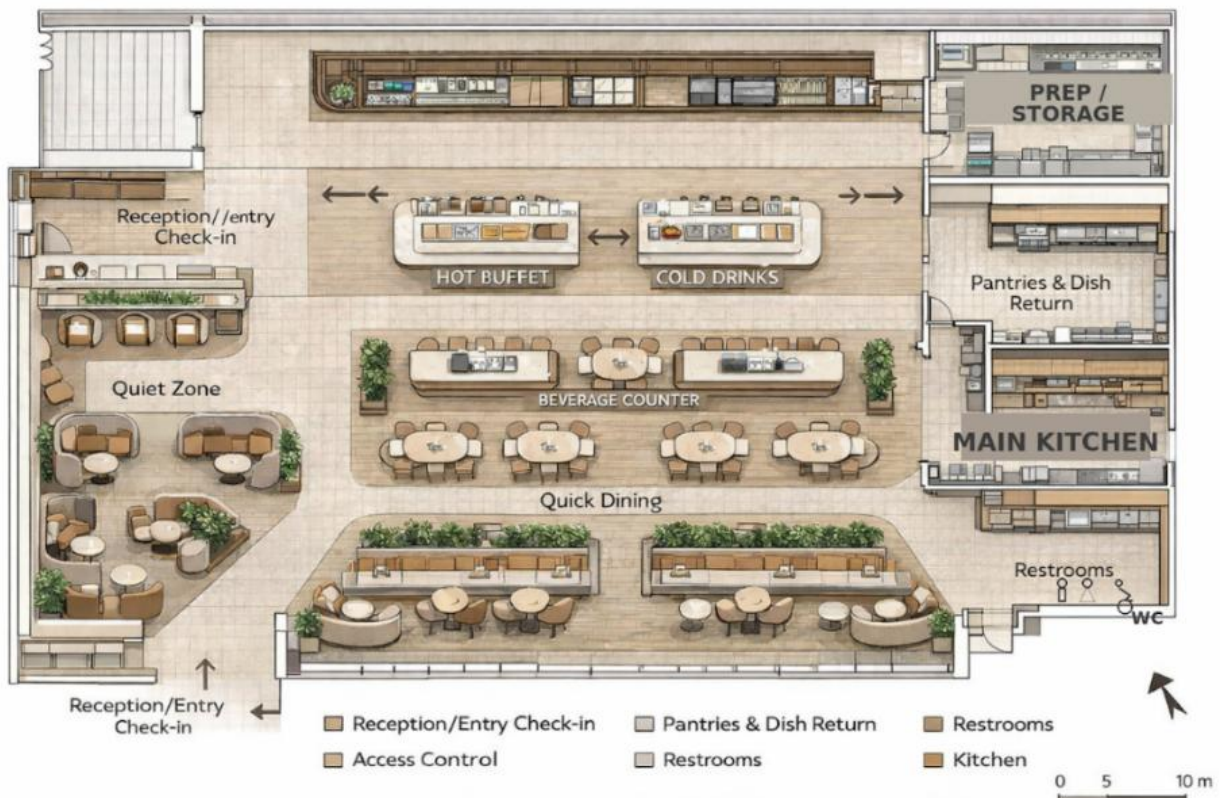


Figure I-1: Le Saigonnais Business Lounge

The buffet lounge service chain can be divided into three main operational units. The Front-of-House (FOH) unit includes Reception and Seating (RS), which handles guest arrival, check-in, access control, queue coordination, and seating support, and Dining Area Support (DAS), which handles table clearing, resetting, and dining-area readiness. The Food Service Line (FSL) unit includes Buffet Line Operations (BLO), which manages buffet replenishment, chafing-dish refill, presentation, and

guest assistance at service stations, and Food Safety and Temperature Control (FSTC), which handles temperature checks, food-safety logging, and corrective actions. The Back-of-House (BOH) unit includes Dishwashing and Sanitation (DS), which manages dirty ware collection, washing, sanitation, and clean dish recovery, and Service Logistics Support (SLS), which supports runner tasks, utensil replenishment, clean-ware redistribution, and movement of supporting materials. Together, these three units form an interdependent service chain that supports buffet continuity, hygiene compliance, and operational readiness under fluctuating airport demand.

Workforce scheduling is crucial for maintaining service quality and operational efficiency in airport buffet lounges, where demand varies sharply due to clustered flight arrivals and departures. Currently, the staffing model for Le Saigonais Business lounges relies on fixed shift templates generated from passenger volume projections. The shift templates ultimately provide a baseline for staffing purposes; however, they lack sufficient adaptability in order to satisfy changing time-sensitive (i.e., flight-associated) demand. In turn, this results in a disparity between labor capacity and workload generated by the actual volume of flight activity.

Multiple interdependent service streams (customer support, buffet replenishment, table turnover, dishwashing, and hygiene maintenance) make up lounge operations; their performance relies not only on the total number of people working in any given period, but also on how many people work on what kinds of tasks and how often they are coordinated in real time. Nevertheless, the current rostering process continues to view staffing as primarily an aggregated headcount issue (as opposed to an integrated system of synchronized task coverage) and thus creates significant inconsistencies when staffing levels for peak times often lead to bottlenecks, congestion, late replenishments, and slower responses to cleanliness, while overstaffing at non-peak times results in a large amount of idle time for employees, translating into higher labour costs.

Designing an improved schedule is further complicated due to multi-skilled employees, labour regulations, fairness constraints and service-level targets; thus making this problem a type of combinatorial optimization problem similar to the Nurse Rostering Problem (NRP). currently does not have an optimization-based rostering framework to jointly identify shift assignments with appropriate task-level coverages considering flight-dependent demand variability and evaluate the roster's performance based upon realistic operational dynamics. This absence of an optimization-based rostering framework has caused Le Saigonais Business lounges to have ongoing inefficiencies; inconsistent customer service; and increasing labour costs.

The main problem addressed in this capstone is that Le Saigonais International Lounge lacks an optimization-based rostering framework that jointly considers flight-dependent demand variability, role-level task coverage, employee qualifications, labour rules, fairness, and service-level performance. As a result, staffing decisions may lead to under-coverage during peaks, over-coverage during off-peak periods, inconsistent service quality, and inefficient labour utilization. This capstone therefore proposes a demand-driven rostering and evaluation framework inspired by nurse rostering logic and adapted to the buffet lounge context.

3. The Design Project Objectives and Requirements.

The main objective of this project is to develop and validate an integrated demand-driven rostering framework for Le Saigonais International Lounge. The framework aims to improve service performance while maintaining labour efficiency, fairness, and compliance with relevant operating constraints. More specifically, the project translates time-varying customer demand into role-based workload, converts workload into staffing requirements, generates a feasible monthly roster, and evaluates the operational performance of the roster under stochastic conditions.

The project has four specific objectives:

- Develop a structured demand and workload generation method that represents customer arrivals, service tasks, and role-specific operational workload by day, shift, and role group.
- Formulate a MILP rostering model that assigns employees to shifts and primary role groups while satisfying coverage, skill eligibility, one-shift-per-day, work-hour, workday, night-shift, rest, leave, and availability constraints.
- Use penalty terms for soft requirements such as target coverage, overstaffing control, workload fairness, employee preferences, role stability, and selected incompatibility or pairing conditions.
- Validate roster performance using discrete-event simulation and use feedback to identify whether coverage levels should be adjusted for service stability.

The expected analytical outputs are: (1) a demand and workload matrix indexed by day, shift, and role group; (2) an optimized roster generated by MILP; and (3) simulation-based performance measures such as waiting time, queue length, staff utilization, service stability, and frequency of KPI violations during peak and off-peak periods. Together, these outputs provide an evidence-based foundation for assessing roster feasibility and service reliability.

4. Scope and Limitations:

4.1. Scope:

This research examines the workforce rostering and task assignment for the buffet lounge operations of Le Saigonnais International lounges, at Tan Son Nhat International Airport. The scheduling horizon is one month, with three fixed 8-hour shifts: S1 (00:00-08:00), S2 (08:00-16:00), S3 (16:00-24:00). For simulation and workload measurement, each day is also divided into 30-minute time slices. Therefore, one operating day contains 48 slices, and each shift contains 16 slices.

Table I-1: Daily Shift Structure and Time Slice Allocation

Shift	Time	The amount of time	Number of slice time
S1 (Night)	00:00-08:00	8 hours	16 slices
S2 (Day)	08:00-16:00	8 hours	16 slices
S3 (Evening)	16:00-24:00	8 hours	16 slices
Total	00:00-24:00	24 hours	48 slices

The research will develop a demand-based methodology to match employees to shifts and tasks to meet time-dependent service requirements, while maintaining labor efficiency (minimizing overstaffing and imbalance).

The model covers six operational role groups: RS, DAS, BLO, FSTC, DS, and SLS. It considers employee qualifications, hard availability or leave, monthly workday and working-hour bounds, night-shift limits, consecutive-work logic, rest requirements, and soft preferences. The project does not make long-term HR decisions such as hiring, firing, wage design, or staffing-capacity investment. It also does not redesign the menu, pricing strategy, facility layout, or lounge access policy.

The demand inputs are generated through simulation and parameterized assumptions that reflect plausible peak and off-peak airport lounge patterns. Therefore, the study emphasizes the design and validation of the rostering framework rather than the development of a full empirical passenger-demand forecasting model.

4.2. Limitation:

First, the roster is modeled using three fixed 8-hour shifts. This structure is practical for monthly planning, but it may not fully capture operational practices such as staggered start times, split shifts, short overtime, planned breaks, or temporary redeployment during flight-wave peaks. These practices can influence real-time service coverage and should be considered in future implementation.

Second, the MILP model assigns employees at the shift and primary-role level. Short-term task switching within a shift is treated as an execution-level activity rather than as a detailed optimization decision. As a result, the model may underestimate temporary within-shift congestion or micro-bottlenecks that occur when demand rises suddenly in a short time window.

Third, access to detailed historical operating data, including actual flight schedules, lounge passenger arrivals, service-choice data, and historical task completion times, is limited. Because of this limitation, demand and service parameters are simulated rather than fully estimated from historical records. The framework can still demonstrate how demand-driven rostering should operate, but the numerical results should be interpreted as scenario-based rather than as final empirical estimates.

Fourth, the simulation represents normal operating variability but does not explicitly model major disruptions such as mass flight cancellations, airport-wide delays, emergency lounge closures, or sudden policy changes in passenger access. Such irregular operations may require contingency staffing rules beyond the baseline rostering framework.

5. Project Plan:

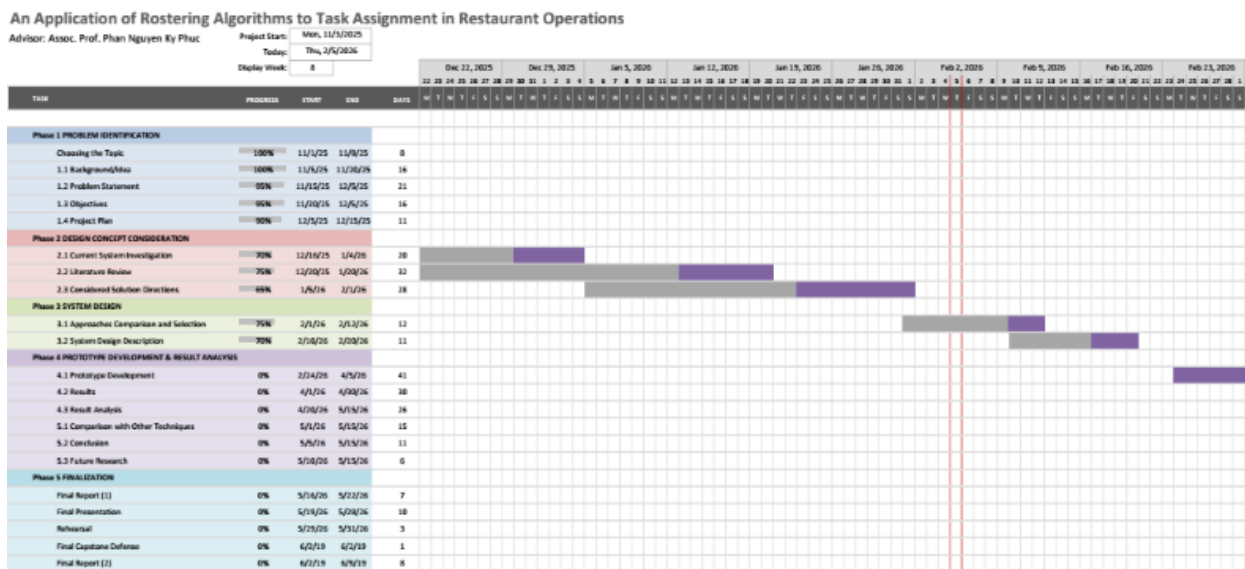


Figure I-2: The Gantt Chart of Project

CHAPTER II: DESIGN CONCEPTS CONSIDERATION

1. Overview:

Southern Airports Services Joint Stock Company (SASCO) is the leading non-aeronautical service provider at Tan Son Nhat International Airport on the operational side of the business, SASCO has a diverse portfolio across both the retail and duty-free spaces, as well as providing higher touch service segments such as food & beverage service (F&B), and business lounges. The lounges in the SASCO ecosystem; specifically, buffet-style F&B service; serve as a premium product contributing to the passenger experience and perceived service quality which ultimately has a direct impact on the commercial performance and brand positioning of SASCO.

The purpose of this research is to analyze the buffet operation of the Le Saigonnais Business Lounge located within the Tan Son Nhat International Airport in Ho Chi Minh City, Vietnam. The lounge is situated behind Burger King in the airside restricted area of the International Departures terminal on the 3rd Floor, and is open 24/7, and guests typically stay in the lounge for a maximum of three consecutive hours. The buffet will play a very important role in the overall lounge experience; in contrast to conventional restaurants, buffets must be continually supplied with food, have clean dining tables and adequate utensils, and allow guests to move about the lounge to create an efficient flow of guests, notwithstanding relatively short passenger dwell times and time-compressed pre-boarding activity.

Conceptually, the buffet operation at Le Saigonnais Business Lounge can be represented as an integrated service chain consisting of three interrelated operational units: Front of House (FOH), Food Service Line (FSL), and Back of House (BOH).

The FOH unit manages guest-facing flow and dining-area readiness through two main role groups: Reception & Seating, which is responsible for queue control, guest reception, access checking, and seating coordination; and Dining Area Support, which focuses on table clearing, resetting, and maintaining the readiness of the dining area.

- The FSL unit manages buffet-facing service activities through Buffet Line Operations and Food Safety & Temperature Control. Buffet Line Operations is responsible for replenishment, chafing-dish or serving-line presentation, and guest support at buffet stations, while Food Safety & Temperature Control ensures routine temperature checks, food-safety logging, and compliance monitoring.
- The BOH unit supports service continuity through Dishwashing & Sanitation and Service Logistics Support. Dishwashing & Sanitation handles warewashing, sorting,



loading/unloading, and sanitation-related support, while Service Logistics Support provides runner and redistribution support to keep buffet items, utensils, and supporting materials moving efficiently across the operation.

Expected service outputs include continuous buffet readiness, timely table turnover, adequate availability of clean utensils and dishware, compliance with hygiene requirements, and smooth guest flow. These service outputs are generated through the coordination of all three operational units rather than through the isolated performance of any single unit.

The day-to-day operating context has a number of structural elements that influence some aspects associated with performance. First of all, demand is closely linked to the availability of flights and has significant peaks that cause rapid accumulation of loads over very short time periods. Secondly, tasks are heavily interdependent among FOH, the food service line and BOH and therefore, a delay in one area or a resource shortfall in one area may propagate and negatively impact other areas (for example, a slow dish turnaround may limit service line readiness; or high arrival volumes during peak periods may place pressure on the tables to turn over rapidly). Thirdly, in terms of work allocation, staff positions may be constrained by knowledge/skill and compliance-related restrictions including hygiene/food safety issues. Thus, the ability to substitute or replace staff assuming duties associated with a specific position may be impossible without adhering closely to established standard operating procedures. Lastly, peak demand periods will require fast responses and short cycle times to ensure the business remains in a state of continuous readiness for delivery of service levels under ever-changing and fluctuating conditions, thereby real-time prioritization and coordination are critical to maintaining service quality during those periods.

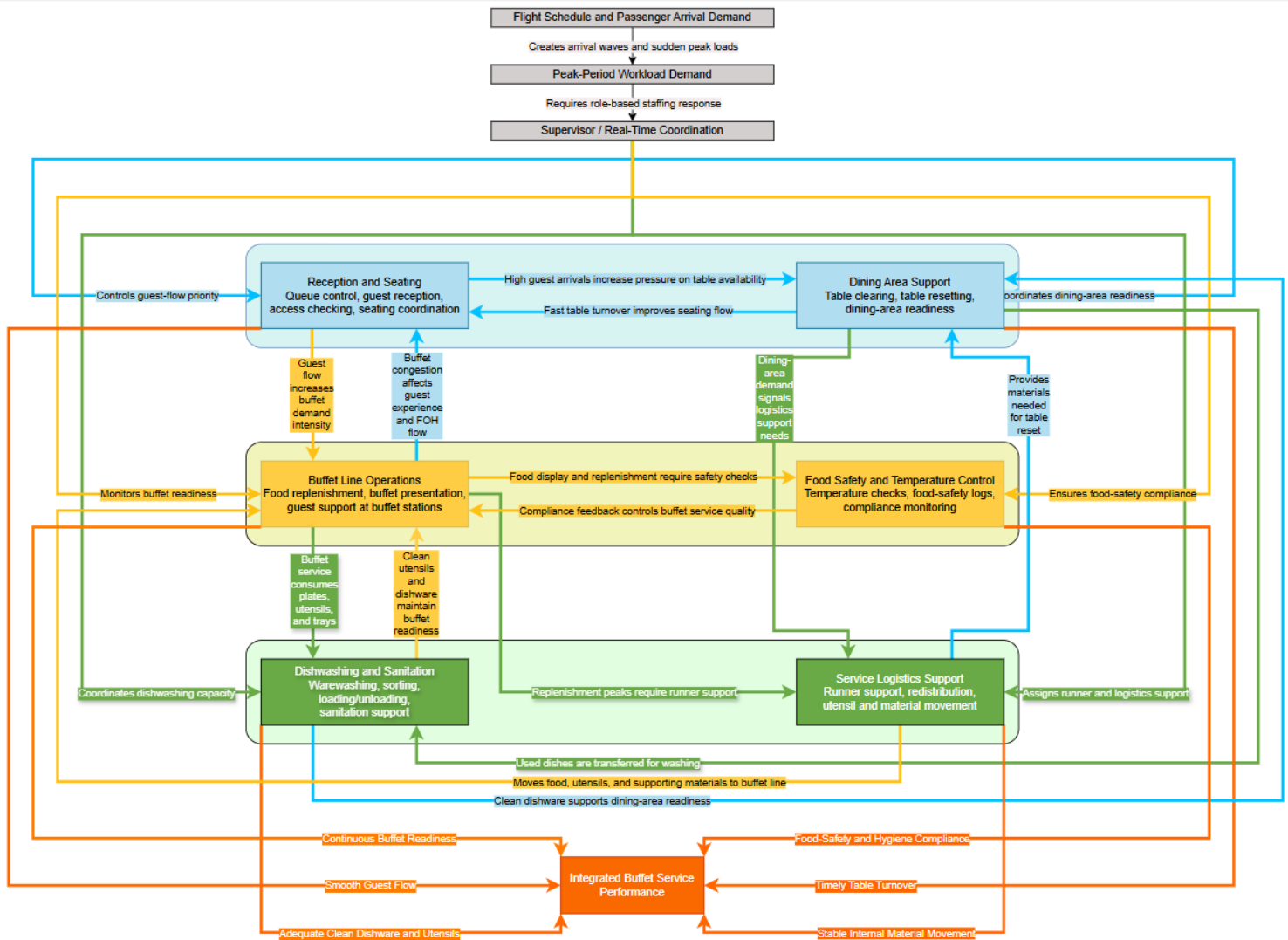


Figure II-1: Operational Staffing Blueprint: FOH - Food Service Line - BOH with Flex Pool

The success of operational delivery comes down to coordinated human resources organized by role groupings using common physical and procedural assets to enable that coordination. A supervisor acts as the main coordinating individual allocating resources to labor in real time during surges in demand, managing escalations, and ensuring compliance with service level standards. The FOH (Front of House) staff perform guest reception, check in guests and provide immediate customer service while the food service line staff oversee replenishment control, execution of live-station service and front line readiness checks for the continuation of inventory on the buffet. The BOH (Back of House) staff execute table clearing and turnover, collect used dishware, wash dishes and perform cleaning tasks that maintain proper hygiene readiness before, during and after peak periods.

The practical sourcing of labour across several roles increases operational flexibility and stabilizes service level outcomes in cases where demand fluctuates significantly.

This conceptualization provides the basis for translating service outcomes into task-level demand and staffing coverage requirements, which will later be encoded in the demand matrix and MILP rostering model and evaluated via discrete-event simulation.

2. Current System Investigation:

Le Saigonnais Business Lounge (SASCO) has a 24-hour buffet that is open seven days a week and offers a rotating schedule based on three shifts (morning, midday, evening). The current system can be summarized as follows elements. The current system can be represented as a managerial control cycle consisting of five main components. Firstly, people: the workforce is structured into functional skill groups, namely front-of-house staff responsible for check-in, guest assistance, and table coordination; buffet support staff responsible for replenishment and line readiness; and back-of-house staff responsible for dishwashing and sanitation, operating within a clear supervisory hierarchy from frontline employees to shift leaders/supervisors and the lounge manager. Secondly, process: service delivery within each shift follows a consistent workflow (check-in → service → refill → clearing → washing → sanitation), implemented through multiple interdependent activity streams that must be coordinated continuously as demand fluctuates. Thirdly, information: rostering and task allocation draw on demand signals such as forecast passenger volumes and flight-wave patterns, complemented by real-time floor observations (queues, table availability, buffet stock-outs, cleanliness conditions) and prior-shift reports, while staff absences and unavailability statuses (public holidays, time-off in lieu, leave, reassignment) represent critical inputs that directly constrain effective capacity. Next, rules and constraints: Staffing decisions are guided by skill-task eligibility and compliance with contractual/labor rules (working time limits, minimum rest periods, limitations on the number of consecutive days worked, and overtime) and practical compatibility (preferred team groupings and prohibited team pairing) as well as requirements for food safety/hygiene that have a minimum level of responsiveness in order to ensure cleaning and buffets are ready to serve. These constraints can be grouped into (i) labor/contract rules (working time limits, minimum rest periods, limits on consecutive working days, overtime) and (ii) food safety and hygiene rules that require minimum responsiveness to keep buffet and dining areas ‘ready-to-serve’ at all times. Furthermore, methods of implementation for tools are reliant upon shift templates, daily assignment sheets, hygiene and operational checklists, and communication channels (i.e., radio/chat), along with point of sale (POS)



data (if available), which can provide supplementary signals about service rhythm and demand but does not constitute a real-time operational control metric by itself. Finally, coverage is the operational requirements expressed as staffing requirements broken down by shift and skill group, and typically defined as minimum-target-maximum in order to alleviate potential service bottlenecks (check-in congestion, replenishment delays, slow table turnover, dish washing backlogs) while avoiding systematic overstaffing during the periods of low customer volume.

Then, the figure below presents the AS-IS framework of workforce planning and control at Le Saigonais Business Lounge (SASCO), illustrating the interaction between the planning layer and the execution layer.

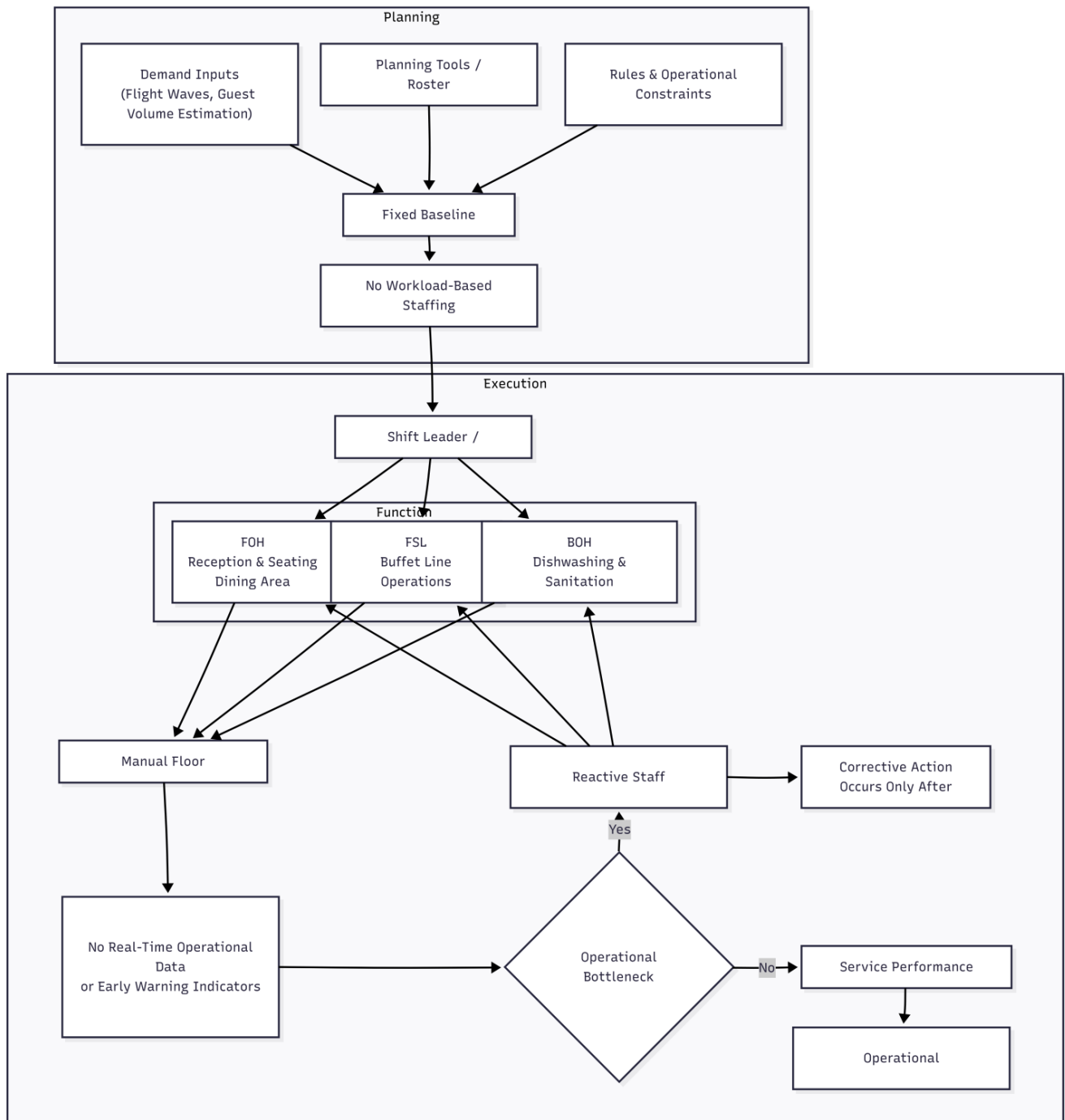


Figure II-2: AS-IS Workforce Planning and Control Framework at Le Saigonnais Business Lounge.

At the planning (rostering) layer, Le Saigonnais Business Lounge (SASCO) maintains baseline by organizing staffing into three fixed shifts (morning, midday, and evening). Rosters are typically prepared by the lounge/restaurant manager (or a designated scheduling officer) and validated by operational supervision, with decisions anchored in forecast passenger volumes and expected flight-wave profiles. These demand signals are commonly supplemented by managerial judgement

informed by prior operating experience and recent service conditions. The resulting plan functions primarily as a stable baseline schedule, typically prepared on a periodic cycle and updated only intermittently most often in response to disruptions such as absenteeism or unexpected demand variation rather than being continuously re-optimized. Consequently, flexibility at the planning stage is limited to incremental headcount adjustments within predetermined shift templates, with relatively constrained capability to dynamically redesign shift structures or skill allocations across time periods. At the execution (intra-shift) layer, operational performance depends on real-time coordination across interdependent service streams, including front-of-house guest handling, buffet replenishment, and back-of-house dishwashing and sanitation. Task allocation is usually directed by the shift leader/supervisor, who assigns primary responsibilities based on functional areas and the skill level of the staff at the beginning of their shift, and then informally reallocates staffing based on changing conditions. Reassignment decisions are informal and reactive, triggered by visible floor events (e.g., check-in congestion, buffet stock-outs, delayed table turnover) rather than predefined thresholds or structured workload metrics. During peak periods for flights, supervisors tend to relieve immediate bottlenecks by moving staff to the area of greatest service need (i.e., applying more staff in the front of the house to reduce queue times or increasing the amount of staff working in replenishment/dishwashing to maintain buffet availability). While this reactive control allows for quick local reactions to changing circumstances, it also leaves execution of tasks at the mercy of individual supervisor experience and situational awareness, especially considering the lack of real-time operational indicators available to supervisors (measured waiting time; lead times for replenishment; turnover time; and stack levels of dishes being washed).

The current system has both advantages and disadvantages of the system. Advantages can be described as follows. Fixed shift templates used in conjunction with experiential judgement for assigning shifts result in a simple, consistent and inexpensive method for assigning shifts and communicating and implementing those shifts. Further grouping staff into FOH, buffet support and BOH creates clear lines of responsibility and contributes to the establishment of routine stability (particularly in BOH (back of house) cleaning and dishwashing, where both sanitary compliance and consistency of service are paramount). Additionally, supervisory discretion gives the ability to provide flexibility to staff during peak hours of flight activity; as multi-skilled employees may be shifted from one function to another to assist with removing immediate bottlenecks (e.g., check-in queues, stock-outs, slow turn of tables, backlog of dish washing) without requiring a complete redesign of the current scheduling or staffing plans.

However, the current system exhibits three key weaknesses. Firstly, loss of demand–capacity alignment because no headcount planning occurs on shift levels. Secondly, lack of task-coverage linking across separate streams when rostering. This leads to a disparate number of shifts covered compared with a corresponding demand for staff by task. Thirdly, the current system has primarily been based on observation and reaction; there are few real-time metrics (or structured coordination routines) available to assist in ensuring that the need for coverage exists before there is an observable loss of service, thus limiting the ability to learn and improve continuously.

In continuing to refine existing strengths and weaknesses identified in the current staffing model; the service outcomes enumerated below will represent the operational implications resulting from the current staffing model as they have been identified through observation at Le Saigonais. The following three recurring patterns reflect the simplified weaknesses of the current staffing model observed at Le Saigonais and illustrate how they are manifesting themselves. First, the demand/capacity mismatch results in prolonged wait times in reception when flight waves are at peak, and excess idle time when demand falls into the troughs. Secondly, the lack of task coverage has resulted in the inability to maintain stocked buffet items, slow table turn times, backlogs in dishwashing, and delays in the cleanliness recovery. Lastly, the lack of real-time information has resulted in the inability to intervene in a timely manner and to consistently evaluate the performance against Key Performance Indicators (KPIs) across the shifts.

During peak flight-wave periods, highly time-dependent passenger arrivals create short but intense demand surges, increasing congestion risks at reception and buffet areas. Under fixed baseline staffing, temporary mismatches between workload and available capacity commonly lead to longer queues, delayed table turnover, slower replenishment, and reduced responsiveness in cleanliness maintenance. Stabilizing service flow often requires rapid on-floor labor reallocation, highlighting the importance of dynamic workforce coordination in time-sensitive airport operations.

Conversely, during the off peak periods between flight waves, the same constant coverage logic will cause wasted space as well as non-uniform utilisation of the workforce. As operators have continuing restrictions on their workforce capabilities the operator may have to use redeployment and sporadically approved overtime during peak periods while the overall staffing levels remain relatively stable but too rigid to adapt to the changes in demand levels.

The three major root causes that have been identified for the observed outcomes utilizing the techniques of Fishbone or 5-Why analysis are: First, shift-based coverage is not reliably aligned with the requirement of flying a wave; Second, the rostering function does not synchronize with task-based

coverage at the various interdependent workstreams; and Third, operational control is primarily reactive and/ or is limited by the availability of real-time metrics, thereby delaying intervention activities and adversely affecting learning. Overall, these three major issues identify the need for a system to link demand - > roster - > task coverage, supported by mutually agreed-to quantitative measures that will allow for proactive; and continuous improvement in operational control.

3. Literature Review:

3.1. Method

From a discipline perspective, recent workforce scheduling research increasingly treats staffing, shift design, and task assignment as interdependent decisions rather than isolated subproblems. This integrated view is especially evident in high-contact service and care settings, where performance depends on short-term coverage adequacy and synchronized task execution under time-varying demand. Within this family of similar systems, nurse rostering is often used as a benchmark because it combines dense feasibility constraints (skills, rest rules, contract restrictions, and preferences) with strict time-indexed coverage. Ceschia et al. (2023) reflects this practice-oriented direction by modeling heterogeneous staff, multi-skill feasibility, and realistic rule sets, which parallels SASCO buffet lounge operations where cross-trained employees must be assigned over time while maintaining feasible coverage for interdependent service tasks. From the objective-structure lens, the literature has shifted away from cost-only formulations toward objectives that encode operational trade-offs in a decision-maker-friendly way. A common approach is penalty-based optimization that separates non-negotiable requirements from flexible preferences through hard and soft constraints. Ceschia et al. (2023) adopts this structure and further represents demand using coverage bands defined by minimum, target, and maximum levels. Such bands are well suited to peak-driven service contexts: minimum coverage prevents service breakdown, target coverage stabilizes throughput and waiting time, and maximum coverage limits inefficiency during off-peak periods directly matching SASCO's operational realities.

These modeling developments motivate current solution approaches that prioritize scalability under realistic constraints. While mixed-integer programming remains valuable for benchmarking and simplified instances, practical systems increasingly rely on metaheuristics and hybrids that produce high-quality solutions within operational runtimes. Ceschia et al. (2023) exemplifies this trend by pairing an exact baseline with a Simulated Annealing engine tailored to coverage-band correction and feasibility preservation. Finally, recent work emphasizes deployment as part of the method itself:

solutions must integrate with data infrastructure and support repeated runs and KPI-based evaluation. In this sense, the key paper provides a coherent, modern template for SASCO, where flight-driven demand scenarios can be translated into time-varying coverage bands and an SA-based solver can generate feasible, near-optimal rosters fast enough for day-to-day decision support under uncertainty.

Table II-1: Summary Method

Classification	Categories	Authors/Year	Summary
Discipline	Nurse rostering (multi-skill, time-indexed)	Ceschia et al. (2023)	Real-world nurse rostering with multi-skill staff and time-varying coverage.
	Personnel task scheduling integrated with rostering (SMPTSP)	Smet & Vanden Berghe (2012)	Studies a finer-grained rostering setting where tasks must be assigned within predefined working times for multi-skilled employees
	Staff scheduling	Aydas et al. (2020)	Determines staffing needs and assigns nurses to shifts.
	Shift scheduling	Bonutti et al. (2016)	Uses simulated annealing to reduce under/overstaffing
	Work assignment	Corominas et al. (2010)	Assigns tasks based on worker experience.
	Project scheduling	Liu & Wang (2012)	Schedules multi-skill crews to shorten duration.
Objective	Penalty-based coverage balancing (min–opt–max)	Ceschia et al. (2023)	Minimize weighted penalties for understaffing, overstaffing, and other soft-rule violations.

	Shift minimization / workforce minimization	Smet & Vanden Berghe (2012)	Minimizes the number of employees used while assigning all tasks to qualified staff.
	Single-objective	Rong (2010)	Minimizes total hiring and scheduling cost.
	Bi-objective	Rerkjirattikal et al. (2022)	Balances cost and workload/satisfaction.
	Multi-objective	Othman et al. (2012)	Integrates cost and human-factor objectives.
		Hematian et al. (2020)	Minimizes time and cost in multi-project staffing.
Method	MIP + Simulated Annealing (swap/change moves)	Ceschia et al. (2023)	Use SA with swap/change neighborhoods, supported by a MIP formulation, to produce high-quality rosters fast.
	Matheuristic local search + constructive heuristic	Smet & Vanden Berghe (2012)	Builds a feasible initial solution using a constructive heuristic, then applies a very large-scale neighborhood search matheuristic by solving selected subproblems to optimality with a solver.
	Exact	Chen et al. (2014)	MIP solved by commercial solvers under uncertainty.
	Heuristics	Walter & Zimmermann (2016)	Construction heuristics for large multi-skill cases.
	Metaheuristic	Zuo et al. (2019)	Tabu + simulated annealing to cut operating cost.

		Akbari et al. (2013)	SA/VNS considering preferences and fatigue.
		Dean (2008)	Genetic algorithm for nurse rostering (improved encoding).
		Porto et al. (2019)	Hybrid staffing with contract types and flexibility

3.2. Simulation-Based Workforce Planning for Airport Lounge Staffing

Airport lounge workforce planning requires a modeling approach that can reflect the variability of passenger demand and the operational workload created by that demand. In airport service environments, customer arrivals are not evenly distributed throughout the day because passenger flow is shaped by flight schedules, terminal behavior, and time-of-day patterns. Postorino et al. (2019) support this view by studying airport passenger arrival processes and estimating arrival-rate functions for different passenger types and periods of the day. This provides the theoretical basis for constructing a time-dependent demand profile rather than assuming a constant arrival rate across the full operating day.

To represent this uncertainty, the project adopts a stochastic arrival logic based on nonhomogeneous Poisson process theory. Kim and Whitt (2014) discuss the use and testing of nonhomogeneous Poisson process models for service systems, which is relevant because airport lounge arrivals vary over time but can be approximated within short time intervals. In this project, the operating day is therefore divided into smaller time slices, where each interval has its own expected arrival rate. This allows the model to generate customer arrivals in a way that is both realistic and manageable for a student capstone simulation.

The arrivals generated are then extended into a discrete-event simulation framework, since staffing decisions depend not only on the number of customers but also on how customers create work inside the lounge. Prior airport terminal simulation studies show that discrete-event simulation can be used to model passenger flow, predict delays, and support rational resource management in airport service processes. This supports the project's decision to simulate the customer journey from arrival to service completion, including reception, lounge stay, buffet activity, table reset, dishwashing, food safety checks, and stock refill operations.

Through this simulation logic, passenger demand is translated into role-specific workload rather than simple staff ratios. This transition is important because each airport lounge role is affected by a different operational driver: reception workload depends on arrivals, dining-area workload depends on table reset tasks, dishwashing workload depends on dirty-dish batches, and logistics workload depends on refill events. The workload output then becomes the foundation for the MILP rostering model, which assigns employees to shifts and roles while considering coverage, skills, availability, workday limits, night-shift rules, and employee preferences. To further improve the roster after feasibility is achieved, Simulated Annealing is applied as a local-search improvement method to reduce soft-constraint penalties. Overall, the literature supports the project's integrated framework, where airport arrival-pattern theory informs demand generation, stochastic process theory supports arrival simulation, discrete-event simulation converts demand into workload, and optimization transforms workload into a practical workforce roster.

4. Considered Solution Directions

4.1. Model overview

Mathematical programming models describe feasible decisions using variables and select the best option by optimizing an objective function, enabling schedules to be compared consistently. For the Le Saigonais Business Lounge buffet operation, a mixed-integer linear programming model is suitable because key decisions assigning employees to shifts and zones are binary, while coverage targets and labor costs can be represented linearly. This framework allows labor rules, zone eligibility, and time-indexed coverage requirements to be handled within one model.

Model development is typically iterative. It starts with core assignment variables, then adds feasibility constraints such as working-hour limits, rest requirements, shift legality, and zone-based flexibility (staff can multitask within a zone but cannot substitute across specialized areas). The objective is then defined to minimize labor cost while penalizing peak-period understaffing more strongly than off-peak deviations, using coverage targets by time block and zone. Additional variables may be introduced as needed to represent coverage deviations, overtime, or workload balance, ensuring the final model remains both realistic and solvable.

4.2. Solution method

To solve the workforce scheduling problem under highly variable demand and complex operational constraints, this study adopts solution methods that are widely used in workforce scheduling and nurse rostering research. The goal is to obtain schedules that remain feasible under labor regulations

and operational rules, while achieving strong coverage during peak periods and avoiding inefficient overstaffing off-peak. Consistent with the literature summarized in Table 1, solution methods can be positioned along a spectrum from exact optimization to heuristic and metaheuristic approaches, as well as hybrid strategies that combine their advantages.

4.2.1. Exact Method (Mixed-Integer Programming)

Mixed-Integer Programming (MIP) is one of the most established exact optimization frameworks for workforce scheduling and rostering. In an MIP model, binary decision variables represent assignment choices such as whether an employee is scheduled for a particular shift or duty, while linear constraints enforce feasibility requirements including labor regulations, coverage demands, and skill compatibility. The objective function typically minimizes staffing cost and penalty terms for coverage of deviations or rule violations. Exact methods are valuable in workforce scheduling because they provide transparent, auditable decision logic and can serve as benchmarks for evaluating heuristic solutions. However, in realistic instances with many structured constraints and large planning horizons, exact solving can become computationally expensive, which motivates complementary heuristic and metaheuristic approaches for operational use.

4.2.2. Construction Heuristics

Heuristic methods are frequently used to obtain feasible schedules quickly when exact optimization is slow or impractical. In workforce scheduling, construction heuristics build a roster incrementally using priority rules, for example filling the most critical coverage requirements first or assigning scarce multi-skilled employees to high-need periods. The main advantage of construction heuristics is speed and reliability: they can generate a workable schedule within tight operational timelines. Although they do not guarantee optimality, they are widely used as standalone practical tools and, importantly, as initialization procedures for more advanced improvement algorithms such as local search and matheuristics.

4.2.3. Simulated Annealing

Simulated Annealing (SA) is a metaheuristic inspired by the physical annealing process, where a material is heated and then slowly cooled to reach a low-energy stable state. In optimization, SA starts from an initial feasible schedule and iteratively explores neighboring solutions. A key feature of SA is that it accepts not only improving moves but also, with a probability that decreases over time, certain non-improving moves. This acceptance mechanism helps the search escape poor local minima, which is especially useful for rostering problems where the solution space is highly discrete and constrained. In workforce scheduling, SA is typically implemented with simple but powerful

neighborhood operators such as swapping assignments between two employees or changing an employee's assignment to correct coverage shortfalls. Cooling schedules and stopping criteria are used to control runtime and solution quality, making SA attractive for large, real-world rostering instances.

4.2.4. Tabu Search

Tabu Search (TS) is a metaheuristic developed by Glover that improves a solution through iterative local moves while using memory structures to avoid cycling and to promote exploration. The defining element of TS is the tabu list, which temporarily forbids certain moves that would return the search to recently visited solutions. This short-term memory helps the algorithm escape local traps and explore new regions of the solution space. In workforce scheduling, TS is often applied with neighborhood moves similar to other local-search methods, such as swapping employees between shifts or reallocating assignments to repair coverage deficits. Beyond short-term memory, TS can incorporate longer-term strategies to intensify the search around promising areas or diversify when progress stalls. Because scheduling problems often involve many constraints and repeated patterns, TS has remained a popular choice for producing high-quality rosters within practical computational limits.

4.2.5. Variable Neighborhood Search

Variable Neighborhood Search (VNS) is a metaheuristic framework that systematically changes the neighborhood structure during the search. Rather than relying on a single move type, VNS alternates between different neighborhoods of increasing size or complexity. The logic is that a solution that is locally optimal under one neighborhood may still be improved when explored through another. In workforce scheduling, VNS typically combines small local adjustments (e.g., swapping two assignments) with larger perturbations (e.g., reassigning several employees across a peak period) to overcome local optimality barriers. This flexibility makes VNS well-suited for problems with complex constraints, because different neighborhoods can target different types of violations, such as coverage deficits, fairness imbalances, or infeasible work patterns.

4.2.6. Genetic Algorithm

Genetic Algorithms (GA) are population-based metaheuristics inspired by evolutionary processes. A GA maintains a set of candidate schedules and improves them over successive generations using selection, crossover, and mutation. Selection favors higher-quality schedules; crossover combines parts of two parent schedules to create new offspring; and mutation introduces random changes to maintain diversity. In workforce scheduling and nurse rostering, GA performance depends heavily



on representation and operators, because naïve crossover or mutation can easily break feasibility constraints. For this reason, many applications embed repair mechanisms or constraint-aware operators to maintain valid schedules. GA is attractive for scheduling because it explores a broad search space and can discover diverse high-quality solutions, though it often requires careful tuning to balance feasibility, convergence speed, and solution quality.

CHAPTER III: SYSTEM DESIGN

1. Approaches Comparison and Selection:

Table III-1: Approaches comparison

Method	Definition	Main strengths	Typical limitations	Similar to	Best-fit use cases
Exact optimization (MIP/MILP)	Exact mathematical programming with binary assignment variables and linear constraints	Transparent and auditable decision logic; strong for benchmarking; enforces feasibility precisely	Can become computationally expensive for large, highly constrained, long-horizon instances	Matheuristics (both use solvers)	Small medium instances; baseline/benchmark; contexts requiring explainability and compliance checking
Construction heuristics (greedy/priority rules)	Fast rule-based procedures that build a schedule incrementally	Very fast; reliable at producing feasible schedules; easy to implement	Lower solution quality; can be myopic; depends heavily on rule design	Often used with SA/TS/VNS or matheuristics as initialization	When schedules must be produced quickly; when data are imperfect; as an initial solution for improvement methods
Simulated Annealing (SA)	Single-solution metaheuristic that explores neighbors and occasionally accepts worse moves (cooling)	Escapes local minima; effective in discrete, constrained spaces; flexible neighborhood design	Requires parameter tuning; quality depends on neighborhood and penalty structure; no optimality guarantee	TS/VNS (local-search family)	Large real-world rostering; tight time limits; strong when coverage repair and feasibility moves can be defined clearly (swap/change)

Tabu Search (TS)	Local search with memory (tabu list) to avoid cycling and encourage exploration	Stable improvement; avoids loops; supports intensification/diversification strategies	Needs tuning (tabu tenure); may block promising moves if misconfigured	SA/VNS (local-search family)	Large instances with repeated patterns; situations where the search tends to cycle; strong for operational-quality rosters
Variable Neighborhood Search (VNS)	Metaheuristic that systematically changes neighborhood structures during search	Effective against different violation types; flexible; can break local optimality barriers	Requires designing multiple neighborhoods; runtime can increase	SA/TS (local-search family)	Problems with heterogeneous constraints (coverage, skills, fairness, rest); when different move types target different issues
Genetic Algorithms (GA)	Population-based metaheuristic using selection/crossover/mutation	Broad exploration; produces diverse high-quality solutions	Feasibility is fragile; needs careful encoding and repair; tuning-heavy	Can resemble hybrids if repair/solver is embedded	When diverse alternatives are desired; when global exploration is important and constraint-handling mechanisms are mature
Matheuristics (exact + local search hybrid)	Hybrid approaches that solve selected subproblems optimally with a solver inside a heuristic search	Strong solution quality; exploits solver precision locally while maintaining	More complex to implement; relies on solver; requires subproblem design	Bridge between MIP and local search	When full MIP is too large but exact enforcement is still valuable; highly constrained production settings

		global exploration			needing strong quality within reasonable runtime
Hybrid staffing models (contract types + flexibility)	Modeling extension (not a solver by itself): multiple contract types, overtime, flex coverage, limited cross- coverage	Captures real operational levers; improves robustness to peaks and disruptions	Adds modeling complexity and data requirements	Can be combined with MIP, SA, or matheuristics	Peak-driven operations; environments needing buffers (part-time, overtime, flex pools) to absorb uncertainty without inflating baseline staffing

The final approach adopted in this study is a hybrid simulation-optimization framework. First, stochastic demand generation and Discrete-Event Simulation are used to model airport lounge operations under time-varying customer arrivals. The simulation captures the main operational processes, including customer arrival, reception, lounge stay, buffet activity, table reset, dishwashing, food safety checks, and stock refill. The resulting operational workload is then converted into role-shift staffing requirements through workload-driver formulas. Based on these staffing requirements, the MILP model is developed to generate a feasible workforce roster. The MILP formulation assigns employees to specific days, shifts, and roles while enforcing key operational and labor constraints, including staffing coverage, skill eligibility, employee availability, monthly workday limits, night-shift limits, rest rules, and role assignment feasibility. This stage ensures that the roster satisfies the essential hard constraints required for practical implementation.

After the MILP solution is obtained, Simulated Annealing is applied as a post-optimization improvement phase. The purpose of SA is to refine the initial feasible roster by exploring local assignment changes, such as employee swaps and reassignment moves. This allows the model to reduce soft-constraint penalties related to overstaffing, employee preferences, workload imbalance, and roster quality. Therefore, the selected framework follows a structured sequence: simulation generates realistic staffing demand, MILP constructs a feasible roster, and SA improves the final roster quality.

2. System Design Description:

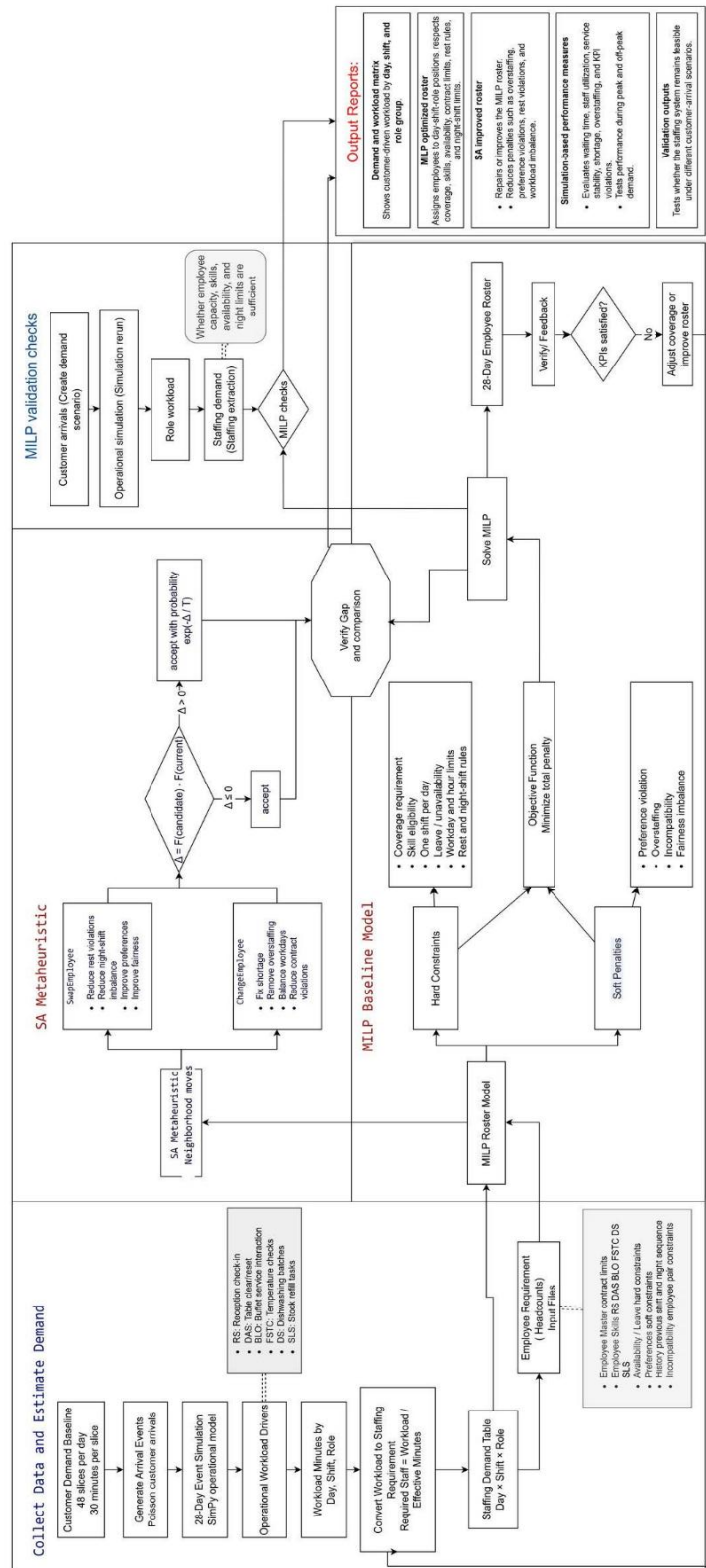


Figure III-1: System Description

The system description phase outlines the proposed workforce planning system for an airport lounge as an integrated Optimization-Simulation-Feedback (OSF) framework. The workforce planning function does not treat workforce planning as a static scheduling problem, but rather a series of connected decisions based on a customer's demand and an employee's verified staffing performance. To generate customer arrival patterns over a 48-slice daily demand baseline (where each slice represents a 30-minute time interval), the arrival rates will be simulated over a 28-day operational simulation, which includes all the activities in the lounge (i.e., checking in customers at the reception, clearing tables, interacting with the buffet, conducting food safety checks, washing dishes, refilling stock, and so on) in order to create the workloads for six operational role groups: RS, DAS, BLO, FSTC, DS and SLS. After generating workload data, shift level staffing requirements will be calculated based on effective working minutes per employee for each shift. The total number of shifts currently required for the 38 employees covering the current workload equates to 1,064 employee shifts in total over the 28-daily demand, given that the practical staffing pool available for work is estimated to be around 50 employees. The staffing demand will then be assigned by the MILP model to specific days, shifts, and role groups, while taking into account all required coverage, skill eligibility, availability, contract limits, one shift per day constraints, night shift limits, rest period rules, preferences, and incompatibilities. The last stage of the process is to apply a simulated annealing post-optimization repair that begins with the MILP roster and uses local change (e.g., employee swap or reassignment) to reduce penalties for employee shortage, overstaffing, rest period violations, night shift/employee imbalance, preference violations, and inaccessibility. Finally, customer arrivals will be scaled, the simulation will be rerun, and then the previously generated workloads will be placed against the workload-generated roster to determine if the workload/employee roster is feasible/reliable across a number of demand scenarios. Collectively, this evidence-based process allows for the conversion of customer arrivals to workload, the workload to staffing demand, the staffing demand to an optimized employee roster, and the optimized roster to validated service performance.

2.1. Formal Definition of the Buffet Lounge Workforce Planning Problem

The buffet lounge workforce planning problem is defined as the problem of determining how many employees are required for each operational role and assigning available employees to feasible shifts and roles while maintaining service-level performance under time-varying demand.

Unlike a conventional static staffing problem, the workload in this study is not assumed to be proportional only to total customer volume. Instead, customer arrivals generate different types of operational tasks across six role groups. Some tasks occur immediately when customers arrive, while others occur after a delay or when a system state reaches a threshold. Therefore, the problem requires a structured flow from demand generation to workload estimation, staffing requirement calculation, roster construction, and performance validation.

The problem consists of four connected layers:

1. **Demand layer:** customer arrivals are recored by time intervals based on flight-wave demand patterns.
2. **Process layer:** customer and operational activities are mapped into role-based tasks.
3. **Planning layer:** role-based workload is converted into shift-level coverage requirements and used by the MILP rostering model.
4. **Validation layer:** The generated schedule will be tested in a discrete-event simulation environment to evaluate its ability to maintain feasibility and model stability as customer arrivals increase or decrease according to different demand scenarios.

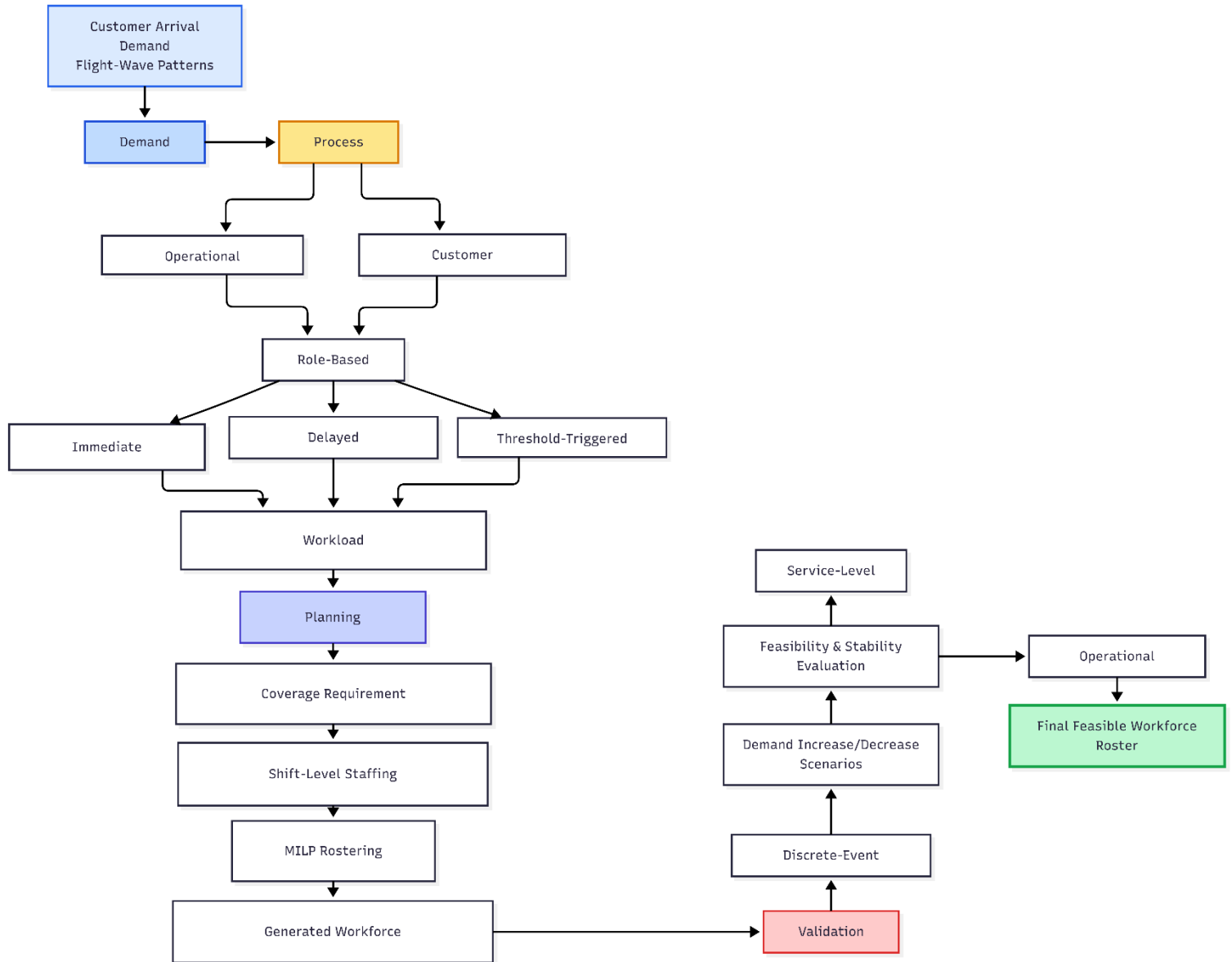


Figure III-2: Buffet Lounge Workforce Planning Layer

This structure ensures that the final roster is not only mathematically feasible but also operationally reliable when exposed to stochastic arrivals, task-time variation, and interdependent service bottlenecks

2.2. Operational Entities in the Intergrated Framework

The proposed framework defines the buffet lounge as an integrated workforce planning system consisting of six main operational entities. Customer arrivals generate service demand, but this

demand is not converted directly into staffing requirements. Instead, it is first translated into role-based operational tasks, then into workload and coverage requirements for roster optimization.

a) Shifts and Periods

A shift is an employee's scheduled working assignment defined by a fixed start time and duration. In this study, three standardized 8-hour shifts are used to cover the full 24-hour cycle: S1 (00:00-08:00), S2 (08:00-16:00), and S3 (16:00-24:00). A period is defined as a coarse day-part (night vs. day) to support pattern-based rostering rules such as consecutiveness and forbidden shift successions. Specifically, S1 is treated as a night period, while S2 and S3 represent daytime/evening periods. To ensure feasibility and compliance with labor regulations, the following rules are applied:

- Night consecutiveness: Employees can work a maximum of 3 consecutive night shifts (S1).
- Minimum rest and forbidden successions: A minimum rest period of 12 hours between shifts is enforced. Therefore, transitions such as S3 (ending at 24:00) followed by S2 (08:00) on the next day are prohibited.

These rules ensure fatigue-safe scheduling and regulatory compliance, while shifts are modeled as fixed 8-hour units for tractability in monthly rostering

b) Absences and non-working statuses

On each day of the planning horizon, an employee is assigned either a working shift or a non-working status. The main non-working statuses include day-off, leave/holiday, rest/recovery day (particularly following night sequences, if required by policy), and unavailability (e.g., reassigned duty, training, or temporary deployment). These statuses must be explicitly modeled because they affect feasibility and accounting rules differently. In particular, the model distinguishes between statuses that may contribute to working-day counters (e.g., paid leave, if applicable) and those that do not (e.g., unavailability due to redeployment), depending on organizational policy.

- Day-off: An unpaid day off automatically scheduled by the system to ensure employees have 8 days off within a 28-day cycle.
- Recovery Day: A mandatory day off immediately following a series of night shifts. This status takes higher priority than regular day-offs to ensure workplace safety.
- Paid Leave: Employees are absent from work but their absence is counted towards their total monthly working hours (to meet the 160-192 hour limit).
- Unpaid Leave: Employees take time off for personal reasons; this is not counted towards their working hours.

c) Skills and operational roles

In the buffet lounge setting, “skills” are represented as operational role profiles aligned with the lounge’s service chain. The main Function Areas are:

- **FOH (Front-of-House):** Reception & Seating, Dining Area Support
- **Food Service Line:** Buffet Line Operations, Food Safety & Temperature Control
- **BOH (Back-of-House):** Dishwashing & Sanitation, Service Logistics Support

The following table outlines the specialized role groups, their corresponding symbols, and the scope of duties required to maintain seamless service continuity across the Front-of-House, Food Service Line, and Back-of-House units

Table III-2: Role Groups Responsibility

Role Group	Symbol	Main Responsibility
Reception & Seating	RS	Guest arrival, check-in, access control, queue coordination, and seating support
Dining Area Support	DAS	Table clearing, table reset, dining area readiness, and guest-area support
Buffet Line Operations	BLO	Buffet replenishment, chafing-dish refill, line presentation, and station readiness
Food Safety & Temperature Control	FSTC	Temperature checks, food-safety logging, and corrective actions
Dishwashing & Sanitation	DS	Dirty ware collection, washing, sanitation, and clean dish recovery
Service Logistics Support	SLS	Runner support, utensil replenishment, clean ware redistribution, and support-item movement

Employees may be cross-trained, but cross-training does not imply full substitutability. An employee can only be assigned to a role for which he or she is qualified and operationally authorized. In the MILP model, each employee is assigned to one primary role group per shift, while short-term task switching within a shift is treated as an execution-level control decision.

d) Cross-Training and Controlled Substitutability.

In the airport buffet lounge context, cross-training serves as an important source of operational resilience because flight-wave demand may create short but intense workload imbalances across interdependent service streams. However, cross-training should not be interpreted as full interchangeability across all operational role groups. In practice, buffet lounge activities are governed

by role-specific qualification requirements, food-safety and hygiene procedures, service protocols, and execution-readiness standards, which limit the range of feasible substitutions. Accordingly, each employee is characterized not only by a primary or core role profile, but also, where applicable, by a restricted set of secondary trained roles that may be activated only within predefined qualification boundaries. This distinction is especially important in the FOH-Food Service Line-BOH service chain, where the ability to support another stream may be valuable during peaks, yet unrestricted reassignment could undermine compliance, coordination, and service consistency. Therefore, the rostering model adopts a controlled-substitutability perspective: employees may be assigned across role groups only when they are explicitly qualified and when such reassignment remains operationally acceptable under the approved substitution structure. In the monthly roster, each employee is assigned to one primary role group per shift, while finer intra-shift redeployment across tasks or zones is treated as an execution-level control decision supported by supervisory judgment and the skills matrix. This modeling view preserves realism by recognizing cross-training as a bounded flexibility mechanism that improves robustness under fluctuating demand, without imposing the unrealistic assumption that all employees can freely substitute for one another across FOH, Food Service Line, and BOH activities.

e) Contracts and labor rules

Each employee is subject to an employment contract and the applicable labor regulations, which jointly determine the feasibility boundaries of the rostering problem in a 24/7 airport buffet lounge characterized by flight-driven demand fluctuations. These rules establish the legal and organizational framework within which staff schedules must be constructed.

- **Workload limits (Hard constraints):** In accordance with the Vietnam Labor Code 2019 (Law No. 45/2019/QH14), regular working time must not exceed 8 hours per day or 48 hours per week, while total daily working time, including overtime, must remain within 12 hours. In addition, overtime is limited to a maximum of 40 hours per month. Employees are also entitled to at least 24 consecutive hours of rest per week, together with statutory leave entitlements. Because these provisions represent mandatory legal requirements, they are modeled as hard constraints and must be strictly satisfied in all feasible rosters.
- **Sequence limits (Soft constraints):** Sequence-related rules include limits on consecutive working days, consecutive days off, and broader work–rest patterns. In the airport buffet context, where staffing demand may temporarily intensify due to flight-wave peaks, these conditions are more appropriately treated as soft constraints. This modeling choice preserves

a degree of operational flexibility, allowing the roster to absorb short-term demand surges while penalizing undesirable outcomes such as excessive workload concentration, insufficient recovery patterns, or inequitable scheduling structures.

- **Night and rest rules (Hard constraints):** Minimum rest between consecutive shifts is enforced as a hard constraint. Under the Vietnam Labor Code 2019 (Article 110), employees working in shifts must receive at least 12 consecutive hours of rest before moving to a subsequent shift. This requirement is especially critical in a 24-hour lounge environment involving night-shift operations, where fatigue risk is inherently higher. Accordingly, rest-related rules are treated as non-negotiable feasibility conditions.

Taken together, these contractual and regulatory provisions ensure that the roster remains legally compliant and operationally feasible, while allowing limited flexibility through soft sequence-related rules to support service continuity and staffing responsiveness in the airport buffet environment.

f) Employees and history

Employees are characterized by their eligible roles/skills, feasible shifts under contract, and (where relevant) a standard role/shift profile used to support operational stability. The model also incorporates history information from the end of the preceding horizon in order to evaluate consecutiveness, forbidden successions, and night-recovery requirements consistently across month boundaries and to avoid infeasible patterns at the start of the month.

g) Pre-assignments, requests, and preferences

The rostering system supports employee-specific requirements and desirability statements, including pre-assignments, approved fixed requirements, and preference-based requests.

- **Pre-assignments and fixed requirements (Hard constraints):** These include assignments that are operationally determined or formally approved in advance, such as mandatory working days/shifts (e.g., “must-work” due to staffing needs or approved commitments). These are treated as hard constraints and must be strictly satisfied.
- **Requests and preferences (Soft constraints):** Employee requests, including preferred working days/shifts or time-off requests (day-off or shift-off), are treated as soft constraints. These requests reflect individual preferences rather than mandatory requirements; therefore, violations are allowed but penalized in the objective function based on priority levels.

This distinction ensures that only validated operational requirements are enforced strictly, while maintaining flexibility to accommodate employee preferences where feasible..

h) Workteams and incompatibilities

To capture coordination effects and interpersonal constraints, employees may be grouped into workteams (preferred to work together) and/or specified as incompatible pairs (to be avoided within the same shift). These relations may be assigned priority levels and are typically modeled as soft constraints unless organizational policy requires strict separation.

i) Coverage Requirements

Coverage requirements are specified for each day of the month, each shift (S1-S3), and each role group in the airport buffet lounge. In this study, coverage is measured as the number of employees assigned to a role group within a given shift on a given day. The six role groups are Reception & Seating (RS), Dining Area Support (DAS), Buffet Line Operations (BLO), Food Safety & Temperature Control (FSTC), Dishwashing & Sanitation (DS), and Service Logistics Support (SLS). These requirements are derived from key service level targets rather than being set arbitrarily. In airport operations, IATA states that service level agreements should define key operational service standards and performance measurements, while the exact thresholds should be determined locally. IATA's Level of Service framework also emphasizes queueing time and capacity sufficiency at passenger touchpoints, and ACI similarly uses ASQ/SLA frameworks to define and monitor airport service quality.

Table III-3: Service-Level KPI Targets Used for Evaluation

Role Group	Time - Based KPI	Unit	Max Allowed	Service Level (%)
Reception & Seating (RS)	Guest wait time from arrival at reception to check-in/seating completion	minutes	5	95%
Dining Area Support (DAS)	Table turnover time (clear, wipe, reset, ready)	minutes/table	4	90%
Buffet Line Operations (BLO)	Replenishment response time after low-stock / stock-out alert	minutes	6	95%
Food Safety & Temp Control (FSTC)	Frequency of regular temperature checks and food incident handling.	minutes/time	30	100%

Dishwashing & Sanitation (DS)	Time from urgent clean-dish shortage signal to first replenished batch available	minutes	10	95%
Service Logistics Support (SLS)	Response time to replenish clean wares, utensils, and service support items at point of use	minutes	5	95%

After the service-level targets are defined, they are converted into staffing requirements through three coverage bands:

- Minimum coverage to prevent service disruption.
- Target/ optimal coverage to stabilize service quality.
- Maximum coverage as a managerial tolerance for overstaffing or a soft upper bound for labor-efficiency control.

Accordingly, these coverage bands represent the operational trade-off between maintaining reliable service during peak flight-wave demand and limiting unnecessary labor cost during off-peak periods. In this sense, coverage requirements are the staffing translation of service-level commitments for each role group in the airport buffet lounge.

2.3. Structural Properties

The buffet rostering problem exhibits several structural characteristics that shape model formulation and motivate the inclusion of specific constraint families.

- **24/7 operations with mandatory night staffing:** Continuous service requires feasible staffing across all shifts, including the overnight shift S1, which typically induces stricter recovery and rest considerations.
- **Interdependent service chain:** FOH, Food Service Line, and BOH activities are tightly coupled; shortages in BOH (dishwashing and hygiene) can propagate to buffet readiness, clean-ware availability, and table turnover, implying that staffing adequacy must be evaluated by role group rather than total headcount alone.
- **Restricted substitutability under multi-skilling:** Cross-training increases flexibility but does not imply full interchangeability; role eligibility and compliance practices constrain feasible assignments and must be explicitly modeled.
- **Time-varying flight-wave demand:** Airport lounge demand is highly time-dependent, with customer arrivals concentrated around short flight-wave peaks rather than distributed evenly

across the operating day. This requires staffing requirements to be estimated from short-interval workload patterns rather than from average shift-level demand alone.

- **Using Discrete-Event Simulation (DES) to Validate Reaction to Variability:** The model is creating a replica of what passenger volumes may look like by utilizing Discrete Event Simulation (DES) to simulate fluctuations in passenger volumes based upon flight/timetable waves and associated service demands for each operational role group. This study, then, evaluates how well the optimally developed roster will provide KPIs of service quality within uncertain operational conditions (e.g., passenger arrival variations, service time variations, and role-specific bottlenecks). Simulation results test the robustness of the overall model and give further direction for modifying coverage requirements as needed.

2.4. Constraint Classification

To make the constraint set interpretable and operationally meaningful, constraints are grouped by the stakeholder most directly affected: Administration/Operations (A), Employees (N), and Service Quality/Guests (P). This classification clarifies trade-offs between cost control, fatigue-safe staffing, and service stability across the FOH-Food Service Line-BOH service chain.

Hard and Soft Constraints (Implementation Rule)

Constraints are separated into hard constraints (non-negotiable feasibility and compliance rules) and soft constraints (violations allowed but penalized). In this capstone, the hard-constraint set is explicitly defined in the modeling/solution section, while remaining constraints are penalized using configurable weights to reflect managerial priorities.

Table III-4: Staffing Constraints Classification (Hard vs. Soft)

Code	Constraint name	Meaning	Type
A1	Minimum Coverage by Day, Shift, and Role Group	For each day and shift, staffing must satisfy the minimum required coverage for each role group to ensure basic operational feasibility. Understaffing in the night shift may be penalized more heavily where operational redundancy is lower.	Hard
A2	Maximum Coverage/Overstaffing Control	Staffing should remain within the maximum desirable coverage for each role group in order to avoid unnecessary labor cost and inefficiency.	Soft



A3	Assignment Feasibility and Qualification Compliance	An employee may only be assigned to shifts and role groups for which they are qualified, contractually admissible, and operationally eligible.	Hard
A4	Single Assignment per Day	Each employee can be assigned to at most one shift per day.	Hard
A5	Monthly Working-Day Bounds	Each employee must satisfy the minimum and maximum number of working days over the planning horizon in accordance with company policy or contract rules.	Hard
A6	Monthly Working-Time Bounds	Each employee must satisfy the minimum and maximum total working hours over the planning horizon.	Hard
N1	Work-Rest Structure and Fatigue Control	The roster must regulate consecutive working days, consecutive days off, consecutive night shifts, minimum rest between shifts, recovery after night-shift blocks, and forbidden high-fatigue shift successions. Historical information from the previous planning horizon should be incorporated to ensure consistency across month boundaries.	Primarily Hard
N2	Fair Distribution of Undesirable Duties	Undesirable duties, such as night shifts or other demanding assignments, should be distributed as fairly as possible across employees.	Soft
N3	Fixed Availability and Leave Requirements	Fixed leave requests, declared unavailability, and mandatory non-working periods must be respected.	Hard
N4	Controlled Flexibility Across Role Groups	Cross-role assignment flexibility may be allowed only within predefined qualification boundaries and approved substitution structures; full interchangeability across role groups is not assumed.	Soft

N5	Assignment Stability and Standard-Pattern Consistency	The schedule should discourage excessive switching across role families and unnecessary deviation from standard role or shift patterns in order to improve execution consistency and reduce supervision burden.	Soft
N6	Workload Fairness/ Concentration Control	If overtime is operationally permitted, it must remain bounded and should not be excessively concentrated on a small subset of employees.	Soft
N7	Employee Preferences	Non-fixed requests and employee preferences should be considered where feasible, with violations treated through penalty terms.	Soft
N8	Employee Pairing / Incompatibility	Certain pairs of employees may be restricted from working together in the same shift due to operational, interpersonal, or policy considerations. This constraint prevents incompatible employee combinations from being assigned simultaneously within the same shift.	Soft
P1	Target Coverage for Service Stability	Beyond minimum feasibility, the model should aim to achieve target staffing levels by day, shift, and role group in order to reduce congestion, stock-outs, dishwashing backlogs, delayed cleaning recovery, and general service instability.	Soft
P2	Role-Balanced Coverage Across the Service Chain	The model should discourage staffing patterns that satisfy total headcount requirements but under-resource a critical stream in the service chain, such as FOH, FSL, or BOH.	Soft

2.5. Design Structure System

The System Structure Design phase defines how the proposed airport buffet lounge workforce planning system should operate after redesign. Based on the system description, the system is not designed as a simple manual staffing or static rostering tool. Instead, it follows an integrated Optimization-Simulation-Feedback structure, where customer demand is first translated into

operational workload, then into staffing requirements, then into a feasible monthly roster, and finally validated through simulation. This design is necessary because airport lounge demand changes by time of day and is strongly affected by flight-wave peaks. Therefore, the system must capture short-term operational congestion while still producing practical 8-hour shift rosters for management use. The structure of the system consists of four connected layers: the demand layer, process layer, planning layer, and validation layer. In the demand layer, customer arrivals are generated by time intervals. In the process layer, customer and system events are converted into role-based operational tasks. In the planning layer, workload is converted into day–shift–role staffing requirements and optimized through MILP. In the validation layer, the roster is tested through discrete-event simulation to check whether MILP model remain acceptable under stochastic arrivals and variable task times.

2.5.1. To- Be Framework (Operational Logic)

The To-Be framework represents the future operating logic of the Workforce Planning System. The current/traditional approach to staffing relies primarily upon fixed assumptions, manager experience, and peak capacity estimates; whereas the To-Be model relies primarily upon workload evidence to drive staffing decisions. Customer arrivals do not translate directly into staff numbers; they trigger different operating tasks that occur across the lounge service chain and are then assigned to the appropriate role group.

The To-Be operational logic can be described as:

Customer Demand → Arrival Events → Operational Drivers → Workload Minutes → Staffing Requirements → MILP Roster → Simulation Feedback.

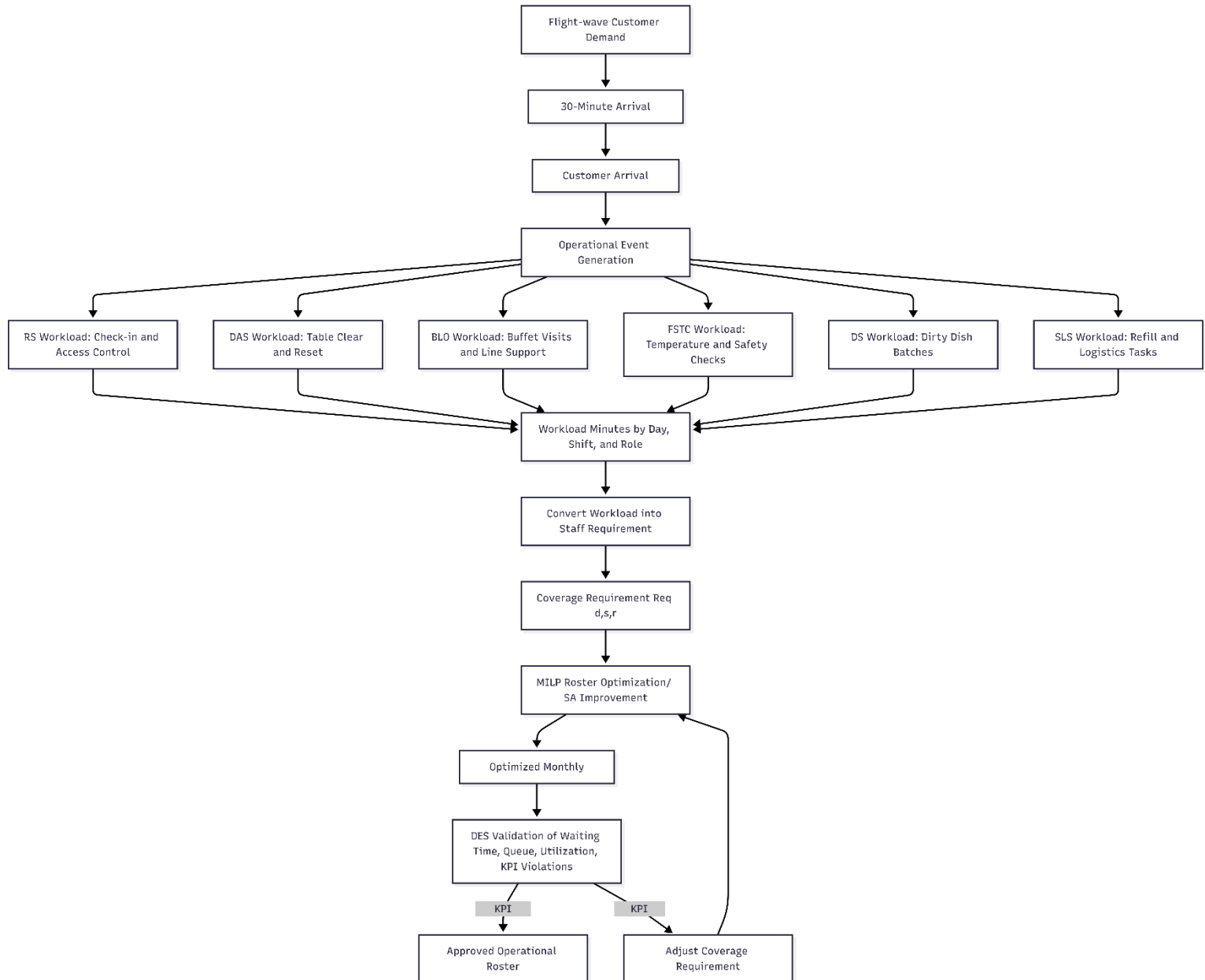


Figure III-3: Integrated Optimization-Simulation-Feedback (OSF) Framework for Demand-Driven Lounge Rostering and Task Coverage

This means that the process begins with a time-based customer demand in 30-minute intervals. Once customers arrive, they create operational events such as reception check in, buffet visits, clearing tables, dishwashing demand, food safety checks and refilling activities. These events lead to workload minutes for 6 different operational role groups: Reception & Seating (RS), Dining Area Support (DAS), Buffet Line Operations (BLO), Food Safety and Temperature Control (FSTC), Dishwashing

and Sanitation (DS), and Service Logistics Support (SLS). The workload driver report indicates that staffing is not allocated by fixed percentages; each role is calculated based on its specific workload driver such as arrivals for RS, table reset tasks for DAS, dish batches for DS, and refill events for SLS.

Thus, the operating logic will utilize a workload-based calculation. For each role and shift, workload will be calculated by multiplying the number of tasks by their average service time per task; then the number of staff required will be calculated by dividing the total workload minutes by the effective working minutes for each employee for each shift; while also observing minimum staffing requirements and KPI validation specifications. For example, the employee input logic utilizes an effective capacity of 408 minutes per staff shift based on an 8-hour shift at target utilization of 0.85. The MILP Model serves as the decision engine within the To-Be framework. The MILP model receives staffing requirements by day, shift and role and assigns employees to the feasible shift-role combinations that meet the skill eligibility and availability as well as the contract rules, rest rules, maximum working hours per month, and one shift per employee per day restrictions. Preferences, overstaffing, fairness and stability of employees will be treated as soft constraints when submitted managerial flexibility is acceptable. This distinction between "hard constraints" and "soft constraints" enables the system to minimize legal and operational feasibility while achieving a balance between service quality, employee fairness and labour efficiency.

The To-Be framework also has an element of feedback. Once the MILP roster is developed it is not considered acceptable solely for mathematical feasibility; it must be validated in a simulated environment to assess waiting time, queue length, utilization, service stability and KPI violations. If the simulation indicates weak service during peak periods, then staffing requirements can be adjusted and the optimization process performed again. The feedback loop creates a more robust staffing system than a static roster as it validates whether a given schedule can provide acceptable service levels under varying levels of customer arrival uncertainty and operational variability.

2.6. Techniques Justification

The selected hybrid technique is justified by the need to balance explainability, computational scalability, and operational realism. The MILP-style formulation provides a transparent structure for encoding feasibility logic decision variables, constraints, and penalty-weighted objectives while also supporting benchmarking and systematic violation of accounting. SA is adopted as the primary operational engine because it can explore the discrete, highly constrained roster space efficiently using practical neighborhood moves, accepting occasional non-improving steps to escape local

minima and improve solution quality within acceptable runtime. After the optimization model generates a mathematically feasible roster, Discrete-Event Simulation (DES) is employed to evaluate whether the roster can operate effectively under realistic operational conditions. Specifically, DES reproduces the stochastic fluctuations of customer arrivals associated with flight-wave patterns, which in turn generate time-varying service demand across different operational role groups, such as Reception & Seating, Buffet Line Operations, and Dishwashing & Sanitation. During periods of sudden demand surges, the workload of each operational unit changes dynamically and may create service bottlenecks, including shortages of clean dishes, delayed buffet replenishment, or increased customer check-in waiting times. DES enables the observation of how the roster responds to these fluctuations through service-level KPIs such as customer waiting time, table turnover time, and replenishment response time. Consequently, the study evaluates not only the mathematical feasibility of the roster but also its operational robustness under uncertain conditions. If the simulated KPIs fail to satisfy the targeted service levels, the DES results provide structured feedback for adjusting coverage requirements or re-optimizing the roster to improve operational performance.

2.7. Key Advantages

The proposed integrated framework provides key benefits to managers, in terms of workforce planning in a 24/7 airport buffet lounge environment with fluctuating flight-wave demand and operational uncertainty. The framework provides managers with a greater level of decision transparency, by combining a MILP-style formulation with explicit constraints and penalty structures. This allows managers to clearly see how their staffing decisions are arrived at, which constraints were violated, and the trade-offs involved in creating the final roster. The framework enhances the practical scalability and computational efficiency of the model. The underlying SA optimization engine allows the model to efficiently explore large, heavily constrained roster spaces and produce high-quality schedules within acceptable time limits. In addition, the logic behind the constraints remains interpretable and operationally meaningful. The framework increases the operational robustness of the generated rosters, through the process of DES-based validation. Rather than evaluating rosters solely based on mathematical feasibility, the model further tests whether the generated roster can meet targeted service level KPI's, given stochastic customer arrivals, fluctuating workloads, and role-specific bottlenecks. This reduces the likelihood of deploying rosters that appear feasible in theory but will fail to meet the operational requirements of the business. Lastly, the framework supports continuous improvement, through the use of a feedback-driven optimization process. Deviations from KPI's, identified during DES simulation, can be systematically converted into adjustments to



coverage requirements or optimization parameters, allowing the roster to be refined iteratively and the operational performance of the business to improve progressively over multiple planning cycles.

CHAPTER IV: PROTOTYPE DEVELOPMENT AND IMLEMENTATION

1. Prototype Development:

1.1. Data Collection

1.1.1. Airport Lounge Staffing Estimation

a) Customer Arrival Estimate

First, the daily passenger demand profile is constructed using a combination of Gaussian demand peaks, which represent airport flight-wave patterns across different periods of the day. The resulting demand weights are normalized so that the total expected number of customers equals 900 per day.

Table IV-1: Total Demand per Month

Day	Day type	Customer demand
1	Weekday	921
2	Weekday	1,007
3	Weekend	960
4	Weekend	1,055
5	Weekday	880
6	Weekday	928
7	Weekday	874
8	Weekday	943
9	Weekday	904
10	Weekend	1,055
11	Weekend	1,048
12	Weekday	845
13	Weekday	829
14	Weekday	903
15	Weekday	920
16	Weekday	876
17	Weekend	935
18	Weekend	1,102
19	Weekday	919
20	Weekday	854

21	Weekday	821
22	Weekday	889
23	Weekday	870
24	Weekend	1,042
25	Weekend	1,012
26	Weekday	912
27	Weekday	900
28	Weekday	889
Total	-	26,093

Then, each 30-minute interval is treated as a piecewise-constant segment of a Non-Homogeneous Poisson Process, where the arrival rate λ_i varies by time slice. Actual customer arrivals are generated using a Poisson distribution, and individual arrival times are subsequently assigned within each interval using a uniform distribution. As a result, the model produces both slice-level arrival data and event-level customer arrival records suitable for downstream simulation and staffing analysis.

Table IV-2: Customer Arrival per Slice (Sample Data)

slice_id	start_time	end_time	shift_id	slice_in_shift	demand_level	actual_arrivals
1	00:00	00:30	1	1	Low	14
2	00:30	01:00	1	2	Low	9
3	01:00	01:30	1	3	Low	3
4	01:30	02:00	1	4	Low	6
5	02:00	02:30	1	5	Low	4
6	02:30	03:00	1	6	Low	8
7	03:00	03:30	1	7	Low	1
8	03:30	04:00	1	8	Low	7
9	04:00	04:30	1	9	Low	8
10	04:30	05:00	1	10	Low	6
11	05:00	05:30	1	11	Low	9
12	05:30	06:00	1	12	Medium	18
13	06:00	06:30	1	13	High	27
14	06:30	07:00	1	14	Peak	37
15	07:00	07:30	1	15	Peak	40
16	07:30	08:00	1	16	Peak	29

b) Operational Capacity

Table IV-3: Summary of Data Collection

Data group	Description	Source / basis	Purpose
Customer arrival pattern	The operating day was divided into 48 time slices, with each slice representing 30 minutes.	Airport passenger-flow logic inspired by Postorino et al. (2019).	To create the baseline daily demand pattern.
Baseline demand	The base scenario used 900 expected lounge customers per day.	Scenario-based airport lounge demand Tan Son Nhat airport.	To estimate expected arrivals by time interval.
Stochastic arrivals	Actual arrivals were generated using a Poisson distribution, and arrival times were assigned within each slice.	Nonhomogeneous Poisson process logic based on Kim and Whitt (2014).	To represent uncertainty in customer arrivals.
Shift structure	The day was divided into three 8-hour shifts: 00:00-08:00, 08:00-16:00, and 16:00-24:00.	Operational planning SASCO lounge.	To connect customer demand with staffing periods.
Simulation horizon	The one-day baseline was expanded into a 28-day simulation.	Simulation design assumption.	To create the planning horizon for workforce rostering.
Lounge operation process	The simulation included admission, reception, lounge stay, buffet visits, table reset, dishwashing, food safety checks, and stock refill.	Discrete-event simulation logic using SimPy.	To convert customer flow into operational workload.
Service and capacity assumptions	The model used lounge capacity, service times, wait KPI targets, stock	Lounge Buffet SASCO operating conditions and	To reflect practical service constraints.

	rules, dishwashing rules, and minimum staffing.	service-capacity assumptions.	
Role-based workload	Workload was estimated for RS, DAS, BLO, FSTC, DS, and SLS based on each role's operational driver.	Workload-driver staffing logic.	To calculate workload by role and shift.
Staffing demand input	Workload minutes were converted into required staff and summarized by role and shift.	Workload-based staffing calculation.	To provide demand input for the MILP workforce rostering model.

The data collection process combined literature-based arrival logic and scenario-based operational assumptions to create a complete input dataset for simulation and workforce planning. Customer arrivals were first generated through a 48-slice daily demand structure and Poisson-based stochastic arrival logic. The baseline demand was then extended into a 28-day event-driven simulation, where customer activities were converted into role-specific workload. The final simulation outputs were used to estimate staffing demand by role and shift, which became the input for the MILP rostering model.

Table IV-4: Staffing capacity assumptions by role group

Role group	Symbol	Recommended planning capacity	Unit
Reception & Seating	RS	100	guests/hour/staff
Dining Area Support	DAS	24	guests/staff
Buffet Line Operations (guest-based)	BLO	25	guests/staff
Food Safety & Temp Control	FSTC	90	guests/staff
Dishwashing & Sanitation (standard machine)	DS	200	guests/hour/staff
Service Logistics Support	SLS	3	dishes/staff

Table IV-2 shows the assumed service capacity for each staff role in the airport lounge. These assumptions help estimate how many employees are needed based on the number of guests or tasks. Each role has a different capacity because the workload of each position is not the same. For example,

Reception and Seating staff (RS) can serve around 100 guests per hour, while Dining Area Support (DAS) is assumed to handle 24 guests per staff member. Buffet Line Operations (BLO) can support 25 guests per staff member, and Food Safety and Temperature Control (FSTC) can cover 90 guests per staff member. Dishwashing and Sanitation (DS) has a higher capacity because the work is partly supported by a standard machine, while Service Logistics Support (SLS) is measured by the number of dishes handled. Overall, this table provides the basic capacity assumptions used to calculate staffing demand for each role group.

Table IV-5 Baseline Workload Limits by Role Group (28 Days)

Role	Working Hour/ Months	Working Hour/ Months	Equivalent Shifts/ Working Days	Std. Hours / Day	Max Hours / Day (incl. OT)	Max OT / Month
Reception & Seating (RS)	160	192	20 - 24	8hrs	12 hrs	40 hrs
Dining Area Support (DAS)	160	192	20 - 24	8 hrs	12 hrs	40 hrs
Buffet Line Operations (BLO)	160	192	20 - 24	8 hrs	12 hrs	40 hrs
Food Safety & Temp Control (FSTC)	160	192	20 - 24	8 hrs	12 hrs	40 hrs
Dishwashing & Sanitation (DS)	160	192	20 - 24	8 hrs	12 hrs	40 hrs
Service Logistics Support (SLS)	160	192	20 - 24	8 hrs	12 hrs	40 hrs

Table IV-3 presents the workload limits for each role group during the 28-day planning period. The same workload policy is applied to all roles, including RS, DAS, BLO, FSTC, DS, and SLS. Each employee is expected to work between 160 and 192 hours per month, which is equal to about 20 to 24 working shifts if one shift is 8 hours. The standard working time per day is 8 hours, while the maximum working time, including overtime, is 12 hours per day. The maximum overtime is limited

to 40 hours per month. These limits are important because they prevent the roster from assigning too many or too few working hours to employees. They also help make sure the staffing plan is realistic and follows the basic labor capacity assumptions.

Table IV-6: Customer Arrival and Simulation Setting

Input	Value	Purpose
Operating hours	24 hours	Defines one full airport lounge operating day
Time slice	30 minutes	Divides customer demand into small time intervals
Number of slices per day	48 slices	Represents 24 hours with 30-minute intervals
Shift length	8 hours	Main staff planning period
Number of shifts per day	3 shifts	S1: 00:00–08:00, S2: 08:00–16:00, S3: 16:00–24:00
Expected daily customers	900 customers	Baseline demand scenario
Simulation horizon	28 days	Represents one monthly planning period
Warm-up day	1 day	Excluded from KPI and staffing summary
Weekday multiplier	1.00	Normal demand level
Weekend multiplier	1.15	Higher demand on weekends
Holiday multiplier	1.35	Higher demand on holidays
Daily random variation	0.05 CV	Adds uncertainty to daily arrivals

The data collection process was developed to provide the input foundation for both the airport lounge simulation and the MILP workforce rostering model. Since actual operational records were not available, this study used a combination of literature-based arrival logic and scenario-based operational assumptions. The customer demand input was created by dividing one 24-hour operating day into 48 time slices, where each slice represents 30 minutes. The baseline scenario assumed 900

expected lounge customers per day, and the demand was later expanded into a 28-day simulation using weekday, weekend, holiday, and random daily multipliers

Table IV-7: Service and Capacity Assumptions

Input group	Value / assumption	Purpose
Lounge capacity	200 concurrent customers	Controls admission capacity
Maximum admission wait	30 minutes	Allows customers to leave if waiting too long
Reception service time	TRIA(0.15, 0.25, 0.40) min	Generates RS workload
Buffet service time	TRIA(0.50, 0.75, 1.20) min	Generates BLO workload
Customer dwell time	min(180, TRIA(45, 90, 180)) min	Defines how long customers stay in the lounge
DAS table reset time	7.1 min/task	Generates DAS workload
FSTC service time	TRIA(2.0, 3.0, 5.0) min	Generates food safety check workload
SLS refill time	TRIA(3.0, 4.5, 6.0) min	Generates stock refill workload
Dirty dishes per customer	Poisson(3.0), at least 1	Generates dishwashing workload
Dish batch size	80 dishes	Triggers dishwashing batch
Stock reorder point	80 units	Triggers stock refill task
Stock refill quantity	220 units	Defines restocking amount

c) Airport Lounge Simulation Model Design

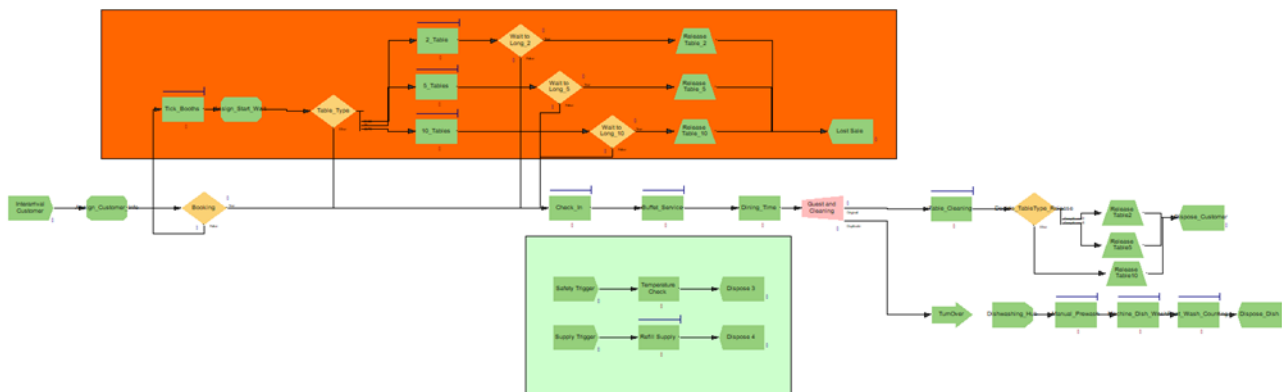


Figure IV-1: Airport Lounge Discrete-Event Simulation Model

Figure IV-1 presents the discrete-event simulation model for the airport lounge operation. The Airport Lounge Simulation Model Design (model) was created to illustrate how passengers and services would operate in an airport lounge. The model describes the full customer flow from arrival, booking, ticket booth/check-in, table assignment, buffet service, dining, table cleaning, and customer departure. In the table allocation section, customers are assigned to different table sizes such as 2-seat, 5-seat, and 10-seat tables. If the waiting time is too long, some customers may leave the system as lost sales. The model was designed to support the assignment of proposed service and capacity assumptions to the relevant operational processes (e.g., reception, buffet service, dining duration, table reset, food safety checks, stock replenishment and dish washing) associated with each parameter's corresponding event in the model so that the study could evaluate if each of the operational feasibility and operational workload assumptions for each role could be produced from the model. For example, occupancy pressure would be evaluated using lounge capacity and customer dwelling times, reception service times would be used to assess workload for the reception staff, table reset times would be used to evaluate demodulated workload for the dining staff and dish generation and batch size would be used to evaluate workload for dishwashing. Therefore, the simulation model serves as a validation mechanism between theoretical assumptions and staffing requirements.

Due to software availability and execution limitations of a complete data simulation in Arena, the model was created as a conceptual simulation model to demonstrate how theoretical assumptions would be incorporated in the logical operation of an actual system without being limited to Arena. As a result, the model will assist the research to identify the justification for input parameters and to explain the conversion of operational workload from passenger arrivals to an operational workload for all staff roles.

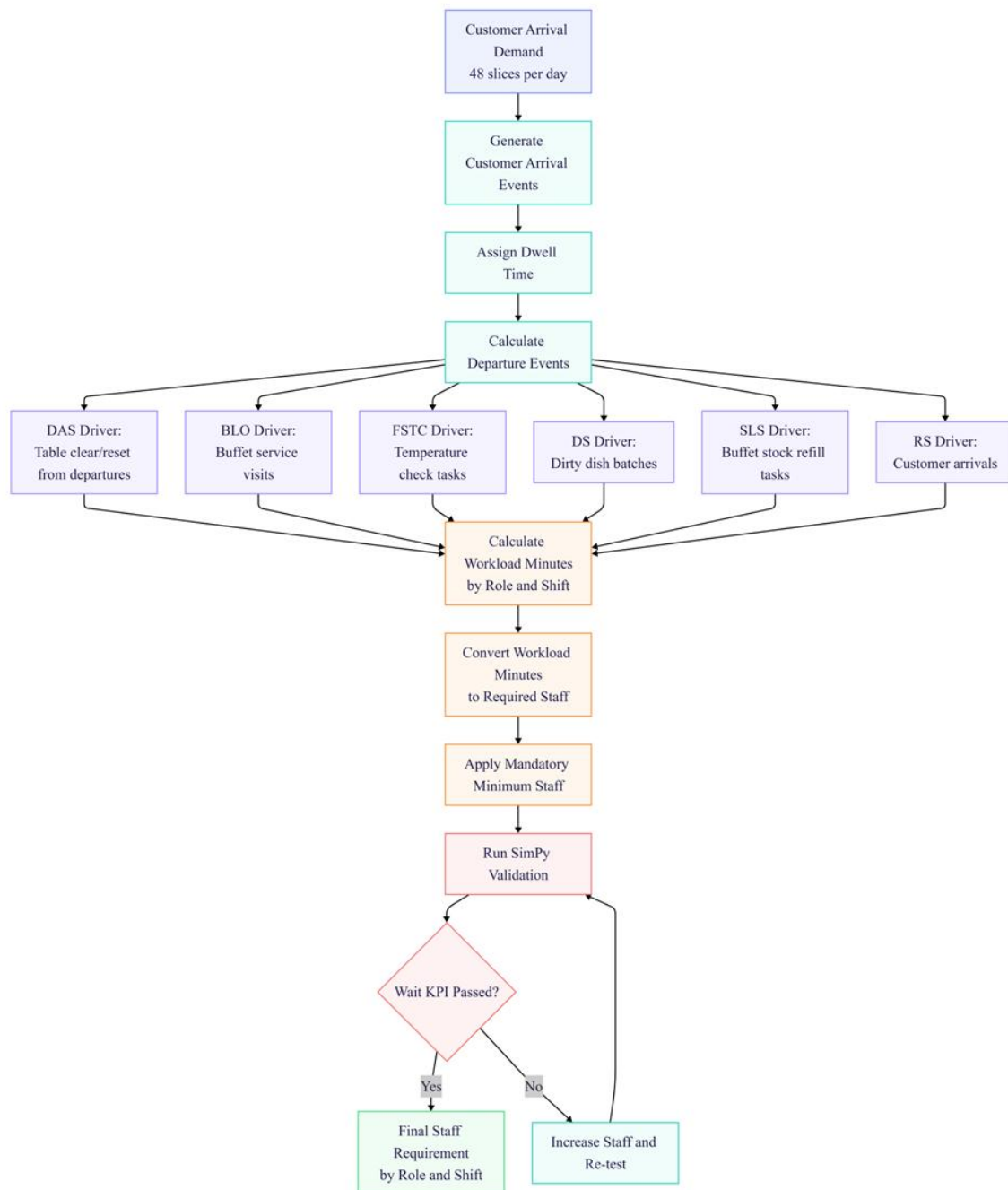


Figure IV-2: Logic for converting customer demand into staff requirements.

Figure IV-2 shows the logic used to convert customer demand into staffing requirements. The process starts from customer arrival demand, which is divided into 48 time slices per day. After arrival events are generated, the model assigns dwell time and calculates departure events. These events are then used to identify workload drivers for each role, such as customer arrivals for RS, table reset tasks for

DAS, buffet visits for BLO, temperature checks for FSTC, dirty dish batches for DS, and stock refill tasks for SLS.

From these drivers, the model calculates workload minutes by role and shift, then converts them into required staff. A minimum staffing rule is applied, and the result is validated using SimPy based on waiting-time KPIs. From this logic, the 28-day active task workload minutes table is generated and used as the main input for staffing demand.

Table IV-8: Workload minute Sample (day x shift x role)

day_index	shift_id	shift_time	role	workload_minutes
1	1	00:00-08:00	BLO	38.47
1	1	00:00-08:00	DAS	497.00
1	1	00:00-08:00	DS	54.75
1	1	00:00-08:00	FSTC	25.70
1	1	00:00-08:00	RS	55.63
1	1	00:00-08:00	SLS	4.71
1	2	08:00-16:00	BLO	78.51
1	2	08:00-16:00	DAS	1732.40
1	2	08:00-16:00	DS	192.62
1	2	08:00-16:00	FSTC	27.36
1	2	08:00-16:00	RS	86.05
1	2	08:00-16:00	SLS	16.75

d) Role-Based Workload Driver Formulas

In this study, customer demand was converted into staffing requirements through a workload-driver approach. Instead of allocating staff by fixed percentages, each role group was calculated based on the operational activity that directly creates its workload. The process starts from customer demand, generates arrival and service events, converts these events into workload minutes, and then estimates the required number of staff by role and shift.

The general workload formula is:

$$W_{r,s} = Q_{r,s} \times T_r$$

where $W_{r,s}$ is the workload minutes for role r in shift s , $Q_{r,s}$ is the workload driver quantity, and T_r is the average processing time per task. After the workload minutes are calculated, the required staff is estimated as:

$$\text{Staff}_{r,s} = \max \left(1, \left\lceil \frac{W_{r,s}}{E} \right\rceil \right)$$

where E is the effective working capacity of one employee per shift. In this model, the effective capacity used for staffing conversion is:

$$E = 8 \times 60 \times 0.85 = 408 \text{ minutes}$$

For each role, the demand is converted into workload as follows:

$$W_{RS,s} = A_s \times 0.267$$

where A_s is customer arrivals in shift s , and 0.267 minutes is the average reception service time.

$$\text{TableTurns}_s = \text{Departures}_s \times 0.65$$

$$W_{DAS,s} = \text{TableTurns}_s \times 7.1$$

where DAS workload is driven by table clear/reset tasks, with a table reset rate of 0.65 and 7.1 minutes per task.

$$\text{BuffetVisits}_s = A_s \times 1.5$$

$$\text{BLOServiceEvents}_s = \text{BuffetVisits}_s \times 0.20$$

$$W_{BLO,s} = \text{BLOServiceEvents}_s \times 0.817$$

where BLO workload is based on buffet visits, but only 20% of visits are assumed to require staff intervention because the buffet is mainly self-service.

$$\text{Checks}_s = \frac{480}{\text{CheckInterval}}$$

$$W_{FSTC,s} = \text{Checks}_s \times 3.333$$

where FSTC workload is based on scheduled temperature checks during each 8-hour shift.

$$\text{Batches}_s = \frac{\text{DirtyDishes}_s}{80}$$

$$W_{DS,s} = \text{Batches}_s \times 13.46$$

where DS workload is based on dishwashing batches, with one batch containing 80 dishes and an average processing time of 13.46 minutes.

$$W_{SLS,s} = \text{RefillEvents}_s \times 4.5$$

where SLS workload is based on buffet stock refill events, with an average refill time of 4.5 minutes.

- **Total staff-shift slots:**

$$\text{TotalSlots} = \sum_d \sum_s \sum_r \text{Req}[d, s, r]$$

After the required staff by day, shift, and role has been calculated, the next step is to estimate how many employees are needed in the employee pool. This step is necessary because the MILP model must know the available number of employees before assigning them to shifts and roles.

Table IV-9: Staff Requirement based on Workload

day_index	shift_id	role	workload_minutes	required_staff_by_workload
1	1	BLO	38.47	1.00
1	1	DAS	497.00	2.00
1	1	DS	54.75	1.00
1	1	FSTC	25.70	1.00
1	1	RS	55.63	1.00
1	1	SLS	4.71	1.00
1	2	BLO	78.51	1.00
1	2	DAS	1732.40	5.00
1	2	DS	192.62	1.00
1	2	FSTC	27.36	1.00
1	2	RS	86.05	1.00
1	2	SLS	16.75	1.00
1	3	BLO	99.13	1.00
1	3	DAS	1838.90	5.00
1	3	DS	194.60	1.00
1	3	FSTC	25.37	1.00
1	3	RS	104.06	1.00
1	3	SLS	24.11	1.00

1.1.2. Data employee input generation logic for Airport Lounge MILP Rostering

a) Logic employee profile input generation for MILP Rostering

The employee profile generation process is a workload-based preparation of inputs to support the MILP rostering model. Current estimates of employee profiles rely heavily upon estimating the maximum capacity of the lounge located in the building. Instead, the employee profile logic begins with the successful completion of determine staffing demand through staffing_pivot. Staffing demand

represents how many employees are needed for each shift and role based on the workload simulation via 28 days of scheduled events.

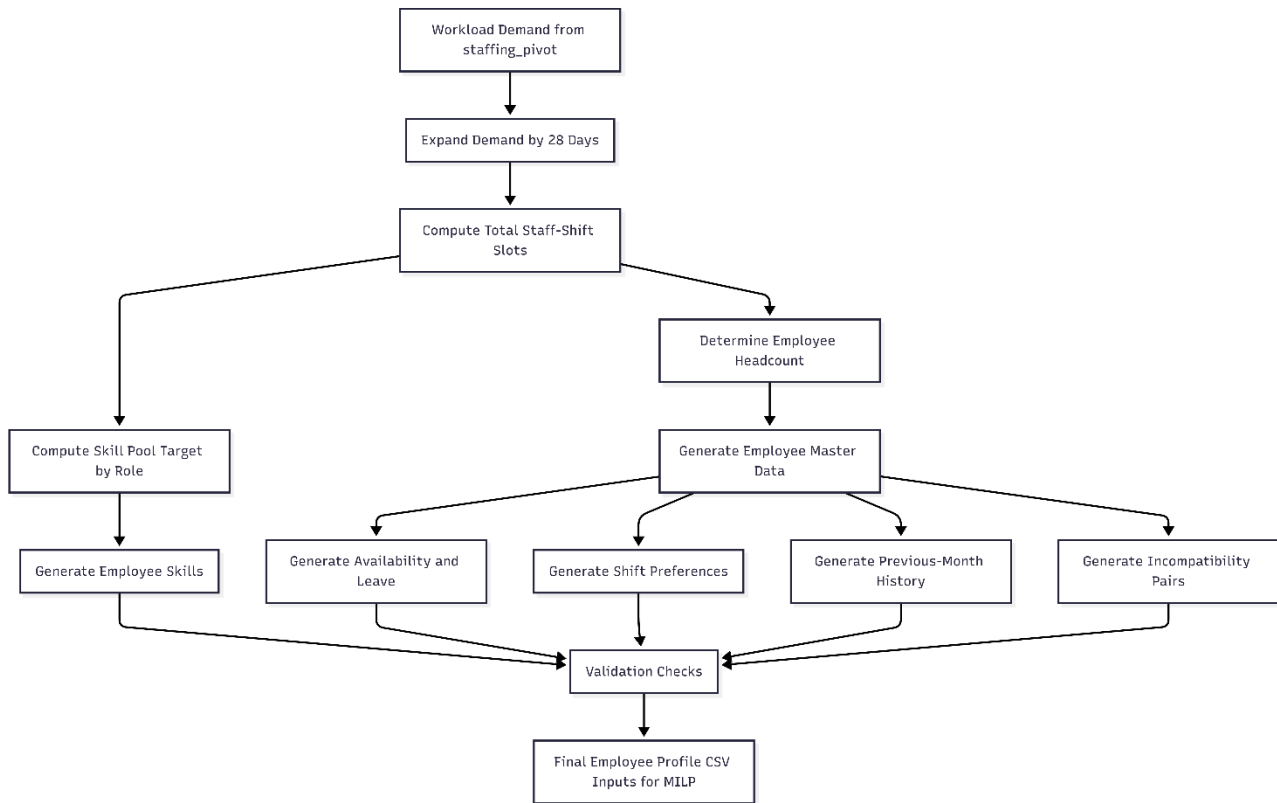


Figure IV-3: Workload-Driven Employee Data Generation Logic for MILP Rostering Input

The workload output indicates that the MILP operation will require 38 staff-shifts to support daily operational activities and will generate a total of 1,064 employee-shift assignments when calculating assignments over a 28-day planning time frame

After the demand is expanded across 28 days, the system computes the total staff-shift slots. This value is then used for two main calculations. First, it determines the feasible employee headcount by comparing total demand with contract limits. With a monthly workday range of 20 to 24 days, the feasible employee pool is between 45 and 53 employees. Therefore, around 48 employees is selected as a practical generation target because it satisfies both minimum and maximum workday constraints. Secondly, the required skill pool per operational role is calculated by the system. This step ensures a sufficient number of employees are generated and a sufficient number of qualified employees exist for each operational role (i.e., RS, DAS, BLO, FSTC, DS, and SLS). The skill pool target is based on demand for the role, peak concurrent need and includes a skill pool buffer.

Once the headcount and skill pool are defined, the system will create the main components of an employee profile. Employees will be created in the employee_master.csv file with employee ID, workday limits, working hour limits, night shift limits, consecutive working limits and so on. All additional files, including employee skills, availability and leave, shift preferences, prior month history, and incompatibility pairs will be created based on this master employee file. All these additional files will have both hard constraints and soft constraints.

All produced worker input data will be checked for validity by using four different criteria before being utilized in the Mixed Integer Linear Programming model. The criteria that will be used for the validation stage are the feasibility of the contract, adequacy of the skill pool, amount of availability, and if every employee has one or more valid skills. This will guarantee that the CSV input files for all worker profiles are realistic, internally consistent and appropriate to develop a practical monthly roster.

Table IV-10: Data input used by generate logic

Data Input group	Description	Purpose
Operational workload / coverage bands	Staffing requirement by day, shift, and role using Copt, Cmin, and Cmax.	Defines coverage requirement and overstaffing bounds.
Shift structure	Defines shift names, time windows, duration, and night-shift indicator.	Controls work hours, shift identity, and night-shift logic.
Rest and transition rules	Defines allowed or forbidden transitions between shifts across consecutive days.	Controls fatigue/rest feasibility.
Employee contract rules	Defines minimum/maximum workdays, hours, days off, night shifts, weekend and public holiday limits.	Controls employee feasibility and fairness bounds.
Skill qualification rules	Binary qualification matrix showing which employee can work which role.	Prevents assignment to unqualified roles.
Availability and leave	Approved leave, unavailable days, and other hard absence conditions.	Blocks assignment on unavailable days.
Employee requests and preferences	Soft day-off, preferred working day, avoid/prefer shift, and role-specific preferences.	Adds soft penalties when violated.
Fixed assignments	Predefined mandatory assignments for specific employee–day–shift–role combinations.	Forces specific assignments if provided.
Previous roster history	Last shift/status and consecutive work/night state from previous horizon.	Controls Day 1 continuity, rest, and consecutive work/night limits.

Relationship constraints	Employee pair rules such as incompatible pairs.	Penalizes or prevents unsuitable employees working together.
Standard role/shift stability	Defines each employee's standard role and standard shift.	Penalizes unnecessary role/shift changes.
Holiday and weekend calendar	Identifies Saturdays, Sundays, weekends, and public holidays.	Controls weekend and holiday workload limits.
Penalty weights	Defines objective-function weights for shortage, overstaffing, preference violation, rest violation, etc.	Determines priority among soft constraints.

b) Logic Employee Profiles Formulas:

(i) Convert operational workload into required staff

The first logic step is to convert operational workload into the number of staff required for each day, shift, and role. In this project, staffing is not calculated by applying a fixed percentage to total customer volume. Instead, each role is estimated based on its own workload driver.

- **Effective capacity per employee per shift:**

$$A = \text{shift_hours} \times 60 \times \text{target_utilization} = 8 \times 60 \times 0.85 = 408 \text{ minutes}$$

- **Daily role-shift staff requirement:**

$$\text{Req}[d, s, r] = \max(\text{min_staff}, \text{ceil}(\text{WorkloadMinutes}[d, s, r] / A))$$

- **Summary role-shift staff requirement:**

$$\text{Req}[s, r] = \max(\text{min_staff}, \text{ceil}(\text{P95_WorkloadMinutes}[s, r] / A))$$

P95 = Percentile 95

(ii) Employee headcount sizing

- **Total staff-shift slots:**

$$\text{TotalSlots} = \sum_d \sum_s \sum_r \text{Req}[d, s, r]$$

- **Coverage lower bound:**

$$N_{\min} = \text{ceil}(\text{TotalSlots} / \text{max_work_days})$$

- **Contract upper bound if every employee must receive minimum days:**

$$N_{\max} = \text{floor}(\text{TotalSlots} / \text{min_work_days})$$

- **Recommended generation headcount:**

After the required staff by day, shift, and role has been calculated, the next step is to estimate how many employees are needed in the employee pool. This step is necessary because the MILP model must know the available number of employees before assigning them to shifts and roles.

$$N = \min(N_{\max}, \text{ceil}(N_{\min} \times \text{buffer_ratio}))$$

(iii) Skill pool sizing

The next logic step is to determine how many qualified employees are needed for each role. This is important because the MILP cannot assign an employee to a role unless that employee has the required qualification. Therefore, total headcount alone is not enough; the model must also ensure that each role has a sufficient qualified pool.

- **Role total slots:**

$$\text{RoleSlots}[r] = \sum_d \sum_s \text{Req}[d, s, r]$$

- **Peak concurrent requirement:**

$$\text{PeakConcurrent}[r] = \max_{d, s} \text{Req}[d, s, r]$$

- **Minimum qualified pool per role:**

$$\text{PoolMin}[r] = \max(\text{PeakConcurrent}[r], \text{ceil}(\text{RoleSlots}[r] / \text{max_work_days}))$$

- **Target qualified pool per role:**

$$\text{PoolTarget}[r] = \text{ceil}(\text{PoolMin}[r] \times (1 + \text{skill_buffer_ratio}))$$

Table IV-11: DAS requires the largest qualified pool

Role	Peak concurrent need	28-day role slots	PoolMin = max(peak, ceil(slots/24))	Recommended PoolTarget with 20% skill buffer
RS	1	84	4	5
DAS	9	588	25	30
BLO	2	140	6	8
FSTC	1	84	4	5
DS	1	84	4	5
SLS	1	84	4	5

Table IV-11 shows DAS requires the largest qualified pool because it has the highest workload demand across the 28-day horizon. If DAS is treated as a basic skill, it may be assigned to all employees. However, if the model is designed with limited qualifications, then DAS should still be generated with a high qualification probability while forcing the minimum target pool of 30 qualified employees

(iv) Availability, leave, and days off

Availability and leave information is used to protect fixed employee absences before the MILP model creates the final roster. In this stage, the model treats approved leave, declared unavailability, training absence, or other fixed non-working conditions as hard constraints. This means that if an employee is recorded as unavailable or on leave on a certain day, the MILP is not allowed to assign that employee to any shift or role on that day.

- **Hard availability constraint:**

$$x[i, d, s, r] = 0 \text{ if } \text{employee_availability}[i, d] \in \{\text{unavailable}, \text{leave}\}$$

After leave and unavailability are blocked, the system must recheck whether enough qualified employees remain available for each role. This is done through the following coverage feasibility check:

- **Coverage check after blocking leave:**

$$\text{AvailableQualified}[d, r] \geq \max_s \text{Req}[d, s, r] + \text{safety_margin}$$

Normal days off should not be pre-marked in `employee_availability.csv`. Regular days off should be decided by the MILP through the minimum and maximum workday constraints, consecutive-work limits, and rest rules. Therefore, the availability file should only contain fixed leave, approved absence, and unavailable-day records. This avoids over-restricting the optimizer and gives the MILP enough flexibility to balance coverage, fairness, and labor rules

(v) Preferences and objective penalties

Employee preferences are included as soft constraints in the MILP model. Unlike leave or unavailability, preferences do not automatically block an assignment. Instead, they create a penalty if the model violates the employee's request. This allows the MILP to consider employee satisfaction while still prioritizing operational feasibility and service coverage.

For an avoid-shift preference, the penalty is applied if the employee is assigned to the undesired shift:

- **Avoid-shift penalty:**

$$\text{PenaltyAvoid} = \text{penalty} \times x[i, d, s, *]$$

- **Prefer-shift penalty:**

$$\text{PenaltyPrefer} = \text{penalty} \times (1 - \sum_r x[i, d, s, r])$$

A preference is not a hard rule unless it is stored as unavailable/leave. The MILP can violate an avoid/prefer request, but the objective function should add the relevant penalty.

(vi) Previous-month history

Previous-month history is required because the new roster does not start from a neutral condition. At the beginning of the planning horizon, some employees may already have worked several consecutive days or night shifts at the end of the previous month. If this historical information is ignored, the first few days of the new roster may violate fatigue, rest, or consecutive-work rules.

- **Consecutive work continuation:**

This ensures that an employee who already worked several consecutive days before the new month is not immediately assigned in a way that exceeds the maximum work-streak limit.

$$\text{previous_consecutive_work_days} + \text{work_on_day_1} \leq \text{max_consecutive_work_days}$$

- **Night-shift continuation:**

This protects employees from excessive night-shift sequences across the boundary between two roster periods.

$$\text{previous_consecutive_night_shifts} + \text{night_on_day_1} \leq \text{max_consecutive_night_shifts}$$

- **Rest after previous night shift:**

$$\text{if previous_last_shift} = S3, \text{ then assign_day1_S1} = 0$$

This means that if an employee worked the evening shift before the new planning horizon, the model should not assign that employee to an early next shift that violates the required rest pattern. Therefore, `employee_history.csv` should include the previous last shift, previous consecutive workdays, and previous consecutive night shifts. This information allows the MILP to produce a roster that is feasible not only inside the current 28-day period but also across month boundaries

(vii) Incompatible employee pairs

Incompatible employee-pair constraints are used to prevent certain employees from being scheduled together in the same shift. This may be required because of interpersonal issues, supervisor rules, training restrictions, or operational team-balance requirements. Although this input is optional, it is useful for making the roster more realistic from an HR and management perspective.

(viii) Incompatibility constraint:

$$\sum_r x[i, d, s, r] + \sum_r x[j, d, s, r] \leq 1 \text{ for every incompatible pair } (i, j), \text{ day } d, \text{ and shift } s$$

This rule can be implemented through an optional file such as `employee_incompatibility.csv`, containing employee pairs and the type or priority of incompatibility. If the rule is treated as a hard constraint, the pair is completely prevented from working together. If management wants more flexibility, it can be modeled as a soft constraint with a penalty. In either case, including incompatible employee pairs improves roster realism because it reflects practical HR and supervisor constraints that are not captured by workload demand alone

c) Data Input

(i) Operational workload / Coverage bands

Table IV-12: Staffing requirement by day, shift, and role

day	shift	shift_time	role	C_min	C_opt	C_max
1	S1	00:00-08:00	RS	1	1	2
1	S1	00:00-08:00	DAS	2	3	4
1	S1	00:00-08:00	BLO	1	1	2
1	S1	00:00-08:00	FSTC	1	1	2
1	S1	00:00-08:00	DS	1	1	2
1	S1	00:00-08:00	SLS	1	1	2
1	S2	08:00-16:00	RS	1	1	2
1	S2	08:00-16:00	DAS	5	8	10
1	S2	08:00-16:00	BLO	1	2	3
1	S2	08:00-16:00	FSTC	1	1	2
1	S2	08:00-16:00	DS	1	1	2
1	S2	08:00-16:00	SLS	1	1	2
1	S3	16:00-24:00	RS	1	1	2
1	S3	16:00-24:00	DAS	5	8	10
1	S3	16:00-24:00	BLO	1	2	3
1	S3	16:00-24:00	FSTC	1	1	2
1	S3	16:00-24:00	DS	1	1	2
1	S3	16:00-24:00	SLS	1	1	2

(ii) Penalty weights

penalty_component	weight	meaning
coverage_shortage	1000	Highest operational penalty
target_shortage	300	Below target but above minimum
overstaff	50	Labor efficiency penalty
skill_violation	100000	Hard infeasibility
availability_violation	100000	Hard infeasibility
duplicate_assignment	100000	Hard infeasibility
rest_violation	800	Fatigue and legal feasibility
consecutive_night_violation	600	Fatigue control
night_excess	300	Monthly night fairness
workday_or_hour_violation	200	Contract balance



preference_violation	10	Employee satisfaction
incompatibility	100	Team compatibility
standard_profile_change	3	Assignment stability

(iii) Employee_master

Table IV-13: Employee_master sample excel data

employee_id	min_work_days	max_work_days	target_work_days	min_days_off	max_days_off	max_night_shifts	standard_role
E001	20	24	22	4	8	6	SLS
E002	20	24	22	4	8	6	DAS
E003	20	24	22	4	8	6	DAS
E004	20	24	22	4	8	6	DAS
E005	20	24	22	4	8	6	BLO
E006	20	24	22	4	8	6	RS
E007	20	24	22	4	8	6	DAS
E008	20	24	22	4	8	6	BLO
E009	20	24	22	4	8	6	DAS
E010	20	24	22	4	8	6	DS
E011	20	24	22	4	8	6	DAS
E012	20	24	22	4	8	6	DAS

The employee_master file is a CSV file that provides the basic employee capacity characteristics needed for the airport lounge MILP roster model. Each record represents a single employee with constraints that are based on their contracts such as: minimum and maximum working days, minimum and maximum working hours, maximum night shifts, and maximum consecutive working days. These constraints are important because they make the transitions between HR and labor policy an official (formal) feasibility constraint for the MILP optimization model. In this sample data file, the employee can work from 17 - 24 days or 136 to 192 hours in a 28- day planning period. Thus the generating roster is capable of satisfying staffing demand while at the same time being realistic with respect to employee workloads, fatigue control, and contract compliance. Therefore, the employee_master file provides guidance for the employee's constraints and provides a boundary for MILP scheduling patterns to avoid creating infeasible or unacceptable scheduling patterns.

(iv) Employee_skills

Table IV-14: Employee_skill sample excel data

employee_id	RS	DAS	BLO	FSTC	DS	SLS
E001	0	0	0	0	0	1

E002	0	1	0	0	0	0
E003	0	1	0	0	0	0
E004	0	1	1	0	0	0
E005	0	0	1	0	0	1
E006	1	0	0	0	0	0
E007	0	1	0	0	0	0
E008	0	0	1	1	0	0
E009	0	1	0	0	0	0
E010	0	1	0	0	1	0
E011	0	1	0	0	0	0
E012	1	1	0	0	0	0
E013	0	0	0	0	1	0
E014	0	1	1	0	0	0

This file will create the framework of qualifications for employee capability for the airport lounge MILP Schedule. For each employee, binary skill indicators are available for the six role groups operationally used in the airport lounge Reception Staff (RCS), Dining Area Staff (DAS), Buffet Line Operator (BLO), Food Safety and Temperature Checker (FSTC), Dishwashing Staff (DS), and Supply and Logistics Staff (SLS). A code of “1” indicates that the employee is qualified for the position and will allow them to work in that role, while a “0” indicates that the employee is not eligible to work in that position/role. As an important piece of data, the employee_skills.csv file converts employee capability into a structured eligibility constraint within the MILP model. Consequently, the optimizer is constrained to assign employees to positions only for which they are explicitly qualified, eliminating the possibility of assigning an employee to a position that is not realistic/operationally valid. Additionally, the employee_skills.csv file ensures the feasibility of the optimized schedule by verifying the number of qualified employees within each position/role prior to optimization. As a result, the employee_skills.csv file supports not just employee capability descriptions but also realistic, feasible, and operationally reliable final rostering.

(v) Employee_availability

Table IV-15: Employee availability sample excel data

employee_id	day	status
E004	11	leave
E006	18	leave
E006	28	leave
E007	13	unavailable
E007	19	leave
E007	26	leave
E008	1	unavailable

E009	1	unavailable
E009	28	unavailable
E012	19	leave
E014	6	unavailable
E014	13	unavailable
E020	2	unavailable

The employee_availability.csv file sets out rules and limitations for how long an airline will have an employee absent from work at an airport lounge (MILP) as per what the airline has told me about the minimum number of employees the airline needs each day to operate the airport lounge MILP. Each employee has been assigned a specific record of missing work to inform the system of when they will not be able to work at the airport; either because they are on leave or do not exist in the pool of employees who can work at that time due to other reasons (e.g. time off). The model will not assign an employee to any shift or operational role for the day that corresponds with an employee's unavailability when the employee is on leave or unavailable. Some employees (i.e. E001, E006, E011) in the example data have been placed into the database at certain times to accurately reflect their unavailability. The employee availability (unavailability) file is vital in ensuring the operational validity of the overall schedule generated by the model, as the schedule must respect the fixed HR constraints that were imposed in the employee availability (unavailability) database. After considering the employee availabilities, the employee qualified worker pool must be validated before creating a feasible employee coverage plan for each shift for each day.

(vi) Employee Preferences and Requests Input Data

Table IV-16: Employee Preferences and Requests Sample Input Data

Source file	Preference level	employee_id	day	shift	role	request / preference type
employee_day_requests.csv	Day-level request	E001	13	—	—	desired_day_off
employee_day_requests.csv	Day-level request	E001	25	—	—	preferred_working_day
employee_preferences.csv	Shift-level preference	E001	5	S1	—	avoid
employee_preferences.csv	Shift-level preference	E001	12	S2	—	prefer
employee_assignment_preferences.csv	Assignment-level preference	E001	20	S2	SLS	prefer

The `employee_preferences.csv` file captures employee-specific shift preferences and converts them into soft constraints in the airport lounge MILP rostering model. Each record specifies the employee, planning day, shift, preference type, and penalty weight. A prefer request encourages the model to assign the employee to the selected shift, while an avoid request discourages such an assignment. These preferences do not create hard feasibility restrictions; instead, violations are penalized in the objective function. In the sample data, employee E001 prefers shift S2 on selected days and avoids shift S3 on other days, with higher penalties assigned to avoid requests. This structure allows the model to improve employee satisfaction while still prioritizing operational coverage, qualification feasibility, and contract compliance. Overall, `employee_preferences.csv` increases the practical realism of the roster by allowing the optimization model to consider employee requests without making the schedule unnecessarily infeasible.

(vii) Employee history

Table IV-17: Employee history sample excel data

employee id	previous_last_shift	previous_last_status	previous_consecutive work days	previous_consecutive days off	previous_consecutive night shifts
E001	S2	WORK	2	0	0
E002	S3	WORK	2	0	0
E003	OFF	OFF	0	1	0
E004	S3	WORK	4	0	0
E005	OFF	OFF	0	2	0

The `employee_history.csv` file contains historical data about the employees who work in the airport lounge rosters, including the employee's last completed shift prior to the new planning period, the employee's total number of consecutive days worked, and the number of consecutive nights worked, as it relates to fatigue, consecutive days worked, and consecutive rest periods. These variables do not automatically reset when a new month begins, thus requiring the use of this data to ensure proper evaluation of all Day 1 assignments within the MILP model as well as to avoid overextending an employee by creating long-term work streaks, controlling the continuation of employees working consecutive nights, and ensuring the employee is provided with the minimum required amount of rest between consecutive days and nights worked. It is clear from the sample data that E004 has already worked a long consecutive work period, while E005 has worked a few nights. The use of these variables in the MILP model will also help to create a roster that is realistic and safe; thus contributing to supporting the development of a monthly schedule that accurately reflects the prior state of the employee's roster.

1.2. Mathematical Model

1.2.1. Sets and Indices

Table IV-18: Indices

Symbol	Definition
$e \in E$	Employee index
$d \in D$	Day index in the planning horizon
$s \in S$	Shift index
$r \in R$	Operational role-group index
$w \in W$	Weekend index / weekend day-set index
$h \in H$	Public-holiday index
$q \in Q$	Non-working status index

Table IV-19: Sets

Symbol	Definition
E	Set of all available employees/staff members.
D	Set of days in the planning horizon (one month:1,...28)
W	Set of weekends
H	Set of public holidays
Q	Set of non-working statuses
R	Set of operational role groups
A	Set of incompatible combinations between shift and task
S	Set of working shifts
$S^N \subseteq S$	Set of night shifts
$S^D \subseteq S$	Set of daytime/evening shifts

Table IV-20: Role groups and shifts

Symbol	Definition
$S=\{S1,S2,S3\}$	Set of fixed shifts. In this case, $\{S1, S2, S3\}$ corresponding to 00:00-08:00, 08:00-16:00, and 16:00-24:00.
$S1$	Night shift (00:00-08:00)

S2	Day shift (08:00-16:00)
S3	Evening shift (16:00-24:00)
S^N = {S1}	Set of night shifts
S^D = {S2, S3}	Set of daytime/evening shifts
R = { RS, DAS, BLO, FSTC, DS, SLS }	Set of operational role groups: Reception & Seating, Dining Area Support, Buffet Line Operations, Food Safety & Temp Control, Dishwashing & Sanitation, Service Logistics Support

Table IV-21: Optional refined sets

Symbol	Definition
T	Set of intra-shift time-slices (e.g., 15–30 minute intervals) used to track flight-wave workload.
R^e ⊆ R	Set of role groups for which employee e is qualified
E_r ⊆ E	Set of employees qualified for role group (r)
F	Set of forbidden shift successions (e.g., night shift followed by morning shift), s1 → s2 → s3 → s1 set of pairs {s1, s2}
I	Set of incompatible employee pairs (e ₁ , e ₂)
P	Set of preferred employee pairs (e ₁ , e ₂)

Table IV-22: Non-working statuses

Symbol	Definition
Q = OFF, LV, RD, UA	Set of non-working statuses
OFF	Regular day-off
LV	Leave / vacation / holiday leave
RD	Recovery / mandated rest day
UA	Unavailability due to reassignment, training, deployment.

1.2.2. Input parameters

Table IV-23: Demand and coverage parameters

Symbol	Definition
C_{dsr}^{Min}	Minimum number of employees required on day (d), shift (s), role (r)
C_{dsr}^{Opt}	Optimal number of employees required on day (d), shift (s), role (r)
C_{dsr}^{Max}	Maximum number of employees allowed on day (d), shift (s), role (r)

Table IV-24: Shift structure parameters

Symbol	Definition
λ_s	Duration of shift (s)
S_s	Start time of shift (s)
$\rho_{s,s'}$	Rest time between shift s on day d and shift s' on day d + 1
$R^{\min}$	Minimum required rest time between two consecutive worked shifts
$\Lambda_e^{\min}$	Minimum working hours for employee e in the planning horizon
$\Lambda_e^{\max}$	Maximum working hours for employee e

Table IV-25: Employee contract and workload parameters

Symbol	Definition
L_e	Average working hours for employee e according to labor contract
$L_e^{\min}$	Minimum number of days off for employee e, if enforced
$N_e^{\min}$	Minimum number of shifts assigned to employee e
$N_e^{\max}$	Maximum number of shifts assigned to employee e
$D_e^{\min}$	Minimum number of working days for employee e
$D_e^{\max}$	Maximum number of working days for employee e
$W_e^{\max}$	Maximum number of working weekends for employee e
$H_e^{\max}$	Maximum number of working public holidays for employee e
γ_e	Indicator parameter equal to 1 if complete weekend constraint applies to employee e
$c^{\min}$	Minimum number of days off after consecutive working shifts

Table IV-26: Pairing and teamwork parameters

Symbol	Definition
I	Set of incompatible employee pairs (e_1, e_2)
P	Set of preferred employee pairs (e_1, e_2)
WT	Maximum attainable teamwork score, if normalized teamwork scoring is used

Table IV-27: Penalty parameters for soft constraints

Symbol	Definition
w^i	Penalty weight associated with constraint (i)
M	Large constant used in mathematical constraints

Table IV-28: Requests, pre-assignments, and availability

Symbol	Definition
ϕ^a	Set of fixed pre-assignments (e,d,s,r)
ϕ^o	Set of fixed requests of not working on shift (s) on day (d) as tuples (e,d,s)
ϕ^w	Set of fixed requests of working on day (d) as tuples (e,d,s)
ϕ_e^h	Set of fixed vacation days for employee (e,d)
Π^a	Set of tuples (e,d,s,r,l) representing preferred assignment of employee e to task (t) during shift (s) on day (d) with priority weight (l)
Π^u	Set of tuples (e,d,s,l) representing undesired shifts for employee e
Π^d	Set of tuples (e,d,l) representing desired days off for employee e
Π^o	Set of tuples (e,d,l) representing preferred working days for employee e

Table IV-29: Assignment stability and standard profile parameters

Symbol	Definition
σ_e^s	Standard shift for employee e
σ_e^r	Standard role of employee e

Table IV-30: Consecutiveness and sequence parameters

Symbol	Definition
$CW_e^{\min}$	Minimum number of consecutive working days for employee (e), if applicable
$CW_e^{\max}$	Maximum number of consecutive working days for employee (e)
$CO_e^{\min}$	Minimum number of consecutive days off for employee (e)

$CN_e^{\max}$	Maximum number of consecutive night shifts for employee (e)
Sep_e^N	Minimum separation between two night-shift blocks for employee (e), if policy applies

Table IV-31: Cross-horizon consistency parameters

Symbol	Definition
h_e^W	Number of consecutive working days of employee (e) immediately before day 1 of the current horizon
h_e^O	Number of consecutive days off of employee (e) immediately before day 1
h_e^N	Number of consecutive night shifts of employee (e) immediately before day 1
h_e^S	Last shift worked by employee (e) before the current horizon
h_e^Q	Last non-working status of employee (e) before the current horizon, if needed

1.2.3. Decision Variables

Table IV-32: Assignment variables

Symbol	Definition
$x_{e,d,s,r} \in 0, 1$	1 if employee (e) is assigned to role group (r) in shift (s) on day (d); 0 otherwise
$z_{e,d,q} \in 0, 1$	1 if employee (e) is assigned non-working status (q in Q) on day (d); 0 otherwise
$y_{e,d} \in 0, 1$	1 if employee (e) works any shift on day (d); 0 otherwise
$n_{e,d} \in 0, 1$	1 if employee (e) works a night shift on day (d); 0 otherwise
$u_{e,w} \in 0, 1$	1 if employee (e) works any shift during weekend (w); 0 otherwise
$g_{e,h} \in 0, 1$	1 if employee (e) works on public holiday (h); 0 otherwise

Table IV-33: Optional support variables

Symbol	Definition
$a_{d,s,r} \geq 0$	Number of employees assigned on day (d), shift (s), role (r)



$t_{e,d}^{\text{start}} \geq 0$	Start time of the assigned shift of employee (e) on day (d); 0 if employee (e) does not work that day
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Table IV-34: Violation variables for soft constraints

Symbol	Definition
$y_{d,s,r}^{\text{under}} \in \mathbb{Z}^+$	Understaffing amount relative to minimum coverage on day d, shift s, role r
$y_{d,s,r}^{\text{opt-}} \in \mathbb{Z}^+$	Shortage relative to target coverage on day d, shift s, role r
$y_{d,s,r}^{\text{over}} \in \mathbb{Z}^+$	Overstaffing amount above maximum or above target, depending on modeling choice
$y_{e,d}^{\text{pref}} \in \mathbb{Z}^+$	Violation of employee preference associated with assignment (e,d,s,r)
$y_{e,d}^{\text{shift}} \in \mathbb{Z}^+$	Violations of maximum or minimum number of assigned shifts for employee e
$y_{e,d}^{\text{role}} \in \mathbb{Z}^+$	Violation of assignment stability due to switching away from the standard role profile
$y_{e,d}^{\text{dayoff}} \in \mathbb{Z}^+$	Violation related to minimum/maximum day-off pattern for employee e starting at day d
$y_{e,d}^{\text{workday}} \in \mathbb{Z}^+$	Violation related to working-day pattern for employee e starting at day d
$y_{e,d}^{\text{night}} \in \mathbb{Z}^+$	Violation of night-shift consecutiveness or night-block rule for employee e starting at day d
$y_{e,d}^{\text{rest}} \in \mathbb{Z}^+$	Violation of minimum rest time between consecutive duties for employee e across days d and d + 1
$y_{e,d}^{\text{succ}} \in \mathbb{Z}^+$	Violation of forbidden or undesirable shift succession for employee e across days d and d + 1
$y_{e,w}^{\text{weekend}} \in \mathbb{Z}^+$	Violation of weekend assignment preference or weekend workload bound for employee e in weekend w
$y_{e,h}^{\text{holiday}} \in \mathbb{Z}^+$	Violation of public-holiday workload preference or bound for employee e on holiday h
$y_{e,d}^{\text{consecShift}} \in \mathbb{Z}^+$	Violations of maximum number of consecutive shifts for employee e starting on day (d)
$y_{e,d}^{\text{consecWork}} \in \mathbb{Z}^+$	Violations of maximum number of consecutive working days for employee e starting on day (d)

$y_{e,d}^{\text{Off}} \in \mathbb{Z}^+$	Violations of minimum required days off for employee e on day d
$y_{e_1,e_2,d} \in \{0, 1\}$	Violation indicator if employees e_1, e_2 should not work together on day (d)
$y_{e,d,s,r}^{\text{Skill}} \in \mathbb{Z}^+$	Violation of controlled flexibility / non-standard qualified assignment, if this is modeled as soft rather than forbidden
$y_{e,d}^{\text{rest}} \in \mathbb{Z}^+$	Violations of minimum rest time between two shifts for employee e
$y_{e,w}^{\text{weekend}} \in \mathbb{Z}^+$	Violations of weekend assignment preference for employee e
$y_e^{\text{fair}} \in \mathbb{Z}^+$	Violations of workload fairness for employee e
$y_{d,s,r,r'}^{\text{bal}} \in \mathbb{Z}^+$	Soft imbalance variable capturing deviation from the desired target staffing ratio between two role groups

1.2.4. Objective Function

The buffet lounge rostering problem is formulated as a weighted-penalty minimization model. The objective is to construct a monthly roster that first protects baseline service feasibility, then stabilizes service quality across the FOH-FSL-BOH chain, and finally improves fairness, preference satisfaction, and assignment stability. In the lounge context, this priority is appropriate because understaffing directly creates guest-facing congestion, replenishment delays, table-turnover problems, and dishwashing backlogs, while overstaffing mainly creates labor inefficiency.

We propose the following objective:

$$\begin{aligned}
 \min Z = & w^{\text{under}} \sum_{d \in D} \sum_{s \in S} \sum_{r \in R} y_{d,s,r}^{\text{under}} + w^{\text{opt}} \sum_{d \in D} \sum_{s \in S} \sum_{r \in R} y_{d,s,r}^{\text{opt-}} + w^{\text{over}} \sum_{d \in D} \sum_{s \in S} \sum_{r \in R} y_{d,s,r}^{\text{over}} \\
 & + w^{\text{bal}} \sum_{d \in D} \sum_{s \in S} \sum_{(r,r') \in B} y_{d,s,r,r'}^{\text{bal}} \\
 & + w^{\text{rest}} \sum_{e \in E} \sum_{d \in D} y_{e,d}^{\text{rest}} + w^{\text{succ}} \sum_{e \in E} \sum_{d \in D} y_{e,d}^{\text{succ}} + w^{\text{conW}} \sum_{e \in E} \sum_{d \in D} y_{e,d}^{\text{consecWork}} \\
 & + w^{\text{off}} \sum_{e \in E} \sum_{d \in D} y_{e,d}^{\text{Off}} + w^{\text{night}} \sum_{e \in E} \sum_{d \in D} y_{e,d}^{\text{night}} + w^{\text{weekend}} \sum_{e \in E} \sum_{w \in W} y_{e,w}^{\text{weekend}} \\
 & + w^{\text{holiday}} \sum_{e \in E} \sum_{h \in H} y_{e,h}^{\text{holiday}} + w^{\text{fair}} \sum_{e \in E} y_e^{\text{fair}} + w^{\text{role}} \sum_{e \in E} \sum_{d \in D} y_{e,d}^{\text{role}} + w^{\text{shift}} \sum_{e \in E} \sum_{d \in D} y_{e,d}^{\text{shift}} \\
 & + w^{\text{pair}} \sum_{(e_1,e_2) \in I} \sum_{d \in D} y_{e_1,e_2,d} + w^{\text{skill}} \sum_{e \in E} \sum_{d \in D} \sum_{s \in S} \sum_{r \in R} y_{e,d,s,r}^{\text{Skill}} \\
 & + w^{\text{pref}} \left(\sum_{(e,d,s,r,l) \in \Pi^a} l v_{e,d,s,r}^a + \sum_{(e,d,s,l) \in \Pi^u} l v_{e,d,s}^u + \sum_{(e,d,l) \in \Pi^d} l v_{e,d}^d + \sum_{(e,d,l) \in \Pi^o} l v_{e,d}^o \right).
 \end{aligned}$$

A reasonable weight hierarchy for the buffet lounge case is

$$w^{\text{under}} \gg w^{\text{opt}} > w^{\text{over}}$$

with rest/succession/fairness/stability penalties placed below minimum-coverage protection but still high enough to prevent fragile rosters. This follows both the nurse-rostering template and your proposal's logic that minimum coverage protects service feasibility, target coverage stabilizes service, and maximum coverage controls inefficiency.

1.2.5. Constraints

a) Assignment and linking constraints

Each employee must receive exactly one daily status: either one working assignment (s, r) or one non-working status q .

$$\sum_{s \in S} \sum_{r \in R} x_{e,d,s,r} + \sum_{q \in Q} z_{e,d,q} = 1 \forall e \in E, d \in D \quad (1)$$

Role eligibility is enforced by qualification.

$$x_{e,d,s,r} = 0 \forall e \in E, d \in D, r \notin R_e \quad (2)$$

If shift-role incompatibilities are explicitly modeled, add

$$x_{e,d,s,r} = 0 \forall e \in E, d \in D, (s,r) \in A \quad (3)$$

The daily work indicator is linked to the assignment variables by

$$y_{e,d} = \sum_{s \in S} \sum_{r \in R} x_{e,d,s,r} \forall e \in E, d \in D \quad (4)$$

Since the proposal defines $S^N = \{S1\}$, the night indicator becomes

$$n_{e,d} = \sum_{r \in R} x_{e,d,S1,r} \forall e \in E, d \in D \quad (5)$$

Coverage by day-shift-role is aggregated through

$$a_{d,s,r} = \sum_{e \in E} x_{e,d,s,r} \forall d \in D, s \in S, r \in R \quad (6)$$

These linking constraints follow the proposal's one-role-per-shift logic and keep the roster at the level of primary shift-role assignment, while intra-shift task rotation remains part of the execution layer in the OSF framework.

b) Coverage-band constraints

Minimum coverage is enforced by role group and shift:

$$a_{d,s,r} + y_{d,s,r}^{\text{under}} \geq C_{d,s,r}^{\text{Min}} \forall d \in D, s \in S, r \in R \quad (7)$$

Target (optimal) coverage is modeled as a soft service stability requirement:

$$a_{d,s,r} + y_{d,s,r}^{\text{opt-}} \geq C_{d,s,r}^{\text{Opt}} \forall d \in D, s \in S, r \in R \quad (8)$$

Maximum coverage limits overstaffing:

$$a_{d,s,r} - y_{d,s,r}^{\text{over}} \leq C_{d,s,r}^{\text{Max}} \forall d \in D, s \in S, r \in R \quad (9)$$

To model P2: role-balanced coverage across the FOH-FSL-BOH service chain:

$$C_{d,s,r'}^{\text{Opt}} a_{d,s,r} - C_{d,s,r}^{\text{Opt}} a_{d,s,r'} \leq y_{d,s,r,r'}^{\text{bal}} \forall d, s, (r, r') \in B \quad (10)$$

$$C_{d,s,r}^{\text{Opt}} a_{d,s,r'} - C_{d,s,r'}^{\text{Opt}} a_{d,s,r} \leq y_{d,s,r,r'}^{\text{bal}} \forall d, s, (r, r') \in B \quad (11)$$

These two constraints penalize deviation from the target staffing proportions across the service chain. They are especially relevant in buffet operations because BOH shortages can propagate into clean-ware shortages and then degrade FOH and line performance, even when total staffing appears acceptable.

c) Fixed assignments, leave, and availability constraints

Fixed pre-assignments are enforced by

$$x_{e,d,s,r} = 1 \forall (e, d, s, r) \in \phi^a \quad (12)$$

Fixed requests not to work a specific shift are modeled as

$$\sum_{r \in R} x_{e,d,s,r} = 0 \forall (e, d, s) \in \phi^o \quad (13)$$

Fixed vacation or leave days are modeled as

$$\sum_{s \in S} \sum_{r \in R} x_{e,d,s,r} = 0 \forall (e, d) \in \phi_e^h \quad (14)$$

If a must-work day request is included, impose

$$\sum_{s \in S} \sum_{r \in R} x_{e,d,s,r} = 1 \forall (e, d) \in \phi^w \quad (15)$$

These constraints are directly aligned with the proposal's treatment of leave, unavailability, pre-assignments, and fixed requests.

d) Contract and workload constraints(hard)

Working-day bounds over the planning horizon are

$$20 \leq \sum_{d \in D} y_{e,d} \leq 24 \forall e \in E \quad (16)$$

Monthly working-time bounds:

$$160 \leq \sum_{d \in D} \sum_{s \in S} \sum_{r \in R} \lambda_s x_{e,d,s,r} \leq 192 \forall e \in E \quad (17)$$

Because the current roster permits at most one shift per day, the number of worked days and the number of assigned shifts coincide. If management wants paid leave to count toward assignment totals, an adjusted accounting equation can be added. This matches the proposal's distinction between statuses that do and do not count in workload accounting.

e) Rest and succession constraints (hard)

Minimum rest between two consecutive worked shifts is enforced using the precomputed rest-time parameter $\rho_{s,s'}$:

$$R^{\min} - \rho_{s,s'} \leq M(2 - \sum_r x_{e,d,s,r} - \sum_{r'} x_{e,d+1,s',r'}) \forall e, d, s, s'$$

$$\forall e \in E, d = 1, \dots, |D| - 1, s, s' \in S \quad (18)$$

Forbidden shift successions are modeled by

$$\sum_r x_{e,d,s,r} + \sum_{r'} x_{e,d+1,s',r'} \leq 1 \forall e, d, (s, s') \in F \quad (19)$$

If management treats these as legal or safety requirements, set the corresponding violation variables to zero and enforce them as hard constraints. This mirrors the nurse-rostering paper's emphasis on rest and forbidden successions while fitting the proposal's labor-rule logic.

f) Consecutive-work and off-day pattern constraints

Maximum consecutive working days can be controlled with a sliding-window form:

$$\sum_{\tau=d}^{d+CW_e^{\max}} y_{e,\tau} \leq CW_e^{\max} + y_{e,d}^{\text{consecWork}} \forall e \in E, d = 1, \dots, |D| - CW_e^{\max} \quad (20)$$

Define the off-day indicator

$$o_{e,d} = \sum_{q \in Q^{\text{off}}} z_{e,d,q} \forall e, d \quad (21)$$

Then the minimum required number of off-days after the start of an off-block can be written as

$$\sum_{\tau=d}^{d+c^{\min}-1} o_{e,\tau} \geq c^{\min}(y_{e,d-1} - y_{e,d}) - y_{e,d}^{\text{off}} \forall e \in E, d \in D \quad (22)$$

This equation activates when a work block ends at day $d - 1$ and an off-block starts at day d . For $d = 1$, the term $y_{e,0}$ is evaluated from history data from the previous horizon. This is consistent with the proposal's cross-horizon logic and the use of previous-month history to evaluate patterns near the horizon boundary.

g) Night-work constraints (hard)

Maximum consecutive night shifts are controlled by

$$\sum_{\tau=d}^{d+CNe^{\max}} n_{e,\tau} \leq CNe^{\max} \quad \forall e, d = 1, \dots, |D| - CNe^{\max} \quad (23)$$

If the policy requires a recovery day RD after the end of a night block, impose

$$z_{e,d+1,RD} \geq n_{e,d} - n_{e,d+1} \quad \forall e \in e, d \quad (24)$$

This buffet-lounge concept with explicit non-working status RD and is especially relevant because the lounge includes a night shift $S1$ in a 24-hour operation.

h) Weekend and public-holiday constraints

Let $Sat(w)$ and $Sun(w)$ denote the two days of weekend w . The weekend work indicator is linked by

$$u_{e,w} \geq y_{e,Sat(w)}, u_{e,w} \geq y_{e,Sun(w)}, u_{e,w} \leq y_{e,Sat(w)} + y_{e,Sun(w)} \quad \forall e, w \quad (25)$$

If the complete-weekend rule is active, penalize split weekends through

$$y_{e,Sat(w)} - y_{e,Sun(w)} \leq y_{e,w}^{\text{weekend}} \quad \forall e, w \quad (26)$$

$$y_{e,Sun(w)} - y_{e,Sat(w)} \leq y_{e,w}^{\text{weekend}} \quad \forall e, w \quad (27)$$

Maximum working weekends are bounded by

$$\sum_{w \in W} u_{e,w} - W_e^{\max} \leq \sum_{w \in W} y_{e,w}^{\text{weekend}} \quad \forall e \in E \quad (28)$$

For public holidays,

$$g_{e,h} = \sum_{s \in S} \sum_{r \in R} x_{e,h,s,r} \quad \forall e \in E, h \in H \quad (29)$$

$$\sum_{h \in H} g_{e,h} - H_e^{\max} \leq \sum_{h \in H} y_{e,h}^{\text{holiday}} \quad \forall e \in E \quad (30)$$

These constraints are adapted from the nurse-rostering weekend/holiday structure but suit the buffet context, where fairness in undesirable time periods is important for employee acceptance and long-run roster stability.

i) Preferences and assignment stability

To keep preference modeling clear, it is better to disaggregate preference-violation variables as follows.

Preferred assignment:

$$1 - x_{e,d,s,r} \leq v_{e,d,s,r}^a \forall (e, d, s, r, l) \in \Pi^a \quad (31)$$

Undesired shift:

$$\sum_{r \in R} x_{e,d,s,r} \leq v_{e,d,s}^u \forall (e, d, s, l) \in \Pi^u \quad (32)$$

Desired day off:

$$y_{e,d} \leq v_{e,d}^d \forall (e, d, l) \in \Pi^d \quad (33)$$

Preferred working day:

$$1 - y_{e,d} \leq v_{e,d}^o \forall (e, d, l) \in \Pi^o \quad (34)$$

Standard role consistency is modeled by

$$\sum_{s \in S} \sum_{r \in R \setminus \{\sigma_e^r\}} x_{e,d,s,r} \leq y_{e,d}^{\text{role}} \forall e \in E, d \in D \quad (35)$$

Standard shift consistency is modeled by

$$\sum_{r \in R} \sum_{s \in S \setminus \{\sigma_e^s\}} x_{e,d,s,r} \leq y_{e,d}^{\text{shift}} \forall e \in E, d \in D \quad (36)$$

These constraints are appropriate for the lounge because the proposal explicitly emphasizes non-fixed requests, standard profiles, and execution readiness, meaning the model should not generate excessive role volatility that increases supervision effort during peaks.

j) Pairing, incompatibility, and fairness constraints

If two employees should not work together in the same shift, use

$$\sum_{r \in R} x_{e_1,d,s,r} + \sum_{r' \in R} x_{e_2,d,s,r'} \leq 1 + y_{e_1,e_2,d} \forall (e_1, e_2) \in I, d \in D, s \in S \quad (37)$$

Define total assigned hours

$$H_e = \sum_{d \in D} \sum_{s \in S} \sum_{r \in R} \lambda_s x_{e,d,s,r} \forall e \in E \quad (38)$$

Then workload fairness can be written as

$$H_e - L_e \leq y_e^{\text{fair}}, L_e - H_e \leq y_e^{\text{fair}} \forall e \in E \quad (39)$$

If L_e is only an average daily figure rather than a true monthly reference, replace it with a dedicated monthly target parameter H_e^{ref} . This fairness construction is already compatible with the proposal and is a good buffet-lounge adaptation because it helps avoid persistent concentration of undesirable workload on the same employees.

k) Optional controlled-flexibility constraint

If management wants to allow emergency cross-role assignments instead of forbidding them completely, controlled flexibility can be modeled as a soft constraint. Let $\chi_{e,r} = 1$ if employee e is qualified for role r , and 0 otherwise. Then

$$x_{e,d,s,r} \leq x_{e,r} + y_{e,d,s,r}^{\text{Skill}} \forall e, d, s, r \quad (40)$$

When strict qualification is required, simply remove this relaxation and keep Eq. (2) as hard feasibility. This matches the proposal's idea of "controlled flexibility" rather than full interchangeability across FOH, FSL, and BOH.

2. Solution Method Development

This methodology includes two main stages: an initial stage wherein the Mixed Integer Linear Programming (MILP) model is solved to produce the primary optimized roster, and a second stage where the output of the MILP serves as the initial solution for the post MILP procedure employing simulated annealing (SA). As explained later in this chapter, simulated annealing is not intended as a substitute for MILP, but as a tool to verify whether local search improvement on a solution produces less soft penalty and continues to provide active solutions when problem size increases.

2.1. MILP Solver Implementation

A multi-level programming (MILP) rostering model allocates employees to their respective shifts, days of the week, and operating roles for a 28-day planning horizon. The primary binary decision variable $x[e,d,s,r]$ indicates whether the employee e is assigned to day d in shift s with operational role r . Only if an employee is mutually qualified for and available to do the work assigned will feasible decision variables exist for that employee. Coverage requirements in the model are defined as a minimum, optimal, and maximum level of coverage ($C_{\min}$, C_{opt} , and $C_{\max}$) in each day-shift-role cell. The default model will target an optimal C_{opt} as the minimum acceptable level of coverage for each employee. $C_{\max}$ is used to control an excessive amount of overstaffing. The objective function is to minimise the total of system-level penalties that may be assessed against employees for coverage shortfalls, target shortfalls, overstaffing, fairness (i.e. distribution of workday), violations of soft start history, violations of employee preferences, incompatibility penalties, and instability relative the standard role/ shift. Hard constraints restrict an employee to working no more than one shift per day, total associated monthly number of workdays worked and total monthly hours worked, total number of days an employee must be off each month, number of consecutive workdays or nights worked, number of consecutive weekends or public holidays off, restrictions on shift transitions, and no fixed assignments. The output produced by the model includes a long format table of employee assignments, an employee-by-day roster matrix, a coverage audit (i.e. number of employees rostered against available on each shift), a summary of employee workloads, and overall model status metrics.

The MILP roster model was implemented in Python using the PuLP modelling package and solved with the CBC mixed-integer programming solver. The solver was configured with a 300-second computational time limit and a relative optimality gap of 0.01. These settings were selected to balance solution quality and computational practicality, particularly because the staff scheduling problem contains a large number of binary assignment variables and operational constraints. The objective sense was set to minimization, where the solver seeks the roster with the lowest total weighted penalty, including coverage shortage, overstaffing, preference violations, rest-rule violations, incompatibility penalties, and fairness deviations. Solver messages were disabled by default to keep the execution output concise, but they can be enabled for debugging infeasibility or convergence behavior.

Table IV-35: Technical Setting

Solver setting	Value
Solver	CBC
Package	PuLP
Time limit	300 seconds
Relative gap	0.01
Solver log	Off by default
Objective sense	Minimize

2.2. MILP Output as the Initial SA Solution

MILP is the primary model, and its generated roster is used as the initial solution for SA. In this research, it is the MILP model that serves as the optimisation Model. The roster produced by this optimisation model is utilized as the starting point for the Simulated Annealing (SA) phase within the scope of this study. Instead of using SA to create a new roster from scratch, the SA phase will utilize the output from the MILP assignment in the either the assignment_long.csv file or the assignment_long worksheet in the milp_roster_output.xlsx workbook file. Each record that exists within the assignment table provides identification of an employee, day, shift and role, as produced by the MILP model. The structure of the assignment table is changed to represent the state, so each employee-day combination will be either classified as being off-duty or classified as being assigned to a shift - role combination. The structure of this state representation allows SA to utilise a pre-defined MILP-generated roster that is organised, and then to perform local search to identify any improvements without losing the feasibility structure produced by the mathematical model.

Beginning with an initial solution of an MILP output is of utmost importance to allow for a feasible and structured roster for the SA process versus a randomly generated roster. Randomly generating a roster may inadvertently violate many hard constraints, and will take an inordinate amount of search time to become feasible. With a known solution to begin the SA process, SA can concentrate its effort on improving the soft-performance characteristics in the roster such as overstaffing, employee preference, fairness, and shifting stability.

2.3. Post-MILP Simulated Annealing Improvement Method

Simulated Annealing, which is described as being used in Phase-2 to improve the solution generated by a Mixed Integer Linear Program (MILP) via local search (neighbour), is based upon the intent of simulating the global search alternatives for the purpose of adapting a MILP solution through a series of controlled changes to the original roster. Examples of neighbourhood actions are defined through the code and include; reassigning coverage for the same shift-role to another qualified employee, swapping assignments between employees, moving an assignment from a cell which is overstaffed to a cell which is under-targeted, dropping an assignment from an overstaffed cell, adding an assignment to an under-targeted cell where qualified resources are available. The primary improvement objectives consist of, reducing shortages below C minimum (Cmin) or C optimal (Copt), reducing overstaffing in excess of C optimal (Copt) or C maximum (Cmax), reducing employee preference violations, improving fairness in relation to target workdays, and improving stability of standard role and standard shift. Simulated Annealing is a beneficial approach when compared with an exact optimization method such as MILP, as MILP becomes increasingly time-consuming and requires more computational resources when the size and complexity of the problem increases e.g. adding employees, days, roles, rules, or demand scenarios. As a result, Simulated Annealing offers a flexibility that allows for alternative rosters to be examined without having to rebuild the mathematical model from the beginning of the process.

2.4. Common Evaluation Score

Common Evaluation Score is used to evaluate both the initial MILP roster and the final SA roster. This score is not the raw MILP objective value. Instead, it is an independent evaluation score designed to compare the MILP-generated roster and the SA-improved roster under the same criteria. The code specifically notes that Cmin and Cmax are treated as hard service feasibility boundaries, while Copt is treated as the main service-quality target.

The score can be represented as:

$$\text{Score}(S) = P_{\text{hard}}(S) + P_{\text{coverage}}(S) + P_{\text{overstaff}}(S) + P_{\text{preference}}(S) + P_{\text{fairness}}(S) + P_{\text{standard}}(S) + P_{\text{rest}}(S) + P_{\text{incompatibility}}(S)$$

Where:

Table IV-36: Common Evaluation Score

Score Component	Meaning
P_{hard}	Penalty for hard feasibility violations
P_{coverage}	Penalty for shortage below C_{min} or C_{opt}
$P_{\text{overstaff}}$	Penalty for overstaffing above C_{opt} or C_{max}
$P_{\text{preference}}$	Penalty for violating employee preferences
P_{fairness}	Penalty for deviation from target workdays
P_{standard}	Penalty for deviation from standard role or standard shift
P_{rest}	Penalty for forbidden rest or shift-transition violations
$P_{\text{incompatibility}}$	Penalty for incompatible employees working together

The comparative penalties of shortages that occur below the level of C_{min} and those caused when workers are overstaffed, which occur at levels above C_{max} , receive a very large penalty. Shortages of less than optimum C_{opt} can incur service-level penalties and labour-based penalties for being above optimum in addition to hard penalties defined for being below C_{min} and overstaffed at greater than C_{max} . Skill violations, availability violations, fixed-style violation contracts, consecutive work violations, consecutive night-shift violations, preference penalties, penalties due to not meeting standard profiles, penalties associated with workers assigned incompatible jobs, and penalties associated with equity among employees will be evaluated as part of the code.

The purpose for developing a common evaluation metric is that the results of MILP solving a mathematical model in accordance with its original objective function, are not necessarily directly comparable to SA evaluating candidate rosters using a combination of penalties. By using a common metric to score both the initial MILP roster and the final SA roster, we can provide an appropriate level of impartiality.

2.5. Move- Neighbourhood

The Move-Neighbourhood approach generates a neighbouring roster solution by modifying one assignment at a time. In the implemented SA model, a move can reassign an existing shift-role coverage from one employee to another qualified employee, relocate an employee from an overstaffed coverage cell to an under-target cell, add a qualified off-duty employee to a shortage cell,

or remove an assignment from an overstaffed cell when contract feasibility is preserved. Each move is accepted as a candidate solution only if the basic feasibility conditions, including employee skill, availability, fixed assignment, and role-shift validity, are satisfied. The resulting neighbour solution is then evaluated using the common evaluation score, which accounts for coverage gaps, overstaffing, preference violations, contract rules, rest constraints, fairness, and standard role-shift stability.

Algorithm: Move-Neighbourhood for Roster Improvement

Input:

Current roster solution S

Employee set E

Planning days D

Shift set H

Role set R

Coverage requirement $C_{\min}, C_{\text{opt}}, C_{\max}$

Employee skills, availability, fixed assignment, contract rules

Output:

A new neighbouring roster solution S'

Step 1: Start from the current roster solution S .

Step 2: Evaluate the current roster to identify:

- Overstaffed cells: $\text{actual_staff} > C_{\text{opt}}$
- Under-target cells: $\text{actual_staff} < C_{\text{opt}}$
- OFF employee-day positions

Step 3: Randomly select one move mechanism:

- a) Reassign same shift-role coverage
- b) Move from overstaffed cell to under-target cell
- c) Add employee to shortage cell
- d) Drop employee from overstaffed cell

Step 4: If $\text{reassign_same_cell}$ is selected:

Randomly select an assigned employee e_1 on day d .

Let the current assignment be (shift s , role r).

Randomly select another employee e_2 who is OFF on the same day.

If e_2 can perform role r in shift s on day d :

Remove assignment from e_1 .

Assign e2 to (s, r).

Return the new solution S'.

Step 5: If move_over_to_short is selected:

Randomly select one overstaffed cell (d, s, r).

Randomly select one employee e assigned in that cell.

Randomly select one under-target cell (d, s', r') on the same day.

If employee e can be assigned to (s', r'):

Move employee e from (s, r) to (s', r').

Return the new solution S'.

Step 6: If add_to_shortage is selected:

Randomly select an under-target cell (d, s, r).

Randomly select an OFF employee e on day d.

If employee e can be assigned to (s, r)

and the assignment does not exceed C_max or max workdays:

Assign employee e to (s, r).

Return the new solution S'.

Step 7: If drop_overstaff is selected:

Randomly select an overstaffed cell (d, s, r).

Randomly select an employee e assigned in that cell.

If employee e is not fixed and remains above minimum workdays:

Remove the assignment from employee e.

Return the new solution S'.

Step 8: If no feasible move is found:

Return no neighbour solution.

Let's assume the following coverage levels for the results:

Table IV-37: Move-Neighbourhood Example

Day	Shift	Role	C_opt	Actual	Status
3	S2	RS	2	3	Overstaff
3	S2	BLO	2	1	Shortage
Role	Before	After			
RS	3	2			
BLO	1	2			

2.6. Swap-Neighbourhood Operator

A roster solution created via the Swap-Neighbourhood method consists of exchanging the shift and role values for two employee/ day assignments that have already been created. For two selected assignments, the algorithm first checks: if the first employee could do the second assignment; and if the second employee could do the first assignment. If both employees are capable of the alternative assignment if they are able to do so, then the two assignments are swapped in order to produce a new candidate roster. The rationale behind this scheme of neighbourhood structures is for the purpose of improving the overall roster through use of the SA model while maintaining the total number of assignments of the roster; therefore, it provides a means to assist in decreasing penalties associated with employee preference, improve shift role stability, and provide a more evenly distributed workload between all employees, while still achieving the original level of coverage provided by the roster.

Algorithm: Swap-Neighbourhood for Roster Improvement

Input:

Current roster solution S

List of assigned employee-day positions

Employee skills, availability, fixed assignment, and feasibility rules

Output:

A new neighbouring roster solution S'

Step 1: Start from the current roster solution S.

Step 2: Create a list A of all assigned employee-day positions.

Each element in A has the form:(employee, day)

Step 3: Randomly select two assigned positions:

$k1 = (e1, d1)$

$k2 = (e2, d2)$

Step 4: Let:

Assignment of $k1 = (s1, r1)$

Assignment of $k2 = (s2, r2)$

Step 5: Check whether:

Employee e1 can be assigned to (s2, r2) on day d1

Employee e2 can be assigned to (s1, r1) on day d2

Step 6: If both assignments are feasible:

Assign employee e_1 to (s_2, r_2)

Assign employee e_2 to (s_1, r_1)

Return the new solution S'

Step 7: If the swap is infeasible:

Try another pair of assignments.

Step 8: If no feasible swap is found after a number of attempts:

Return no neighbour solution.

2.7. Integration with Simulated Annealing

The neighbourhoods of Move and Swap form part of a framework for improving a solution using Simulated Annealing (SA). The starting point for the search is the roster generated by solving an MILP, and from this point the algorithm generates repeated neighbouring solutions to the current solution by using one of the neighbourhood operators. Each new neighbour is then evaluated using a common evaluation (objective) score and either accepted or rejected based on the SA acceptance criteria. Any neighbouring solution that is better than the current solution is automatically accepted, whereas neighbouring solutions that are worse than the current solution may still be accepted depending on a probabilistic acceptance rule that is determined from the current temperature. This process enables the searcher to escape from local optima and/or search the solution space more extensively, and as the temperature decreases, the ability to accept neighbouring solutions becomes progressively more selective.

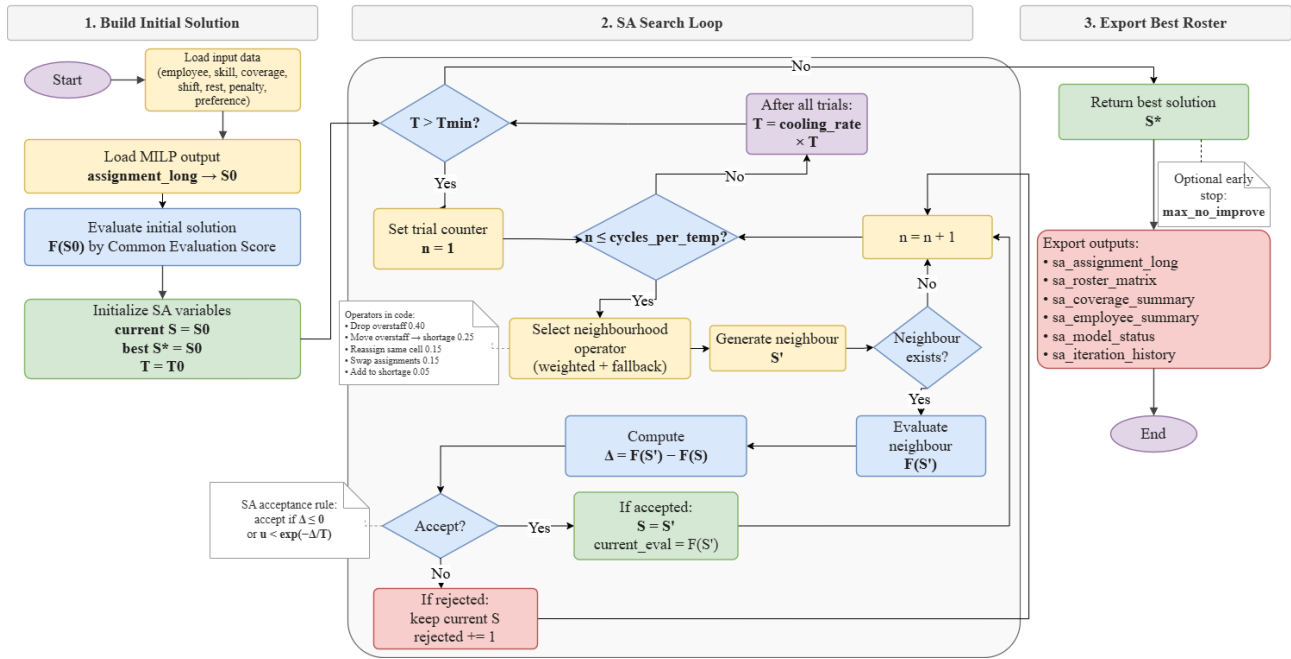


Figure IV-4: Simulated Annealing Algorithm

This workflow represents a post-MILP Simulated Annealing improvement process. The MILP roster is first used as the initial solution S_0 . SA then repeatedly generates neighbouring roster solutions using move, swap, add, drop, and reassignment operators. Each neighbour is evaluated using the Common Evaluation Score. If the neighbour improves the score, it is accepted directly; otherwise, it may still be accepted with probability $e^{-\Delta/T}$. After a fixed number of trials at each temperature, the temperature is reduced. The process stops when $T \leq T_{min}$, and the best roster S^* is exported as the final SA-improved solution.

2.8. MILP-SA Comparative Scalability Workflow

Table IV-38: Two demand scale scenarios

Scale	Expected customers/day	Employee headcount	Days
Small (Baseline)	900	48	28
Medium	1350	70	28

This two-scale validation provides preliminary evidence of MILP–SA performance under increasing demand and workforce size, rather than a full scalability proof. Two demand scale scenarios were created to review the variance in performance between the MILP base case and the post-MILP SA improvement methodology. The smaller of the two scenarios, lower demand scenario, depicts the original operating parameters with an expected customer volume of 900 and a workforce of 48, while the larger of the two represents a greater volume of customers, 1350, and has an employee workforce

of 70. In other words, since the SA model bases its initial solution on the MILP roster, it is more appropriate to view this methodology as a post MILP improvement assessment than as an independent MILP vs. SA competition. Both approaches utilise a single Common Evaluation Score with hard violations, coverage shortages, overstaffing, preference penalties, fairness and stability of role shifts. This two-scale experiment offers potentially initial evidence of whether SA will be achievable while maintaining feasibility and improving roster quality as the size of the problem increases. However, more abortive scenarios would be required to make a more definitive conclusion regarding scalability.

Table IV-39: Technical Setting MILP- SA

Group	Parameter	Small	Medium	Description
Scenario	Expected customers/day	900	1350	Determines staffing demand
Scenario	Employee headcount	48	70	Determines available workforce
Planning	Horizon days	28	28	Affects model size
Planning	Number of roles	6	6	Affects skill-role assignment complexity
MILP	Time limit	300 sec	300 sec	Controls solve effort
MILP	MIP gap	0.01	0.01	Controls optimality tolerance
MILP	Coverage lower bound	C_opt	C_opt	Controls required service level
MILP	Cmax hard	True	True	Controls overstaff feasibility
MILP	Soft coverage	True	True	Allows penalized shortage instead of infeasibility
SA	Initial temperature	500	500	Controls early exploration
SA	Min temperature	0.1	0.1	Controls stopping condition

SA	Cooling rate	0.95	0.95	Controls convergence speed
SA	Cycles per temp	700	700	Controls search depth
SA	Hard penalty	100,000	100,000	Prevents hard violations
SA	Seed	42	42	Keeps results reproducible
SA	Neighbourhood weights	0.40/0.25/0.15/0.15/0.05	Same	Controls search direction

3. Results and Analysis

3.1. MILP Basline Model Output:

The MILP model was applied to generate a 28-day monthly roster for the airport lounge operation. The model assigns employees to one shift and one operational role per working day, using the binary decision variable where each assignment links one employee, day, shift, and role. The model only creates feasible assignment variables for employees who are qualified for the corresponding role, and it blocks assignments on unavailable days. It also evaluates coverage against the staffing bands C_{min} , C_{opt} , and C_{max} , while enforcing employee workload, hour, rest, night-shift, weekend, public-holiday, preference, and incompatibility-related constraints. The output tables exported by the model include `assignment_long`, `roster_matrix`, `coverage_summary`, `employee_summary`, `model_status`, and `demand_input`.

- **Model status result:**

The MILP solver returned an Optimal solution with an objective value of 2,454.0. This means that the solver found the best roster under the defined mathematical constraints and penalty structure. The objective value should be interpreted as a weighted penalty score, not as a monetary cost. A lower value indicates fewer violations of soft goals such as overstaffing, deviation from target workload, preference conflicts, and standard profile changes.

The model used C_{opt} as the lower coverage target, meaning that the roster was designed to satisfy the optimal staffing requirement rather than only the minimum acceptable staffing level. The model

also treated $C_{\max}$ as a hard upper bound, so no role-shift-day cell was allowed to exceed the maximum permitted staffing level.

Table IV-40: Summary of MILP model output

Metric	Result	Interpretation
Solver status	Optimal	The roster is mathematically solved to optimality under the model setting.
Objective value	2,454.0	Total weighted penalty from soft constraints and trade-offs.
Total $C_{\min}$	759	Minimum staffing level required for basic feasibility.
Total C_{opt}	960	Target staffing level used as the main coverage requirement.
Total $C_{\max}$	1,509	Maximum allowable staffing capacity.
Actual assignments	976	Total employee-shift-role assignments generated by the model.
Shortage to C_{opt}	0	No shortage against the optimal staffing target.
Overstaff above C_{opt}	16	Controlled surplus assignments above the target level.
Overstaff above $C_{\max}$	0	No violation of maximum staffing limits.

Overall, the result shows that the model successfully produced a feasible and service-reliable roster. All required staffing cells were covered at or above the optimal level, while no cell exceeded the maximum allowable coverage.

- MILP Roster Output:**

Table IV-41: Employee E018 Shift and Role Allocation Table per Months (Sample Result)

Employee	E018	Total
S1	6	6
DAS	6	6
S2	1	1
DAS	1	1
S3	13	13
DAS	13	13
Grand Total	20	20

The table summarizes the monthly assignment distribution of employee E018 by shift and role. During the planning horizon, E018 was assigned 20 working shifts in total, all under the DAS role. Among these assignments, 6 shifts were allocated to S1, 1 shift to S2, and 13 shifts to S3. This indicates that E018 was mainly scheduled in S3, while S2 was assigned only once. Therefore, the table helps evaluate whether the employee's workload is within the expected working-day range and whether the shift distribution is balanced across the planning period.

Table IV-42: Daily Staff Assignment by Shift and Role (Sample Result)

Shift	Role	Assigned Employees	Headcount
S1	BLO	E014	1
	DAS	E003, E022, E028	3
	DS	E041	1
	FSTC	E048	1
	RS	E024	1
	SLS	E042	1
S2	BLO	E005, E044	2
	DAS	E011, E018, E026, E027, E029, E034, E040, E046	8
	DS	E010	1
	FSTC	E035	1
	RS	E032	1
	SLS	E001	1
S3	BLO	E004, E047	2
	DAS	E002, E019, E020, E021, E023, E031, E037, E045	8
	DS	E013	1
	FSTC	E033	1
	RS	E006	1
	SLS	E025	1

This table explains the structure of the MILP roster schedule table. In this table, each row represents one employee, while the columns from D1 to D28 represent the daily schedule over the 28-day planning horizon. The fields employee_id and employee_name are used to identify each employee in the roster. Each daily cell contains the employee's assigned shift and role, or a non-working status. A working assignment is written in the format Shift-Role. For example, S2-DAS means that the employee works in Shift 2 and performs the Dining Area Support role. Similarly, S1-Role, S2-Role, and S3-Role indicate that the employee is assigned to Shift 1, Shift 2, or Shift 3 with a specific operational role. The value OFF means that the employee has a planned day off, while UNAVAILABLE means that the employee cannot be assigned due to availability restrictions.

Table IV-43: Table sample Schedule result

employee_id	D1	D2	D3	D4	D5	D6	D7
E001	S2-SLS	S2-SLS	S2-SLS	S2-SLS	OFF	S1-SLS	OFF
E002	S3-DAS	S3-DAS	OFF	S3-DAS	S3-DAS	S3-DAS	S3-DAS
E003	S1-DAS	S1-DAS	S2-DAS	S2-DAS	OFF	S2-DAS	S2-DAS
E004	S3-BLO	OFF	S1-BLO	S1-BLO	OFF	S1-BLO	S1-DAS
E005	S2-BLO	S2-BLO	S2-BLO	OFF	S2-BLO	S2-BLO	S2-BLO
E006	S3-RS	S3-RS	S3-RS	S3-RS	S3-RS	OFF	S3-RS
E007	OFF	S1-DAS	S2-DAS	S2-DAS	S2-DAS	S2-DAS	OFF
E008	UNAV AILAB LE	S2-BLO	S2-BLO	S2-BLO	S3-BLO	OFF	S2-BLO

From the table, employee E001 is assigned as S2-SLS from Day 1 to Day 4, which means this employee works in Shift 2 (08:00–16:00) as Service Logistics Support for four consecutive days. On Day 5, the cell is marked as OFF, indicating a planned rest day. On Day 6, the assignment changes to S1-SLS, meaning the employee works Shift 1 (00:00–08:00) but still performs the same SLS role. On Day 7, the employee is again assigned OFF. This example shows that each schedule cell clearly combines three types of information: the working shift, the assigned operational role, and the employee's daily status. Therefore, the roster matrix helps managers quickly identify who works on each day, which shift they work, and what role they perform.

- **Coverage summary result:**

The coverage_summary.csv file presents the coverage audit of the MILP roster solution. Its purpose is to compare the staffing demand required by the model with the actual number of employees assigned in each day, shift, and operational role. The file contains 504 records, representing the full planning horizon of 28 days $\times$ 3 shifts $\times$ 6 role groups. The six role groups include RS, DAS, BLO, FSTC, DS, and SLS, corresponding to the main operational functions of the airport lounge.

Table IV-44: Coverage band (Sample Result day=28)

shift	role	C_min	C_opt	C_max	actual_staff	gap_to_Copt	shortage_to_Cmin	shortage_to_Copt	overstaff_above_Copt	overstaff_above_Cmax	coverage_status
S1	BLO	1	1	2	1	0	0	0	0	0	OK
S1	DAS	2	3	4	3	0	0	0	0	0	OK
S1	DS	1	1	2	1	0	0	0	0	0	OK
S1	FSTC	1	1	2	1	0	0	0	0	0	OK
S1	RS	1	1	2	1	0	0	0	0	0	OK
S1	SLS	1	1	2	1	0	0	0	0	0	OK
S2	BLO	1	1	2	1	0	0	0	0	0	OK
S2	DAS	5	8	10	8	0	0	0	0	0	OK
S2	DS	1	1	2	1	0	0	0	0	0	OK
S2	FSTC	1	1	2	1	0	0	0	0	0	OK
S2	RS	1	1	2	1	0	0	0	0	0	OK
S2	SLS	1	1	2	1	0	0	0	0	0	OK
S3	BLO	1	1	2	1	0	0	0	0	0	OK
S3	DAS	5	8	10	8	0	0	0	0	0	OK
S3	DS	1	1	2	1	0	0	0	0	0	OK
S3	FSTC	1	1	2	2	1	0	0	1	0	OK
S3	RS	1	1	2	1	0	0	0	0	0	OK
S3	SLS	1	1	2	1	0	0	0	0	0	OK

The sample table illustrates how the MILP roster result is evaluated at the coverage level. Each row represents one operational role in one specific day and shift. The columns $C_{\min}$, C_{opt} , and $C_{\max}$ define the staffing requirement range used by the model. $C_{\min}$ is the minimum acceptable number of employees, C_{opt} is the preferred staffing target, and $C_{\max}$ is the maximum allowable staffing level to avoid excessive labor allocation.

The column Actual Staff shows the number of employees assigned by the MILP model. The gap and shortage columns are used to check whether the roster satisfies the required staffing level. A value of 0 in Shortage to $C_{\min}$ means there is no critical understaffing, while a value of 0 in Shortage to C_{opt} means the optimal staffing target is fully achieved. Similarly, Overstaff above C_{opt} and Overstaff above $C_{\max}$ indicate whether the model assigns more staff than the optimal or maximum level.

For example, the staffing coverage validation result for Day 28. The result shows that all shift–role combinations satisfy the required coverage constraints. There is no shortage below either the minimum coverage level $C_{\min}$ or the optimal coverage level C_{opt} . Most assignments exactly match the optimal staffing target. The only deviation occurs in Shift 3 for the FSTC role, where actual staffing exceeds C_{opt} by one employee. However, this value remains within the maximum allowable coverage level $C_{\max}$, so it is not treated as a hard coverage violation. Overall, the Day 28 roster is operationally feasible from a staffing coverage perspective.

- **Employee summary result:**

Table IV-45: Employee-Level Workload and Assignment Summary from MILP Roster

Employee ID	Work Days	Hours	Night Shifts	Standard Role	Shift Allocation	Role Allocation	Contract Status
E001	20	160	6	SLS	S1: 6, S2: 13, S3: 1	SLS: 20	Within range
E002	20	160	0	DAS	S3: 20	DAS: 20	Within range
E005	22	176	6	BLO	S1: 6, S2: 16	BLO: 22	Within range
E012	21	168	5	DAS	S1: 5, S3: 16	RS: 13, DAS: 8	Within range
E016	23	184	4	DS	S1: 4, S2: 19	DS: 23	Within range

In this result, all selected employees are within the required contract range. For example, employees work between 20 and 23 days, equivalent to 160–184 hours, which remains within the allowed range

of 20-24 workdays and 160-192 hours. This indicates that the MILP model successfully assigns employees without violating monthly workload limits. The Night Shifts column is important because S1 represents the overnight period from 00:00 to 08:00. For example, E001, E005, and E012 receive several night shifts, while E002 receives none. This shows that night work is tracked separately so the roster can be evaluated for fairness and fatigue risk. The Shift Allocation and Role Allocation columns explain how each employee contributes to the final schedule. Some employees are assigned mainly to one role, such as E001 working fully as SLS and E005 working fully as BLO. Meanwhile, E012 is assigned to both RS and DAS, showing that the model can use multi-skilled employees flexibly when coverage demand requires it.

Overall, this sample table demonstrates that the MILP roster is not only feasible at the coverage level but also practical at the employee level. It satisfies contract limits, controls night-shift allocation, and distributes employees across roles and shifts according to operational needs

3.2. MILP-SA Comparative Scalability Output

3.2.1. Small Scale (Baseline)

The comparison shows that the SA output is effectively identical to the MILP output in the small-scale baseline case. The number of assignments, coverage performance, shortages, overstaffing, and employee workload all remained unchanged. The SA algorithm was configured with an initial temperature of 500 and a minimum temperature of 0.1. The temperature was reduced by a cooling rate of 0.95 after each temperature level, and 700 neighbourhood solutions were evaluated at each level. This setting allows the algorithm to explore the solution space broadly at the beginning and gradually become more selective as the temperature decreases

Table IV-46: MILP-SA Small Scale Result Compare

Evaluation item	Initial Solution (MILP Baseline)	Final Solution
score	2186	2186
hard_violations	0	0
skill_violations	0	0
availability_violations	0	0
fixed_assignment_violations	0	0
rest_violations	0	0
contract_violations	0	0
consecutive_work_violations	0	0
consecutive_night_violations	0	0
shortage_to_Cmin	0	0
shortage_to_Copt	0	0
overstaff_above_Copt	16	16

overstaff_above_Cmax	0	0
coverage_under_Cmin_cells	0	0
coverage_under_Copt_cells	0	0
coverage_over_Cmax_cells	0	0
preference_penalty	18	18
standard_profile_penalty	1282	1282
incompatibility_penalty	0	0
fairness_penalty	86	86
actual_assignments	976	976

This result suggests that the MILP model already produced a strong solution for the small-scale case. Since the MILP solution was optimal and feasible, the room for SA to improve the roster was limited. Therefore, SA mainly acted as a validation tool in this case rather than producing an actual improvement.

It is important to note that the MILP objective value and the SA score should not be directly compared as if they were the same measurement. The MILP objective value is calculated based on the mathematical optimization model, while the SA score is calculated using the post-processing penalty function. Therefore, the fact that MILP objective = 2454 and SA score = 2186 does not mean that SA performed better. Since the assignment results are unchanged, the difference comes from the scoring structure rather than from an improved roster.

Overall, the small-scale baseline experiment confirms that the MILP model can generate a fully feasible and optimal roster under the original demand setting. The roster satisfies all coverage requirements, avoids understaffing, respects employee workload constraints, and maintains zero hard violations.

The SA phase successfully preserved the feasibility of the MILP roster but did not improve the solution in this small-scale case. This outcome is reasonable because the MILP model already found an optimal solution, leaving limited space for local search improvement. As a result, the value of SA is expected to become more visible in larger-scale scenarios, where the MILP solver may require longer computation time or may not easily reach optimality within the time limit.

3.2.2. Medium-Scale MILP–SA Comparative Results

The medium-scale experiment was designed to test the scalability of the proposed rostering framework under higher operational demand. Compared with the small-scale baseline scenario, the medium scenario increased the expected number of customers per day to 1,350 passengers/customers and expanded the available workforce to 70 employees. This setting represents a higher-demand operating condition and is used to evaluate whether the MILP model and the post-MILP SA

improvement method can maintain staffing feasibility and coverage quality when the system scale increases.

For this medium-scale scenario, both optimization methods were executed under the same computational time setting. The MILP model was given a solver time limit of **300 seconds**, while the Simulated Annealing model was also run for **300 seconds**. This equal runtime setting allows the comparison to focus on how each method performs under the same computational budget.

Table IV-47: Comparative Coverage Results between MILP and SA in the Medium-Scale Scenario

Evaluation item	Initial Solution (MILP Medium)	Final Solution (SA Medium)
score	2560323	1515322
hard_violations	25	15
skill_violations	25	2
availability_violations	57	6
fixed_assignment_violations	3	0
rest_violations	0	0
contract_violations	0	127.75
consecutive_work_violations	111.5181	141.806
consecutive_night_violations	500	564
shortage_to_Cmin	0	11
shortage_to_Copt	0	0
overstaff_above_Copt	0	0
overstaff_above_Cmax	0	0
coverage_under_Cmin_cells	0	2
coverage_under_Copt_cells	0	0
coverage_over_Cmax_cells	0	0

To sum up, The analysis confirms that SA provides a significant improvement to the results generated using an initial roster from a MILP optimization. By using SA methodology, the total penalty score was reduced by approximately 40% from its pre-SA value, with most of this result being due to the improved performance in the area of coverage. The shortage of staff versus Cmin decreased from 25 to 2, while the shortage of staff versus Copt decreased from 57 to 6. There were no overstaffed cells (above Copt) in the final assignment set, which consisted of a total of 1,649 personnel compared to the total required based on the defined Copt of 1,655. The findings support the efficacy of SA in repositioning staff to improve coverage when increasing problem scale limits the ability to efficiently perform an exact MILP optimization.

- **Role-Level Improvement**

The role-level comparison indicates that SA mainly improved the roles that were previously identified as bottlenecks in the MILP solution. In the MILP medium result, the largest shortages occurred in DAS and BLO. After applying SA, the shortage in DAS decreased from 22 to 2, while the shortage in BLO decreased from 18 to 2.

Table IV-48: Role-Level Shortage Improvement after SA

Role	MILP Shortage to C_opt	SA Shortage to C_opt	Improvement
DAS	22	2	-20
BLO	18	2	-16
RS	8	2	-6
SLS	5	0	-5
DS	4	0	-4
FSTC	0	0	0

This result shows that SA was particularly effective in reallocating staff toward high-pressure operational roles. DAS and BLO are workload-sensitive roles because they are directly affected by customer volume, dining area turnover, and buffet service demand. Therefore, the improvement in these two role groups is operationally meaningful.

- **Remaining Shortage After SA:**

Although SA significantly improved the coverage result, the solution was not completely perfect. After SA, only six cells remained below C_opt, and only two of them were below C_min.

Table IV-49: Remaining Coverage Shortages after SA Improvement

Day	Shift	Role	C_min	C_opt	Actual	Status
19	S1	RS	2	2	1	UNDER_CMIN
23	S1	RS	2	2	1	UNDER_CMIN
2	S3	BLO	2	3	2	UNDER_COPT
2	S3	DAS	9	14	13	UNDER_COPT
7	S3	BLO	2	3	2	UNDER_COPT
11	S1	DAS	3	5	4	UNDER_COPT

The remaining shortages are concentrated mainly in RS, DAS, and BLO. This confirms that customer-facing and workload-sensitive roles remain the most difficult areas to fully cover when demand is scaled upward.

- **Trade-Offs of SA Improvement:**

Although SA improved coverage significantly, the result also introduced some feasibility trade-offs. The SA solution reduced the overall hard violation score compared with the initial SA state, but it generated rest and contract violations that did not exist in the MILP solution.

Table IV-50: Trade-Offs of SA Improvement

Metric	MILP as Initial SA Solution	SA Final Solution	Interpretation
Hard violations	25	15	Improved but not fully eliminated
Rest violations	0	11	Worse than MILP
Contract violations	0	2	Worse than MILP
Preference penalty	0	127.75	Worse than MILP
Fairness penalty	111.52	141.81	Worse than MILP
Stability penalty	500	564	Worse than MILP

This indicates that SA improved staffing coverage by adding and reallocating assignments, but some of these changes caused employees to work too close between shifts or exceed contractual workday limits. For example, one employee was assigned 25 working days, exceeding the maximum limit of 24 days. Therefore, although SA improved coverage, it also weakened compliance with some employee-related constraints.

The SA score decreased from 2,560,322.77 to 1,515,322.09, representing an improvement of approximately 40.8% within the SA evaluation framework. However, this SA score should not be directly compared with the MILP objective value, because the MILP objective and SA scoring function use different penalty structures. The correct interpretation is that SA improved the MILP-generated roster according to the SA evaluation score, especially by reducing coverage shortages.

In Conclusion, the medium-scale comparison shows that SA can significantly improve the MILP baseline in terms of coverage performance. Starting from the MILP roster, SA increased total assignments from 1,601 to 1,649 and reduced the shortage to C_{opt} from 57 to only 6 staff-shifts. The number of under- C_{opt} cells decreased from 55 to 6, while under- C_{min} cells decreased from 25 to 2. These improvements were mainly observed in S1 and in the DAS/BLO role groups.

The medium-scale results indicate that SA has stronger scalability potential than MILP under the same 300-second computational budget. While MILP provides a structured and optimization-based baseline roster, SA can further improve the solution by exploring local adjustments around the MILP output. In this scenario, SA substantially reduced coverage shortages and under-covered day–shift–role cells, suggesting that it can be particularly useful when the problem scale increases and exact optimization becomes more computationally constrained.

However, the SA improvement also involved operational trade-offs. Although the final SA solution achieved better coverage recovery, it introduced rest and contract violations and increased preference, fairness, and stability penalties. Therefore, SA should not be interpreted as a fully feasible replacement for the MILP solution at this stage. Instead, it should be viewed as an effective post-MILP improvement method that can enhance coverage performance within a limited runtime.

For future large-scale application, the SA model should be strengthened with stricter hard-constraint control. This would ensure that coverage improvements are achieved without compromising employee feasibility, rest requirements, and labor-rule compliance. In other words, SA demonstrates strong potential for large-scale roster improvement, but additional feasibility protection is required before it can be considered an operationally reliable optimization approach.

4. Validation

4.1. Purpose of Validation

The purpose of the validation phase is to evaluate whether the proposed MILP rostering model remains feasible, logical, and operationally robust under different levels of customer arrival demand. Since the staffing requirements in this study are generated from customer-driven workload, the validation does not directly scale the number of required staff. Instead, customer arrivals are scaled first, and the resulting demand is passed through the event-driven simulation model to regenerate workload-based staffing requirements.

This validation approach is appropriate because the airport lounge operation is affected by time-varying customer arrivals, buffet visits, table reset tasks, dishwashing demand, stock replenishment activities, and food-safety control tasks. Therefore, testing the model under different customer-arrival scenarios provides a more realistic assessment of whether the roster model can respond to changes in operational demand.

4.2. Validation Framework



Figure IV-5: Customer-arrival scaling validation framework for the MILP roster model.

The validation framework follows a customer-arrival scaling approach. First, the baseline customer arrival profile is adjusted using a set of demand scaling factors. Second, the scaled arrival data is used as the input for the 28-day event-driven simulation. Third, the simulation output is converted into role-based staffing requirements by day, shift, and role group. Finally, the MILP model is solved for each scenario to determine whether a feasible monthly roster can still be generated.

4.3. Validation Scenario Design

Seven customer-arrival scaling scenarios were evaluated to examine how changes in passenger demand affect staffing requirements and MILP roster feasibility. The baseline scenario assumes 900 expected customers per day, while the remaining scenarios apply arrival scaling factors from 0.90 to 1.50, corresponding to 810 to 1,350 expected daily customers. For each scenario, the validation table compares the resulting coverage bands (Total C_{min} , C_{opt} , and C_{max}) with the required and actual roster assignments, as well as shortage and overstaffing levels. This structure allows the model to be assessed under low-demand, baseline, and increasing high-demand conditions.

Table IV-51: Customer Arrival Scaling Scenarios

Arrival Scale	Expected Daily Customers	Total C_min	Total C_opt	Total C_max	Required Assignments	Actual Assignments	Overstaff
0.90	810	784	784	2,436	784	960	176
1.00	900	840	840	2,492	840	960	120
1.10	990	840	840	2,492	840	960	120
1.20	1,080	868	868	2,548	868	960	92
1.30	1,170	896	896	2,604	896	960	64
1.40	1,260	980	980	2,688	980	996	16
1.50	1,350	980	980	2,688	980	996	16

To evaluate the feasibility and flexibility of our proposed MILP rostering model, we tested the system across seven customer demand scenarios. By applying arrival scaling factors from 0.90 to 1.50, the expected daily passenger volume was adjusted to range from 810 to 1,350, compared against a baseline of 900 customers. During the pre-check validation phase, we observed a crucial characteristic: in low-demand scenarios, the actual operational workload requirement falls below the minimum shift allocations stipulated in the employees' contracts.

Building upon these established scenarios, our experimental results confirm that the MILP model consistently generates a feasible scheduling solution across all demand levels. As passenger volume increases, the total required assignments grow from 784 to 980 shifts. However, this growth occurs in a stepwise rather than a strictly linear manner, since personnel must be allocated in whole-shift units. Notably, overstaffing is present in all scenarios, meaning the actual assigned shifts consistently exceed the minimum operational requirements. Through our analysis, we can assert that this is not an optimization error within the algorithm, but rather a direct consequence of the model strictly adhering to the hard constraints regarding minimum workdays and hours in the labor contracts. Nevertheless, as customer demand peaks and approaches the workforce's maximum productive capacity, this surplus of shifts drops significantly from 176 down to merely 16. This demonstrates that resource utilization is highly optimized when operational pressure increases.

From a practical operational management perspective, we recognize that the surplus staff in low-demand scenarios inadvertently provides a significant advantage, acting as an "operational buffer." In the specific environment of an airport lounge which frequently experiences unpredictable fluctuations such as flight delays, clustered arrivals, or sudden peak hours this buffer is essential for

maintaining stable service quality. Conversely, in high-demand scenarios, the model demonstrates its ability to closely align with the actual workload, maximizing the utilization of employees' available time while still retaining a necessary margin of flexibility.

Despite achieving positive results regarding schedule feasibility, we frankly acknowledge certain limitations within the scope of this study. Currently, our validation process primarily focuses on evaluating shift coverage and contract compliance, without directly measuring real-time service key performance indicators (KPIs) such as customer waiting time, queue length, or the turnaround speed for cleaning and dishwashing. Furthermore, the simulated scenarios currently account only for steady variations in passenger volume and do not yet encompass severe operational disruptions, such as mass flight cancellations or sudden staff shortages.

In conclusion, the validation results prove that our proposed MILP framework is a reliable workforce planning tool that responds logically to demand fluctuations while strictly complying with labor regulations. To fully complete this research, our next direction is to integrate the output roster generated by the MILP model into a discrete-event simulation system. This subsequent step will allow us to comprehensively and visually evaluate real-time service quality KPIs before considering practical implementation.

CHAPTER V: CONCLUSIONS

1. Summary of results.

This capstone project developed an integrated workforce rostering framework for Le Saigonais International Lounge to improve the connection between customer demand, staffing requirements, and employee assignment decisions. Instead of treating workforce planning as a fixed headcount problem, our group designed a demand-driven framework in which customer arrivals are transformed into operational workload, workload is converted into role-based staffing requirements, and staffing requirements are then used to generate and validate a monthly roster. This approach is consistent with the operating characteristics of an airport buffet lounge, where service quality depends on both staffing quantity and proper role coverage across different shifts.

1.1. Contributions

The main contribution of this project is the development of an integrated Optimization-Simulation-Feedback framework for airport lounge workforce planning. The framework connects demand generation, workload estimation, MILP-based roster construction, Simulated Annealing improvement, and simulation-based validation into one structured decision process. This allows the project to move beyond a traditional staffing approach and create a more systematic method for linking passenger demand with workforce planning decisions.

Another contribution is the workload-based staffing logic. In the proposed model, customer arrivals are not converted directly into staff numbers. Instead, they generate different operational tasks across six role groups, including Reception & Seating, Dining Area Support, Buffet Line Operations, Food Safety & Temperature Control, Dishwashing & Sanitation, and Service Logistics Support. Each role group is estimated using its own workload driver, such as arrivals, table-reset tasks, buffet visits, dishwashing batches, food-safety checks, and refill events. This makes the staffing requirement more realistic because it reflects the actual service chain of the buffet lounge rather than relying on a general staffing ratio.

The project also contributes a MILP rostering model that assigns employees to days, shifts, and role groups while considering operational and labour constraints. The model includes coverage requirements, skill eligibility, availability, working-hour limits, workday bounds, night-shift limits, rest requirements, and one-shift-per-day rules. By separating hard constraints and soft constraints, the model can protect non-negotiable requirements while still allowing flexibility for employee preferences, fairness, overstaffing control, and roster stability.

In addition, the Simulated Annealing phase contributes an improvement mechanism after the MILP solution is generated. Rather than rebuilding the roster from the beginning, SA starts from the MILP roster and applies local changes such as employee swaps and reassignment moves. This helps reduce penalty issues related to coverage imbalance, rest conflicts, night-shift concentration, preference violations, and unnecessary overstaffing. Therefore, the final roster can be improved beyond basic feasibility.

The validation component is also an important contribution. The project does not assume that a mathematically feasible roster is automatically operationally reliable. Instead, the roster is evaluated through simulation-based validation using performance indicators such as waiting time, queue length, staff utilization, service stability, and KPI violations. This feedback logic allows the model to check whether the roster can still support acceptable service performance under uncertain customer arrivals and variable service conditions.

1.2. Benefits

The proposed framework provides practical benefits for airport lounge workforce planning. For SASCO or similar service operators, the model can support more evidence-based staffing decisions by showing how many employees are required by shift and role group. This helps managers justify staffing levels more clearly because the requirements are derived from workload drivers and service targets rather than only from experience or fixed templates.

The framework can also improve service reliability during peak flight-wave periods. Since the model considers time-varying demand and role-specific workload, it can help reduce the risk of under-coverage in critical service areas such as reception, table turnover, buffet replenishment, dishwashing, and logistics support. Better role coverage can contribute to shorter waiting times, faster service recovery, more stable buffet availability, and better hygiene readiness.

At the same time, the model supports labour efficiency. By identifying staffing requirements more accurately, the framework can reduce unnecessary labour capacity during off-peak periods and limit avoidable idle time. The use of maximum coverage and overstaffing penalties also helps the roster avoid assigning more employees than needed when demand is low.

The project also benefits employee scheduling quality. The MILP model considers workday limits, working-hour limits, rest requirements, night-shift limits, availability, and preferences. This helps create a more balanced and compliant roster, reducing the risk of unfair workload concentration or excessive night-shift assignment for a small group of employees.

From a managerial perspective, the framework improves transparency. Each roster output can be explained through demand, workload, constraints, and penalty logic. This makes the scheduling process easier to audit, compare, and improve over time. If validation results show service weakness, managers can adjust staffing requirements and rerun the model instead of relying only on manual judgement.

From a student capstone perspective, the project demonstrates how industrial engineering tools can be applied to a real service operation problem. By combining workload modeling, MILP optimization, Simulated Annealing, and discrete-event simulation, our group developed a structured methodology that is both academically meaningful and practically relevant. Although the numerical results are scenario-based due to data limitations, the framework provides a strong foundation for future development and real-world implementation.

2. Recommendations for future research

Future research can further strengthen this framework by improving the quality and availability of real operational data. In this project, several demand and service parameters were developed through assumptions and simulation logic because detailed historical data were limited. With access to actual flight schedules, lounge entry records, passenger dwell times, buffet consumption patterns, table turnover records, dishwashing cycle times, and employee attendance data, the workload model could be calibrated more accurately. This would allow the staffing requirements to better reflect real operating conditions and make the roster output more reliable for implementation.

Another important direction is to expand the shift structure used in the model. The current framework uses three fixed 8-hour shifts because this structure is simple and suitable for monthly planning. However, airport lounge demand is often concentrated in short flight-wave periods, which may require more flexible staffing arrangements. Future studies could consider staggered start times, overlapping shifts, part-time support, planned breaks, overtime, or temporary peak-hour deployment. These extensions would help the model respond more precisely to demand surges without increasing the full-day staffing level unnecessarily.

The framework can also be improved by modeling intra-shift task assignment in greater detail. In the current version, each employee is assigned to one primary role group per shift, while short-term task switching is assumed to be handled by supervisors during execution. Future research could develop a more detailed task-allocation layer that determines how employees move between tasks within a

shift. This would be useful for capturing micro-bottlenecks such as reception congestion, delayed buffet replenishment, dish shortages, or slow table recovery during peak periods.

The Simulated Annealing phase also provides opportunities for further improvement. Its performance depends on penalty weights, cooling schedule, move selection, stopping criteria, and repair priorities. Future studies could tune these parameters more systematically and compare SA with other solution methods such as Tabu Search, Variable Neighborhood Search, Genetic Algorithms, or metaheuristic approaches. Such comparisons would help identify which method provides the best balance between solution quality, computational time, and practical usability for airport lounge rostering.

For practical deployment, the framework should eventually be supported by a user-friendly decision-support interface. A dashboard could allow managers to upload demand forecasts, employee skills, leave requests, availability data, and policy constraints, then automatically generate staffing requirements, optimized rosters, coverage summaries, and validation reports. This would make the model easier to use in real monthly planning and would help managers monitor staffing performance over time.

Future research should also test the framework under disruption scenarios. Airport lounge operations may be affected by flight delays, cancellations, sudden passenger surges, seasonal peaks, temporary employee absence, or changes in access policy. By including these irregular conditions in the validation stage, the model could evaluate whether the roster remains robust under uncertainty and whether contingency staffing rules are necessary.

In conclusion, this capstone project provides a solid foundation for demand-driven workforce rostering in airport lounge operations. Further development should focus on empirical data calibration, flexible shift design, detailed intra-shift task allocation, improved metaheuristic performance, and practical system deployment. These improvements would increase the accuracy, robustness, and managerial value of the proposed framework, making it more suitable as a real decision-support tool for SASCO and similar airport service organizations.



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APPENDICES

1. [Data Input Dictionary](#)

1.1. Customer Arrival/ Staff Requirement Input Dictionary

The data input files define the customer demand baseline, event-driven simulation outputs, and employee-related MILP input files. The full tables are submitted separately; only a dictionary is included in the report.

Appendice 1-1: Customer Arrival/ Staff Requirement Input Dictionary

File name	Type	What it contains	Purpose
arrival_baseline_48_slices.xlsx	Excel input	One-day customer-demand baseline divided into 48 half-hour slices; includes demand weight, expected arrivals, lambda per hour, generated arrivals, event samples, and shift summary.	Starting demand shape for the 28-day event-driven simulation.
airport_lounge_28day_event_sim_output.xlsx	Excel output / downstream input	Main DES workbook containing run parameters, formulas, daily wave plan, 28-day arrival slices, customer summaries, role workload, role-shift recommendation, and staffing pivot.	Main workload and staff-requirement source used to generate MILP demand inputs.
airport_lounge_28day_event_sim_output_arrival_events_28d.csv	CSV detail log	Individual customer arrival records over the 28-day horizon with arrival time, day, slice, shift, and demand level.	Provides traceability from the 48-slice demand baseline to event-level simulation.
airport_lounge_28day_event_sim_output_customer_log.csv	CSV detail log	Customer-level admission, waiting, dwell time, departure, buffet visits, table reset, and dirty-dish indicators.	Supports validation of customer flow and operational behavior.
airport_lounge_28day_event_sim_output_service_events.csv	CSV detail log	Role-based service events for RS, DAS, BLO, FSTC, DS, and SLS.	Used to calculate workload minutes, waiting time, utilization, KPI pass rate, and role-shift staffing need.
airport_lounge_28day_event_sim_output_stock_events.csv	CSV detail log	Buffet stock consumption, stockout, and refill events.	Supports SLS workload estimation and buffet-continuity analysis.
airport_lounge_28day_event_sim_output_dish_events.csv	CSV detail log	Dirty-dish generation and dishwashing batch events.	Supports DS workload estimation and dishwashing-capacity analysis.



Modeling_Sacco(final).Backup.doe	Simulation model backup	Arena/DOE model backup file.	Provides executable backup of the simulation model if required for submission.
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1.2. Employee Profile_MILP Input Dictionary

Appendice 1-2: Profile_MILP Input Dictionary

File name	Key columns	Purpose in MILP
staffing_demand_daily.csv	day, shift, shift_time, RS, DAS, BLO, FSTC, DS, SLS, total_staff	Direct staff-demand input for MILP coverage constraints; values correspond to the target staffing requirement.
staffing_coverage_bands.csv	day, shift, shift_time, role_min, role_opt, role_max, total_min, total_opt, total_max	Wide-format coverage-band input containing minimum, target, and maximum staffing limits by role.
staffing_coverage_bands_long.csv	day, shift, shift_time, role, C_min, C_opt, C_max	Long-format coverage-band input used for MILP/SA coverage evaluation and reporting.
employee_master.csv	employee_id, employee_name, min/max work days, min/max hours, night-shift limits, consecutive-work limits, standard role, standard shift	Defines employee contracts, labor-rule bounds, and standard employee profiles.
employee_skills.csv	employee_id, RS, DAS, BLO, FSTC, DS, SLS	Defines role-qualification eligibility; prevents assigning employees to unqualified roles.
employee_availability.csv	employee_id, day, status	Defines leave or hard unavailability days.
employee_preferences.csv	employee_id, day, shift, preference_type, penalty	Defines soft shift preferences used in the objective function.
employee_assignment_preferences.csv	employee_id, day, shift, role, preference_type, penalty	Defines role-and-shift assignment preferences at a more detailed level than shift-only preference.
employee_day_requests.csv	employee_id, day, request_type, penalty, is_fixed	Defines employee day-off or day-specific requests; fixed requests are treated more strictly.
employee_fixed_assignments.csv	employee_id, day, shift, role	Stores any pre-assigned employee-day-shift-role assignments; currently header-only in this folder.
employee_history.csv	employee_id, previous_last_shift, previous_last_status, previous consecutive work/off/night values	Carries previous-period history into the model for rest and sequence constraints.
employee_incompatibility.csv	employee_i, employee_j, incompatibility_type, priority, penalty	Defines employee pair restrictions or penalties for same-shift assignment.
employee_pairing.csv	employee_i, employee_j, pair_type, priority, penalty	Defines preferred or incompatible employee pair relationships.
employee_standard_profile.csv	employee_id, standard_role, standard_shift	Defines each employee standard role and shift for stability or preference scoring.
standard_role_shift_preferences.csv	role, shift, preference_type, penalty, note	Defines role-shift pattern preferences reflecting operational workload patterns.
holiday_calendar.csv	day, weekday_index, is_saturday, is_sunday, is_weekend, is_public_holiday	Defines calendar classification for weekend and public-holiday constraints.



shift_structure.csv	shift, shift_time, start_time, end_time, start_minute, end_minute, duration_hours, is_night_shift, min_rest_hours	Defines the three fixed 8-hour shifts and rest-rule parameters.
shift_transition_rest.csv	from_shift, to_shift_next_day, rest_hours, min_rest_hours, allowed, is_forbidden_successession	Defines allowed/forbidden next-day shift transitions based on rest hours.
penalty_config.csv	penalty_component, weight, meaning	Defines objective-function weights for coverage, fairness, preferences, stability, and other soft constraints.
skill_pool_targets.csv	role, role_slots_28d, peak_concurrent_need, pool_min, pool_target	Defines required qualified pool sufficiency by role.
weekend_policy.csv	policy_name, max_working_weekends, complete_weekend_required, saturday_sunday_should_match	Defines weekend-work policy parameters.
generation_validation_report.csv	check, status, detail	Documents validation checks confirming whether generated employee and demand inputs are structurally consistent.

1.3. Supporting Data-Generation Files

Appendice 1-3: Supporting Scripts and Methodology Files

File name	What it contains	Purpose
[1] 48_slice_customer_baseline.py	Generates the 48-slice one-day customer-arrival baseline.	Source generator for arrival_baseline_48_slices.xlsx.
[2] airport_lounge_28day_event_simulation.py	Runs the 28-day event-driven lounge simulation.	Generates DES output workbook and event logs.
[3] Employee_Profile.py	Generates employee profile and MILP-ready input files.	Creates files under employee_milp_inputs/.

2. [MILP Baseline and SA \(Small Scale\) Model- MILP Model Folder](#)

2.1. MILP and SA Small Scale Code

The code files should be submitted electronically. The report should describe each script rather than reproducing the full source code.

Appendice 2-1: MILP - SA (Small Scale) Code

Code file	Main function	Main output
milp_roster_model.py	Solves the 28-day, 3-shift, 6-role MILP workforce roster.	milp_roster_output.xlsx and MILP CSV exports
sa_phase_improve_milp.py	Applies Simulated Annealing after MILP to reduce penalties such as coverage overage, preference violations, and rest-rule issues.	sa_roster_output.xlsx and SA CSV exports

2.2. Small Scale Result

Appendice 2-2: Small Scale Result

Folder	Purpose
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MILP_Result	Stores baseline roster assignment, coverage, employee workload, status, and Excel summary.
assignment_long.csv	Long-format MILP assignment output listing employee, day, shift, role, and assignment flag.
coverage_summary.csv	MILP coverage audit by day, shift, and role, including actual staff, shortage, gap, and overstaff measures.
demand_input.csv	Demand table used by the optimizer, containing C_min, C_opt, and C_max by day, shift, and role.
employee_summary.csv	Employee-level roster summary showing workdays, hours, night shifts, contract bounds, and standard role information.
milp_roster_output.xlsx	Consolidated MILP output workbook combining roster assignments, coverage summary, employee summary, and model status.
model_status.csv	Solver or algorithm status file containing objective value, feasibility status, coverage totals, and key run settings.
roster_matrix.csv	Readable employee-by-day roster matrix showing assigned shift-role codes for presentation and review.
milp_roster_model.py	Implements the baseline MILP roster optimization model and exports assignment, coverage, employee, and solver-status outputs.
SA_Improve.py	Runs the post-MILP Simulated Annealing improvement process using the MILP roster as the initial solution.
sa_assignment_long.csv	Long-format SA roster output after post-MILP improvement, listing employee, day, shift, role, and assignment flag.
sa_coverage_summary.csv	SA coverage audit by day, shift, and role after the improvement process.
sa_demand_input.csv	Copy of the demand table used to evaluate the SA solution.
sa_employee_summary.csv	Employee-level summary after SA, including workdays, hours, night shifts, and shift/role distribution.
sa_iteration_history.csv	Iteration-level SA search history showing temperature, current score, best score, accepted/rejected moves, and violation measures.
sa_model_status.csv	SA run status and final score summary, including initial score, final score, improvement, and violation metrics.
sa_roster_matrix.csv	Readable employee-by-day roster matrix after SA improvement.
sa_small_scale_output.xlsx	Consolidated small-scale SA output workbook.

3. MILP and SA Medium Scale- SA Improvement

3.1. The medium-scale Code

Appendix 3-1: The medium-scale Code

Files	Purpose
milp_roster_model_medium.py	Implements the medium-scale MILP roster optimization model for the scaled demand experiment.
sa_phase2_medium	Runs the post-MILP Simulated Annealing improvement process using the MILP roster as the initial solution.
scale_milp_inputs_medium.py	Scales the baseline MILP input data to create the medium-demand scenario before running MILP and SA.



3.2. The medium-scale Result

Appendice 3-2: The medium-scale Result

Files	Purpose
sa_assignment_long.csv	Long-format SA roster output after post-MILP improvement, listing employee, day, shift, role, and assignment flag.
sa_coverage_summary.csv	SA coverage audit by day, shift, and role after the improvement process.
sa_demand_input.csv	Copy of the demand table used to evaluate the SA solution.
sa_employee_summary.csv	Employee-level summary after SA, including workdays, hours, night shifts, and shift/role distribution.
sa_iteration_history.csv	Iteration-level SA search history showing temperature, current score, best score, accepted/rejected moves, and violation measures.
sa_model_status.csv	SA run status and final score summary, including initial score, final score, improvement, and violation metrics.
sa_roster_matrix.csv	Readable employee-by-day roster matrix after SA improvement.
sa_small_scale_output.xlsx	Consolidated small-scale SA output workbook.
assignment_long.csv	Long-format MILP assignment output listing employee, day, shift, role, and assignment flag. In this folder, it represents the medium-scale MILP result.
coverage_summary.csv	MILP coverage audit by day, shift, and role, including actual staff, shortage, gap, and overstaff measures. In this folder, it represents the medium-scale MILP result.
demand_input.csv	Demand table used by the optimizer, containing C_min, C_opt, and C_max by day, shift, and role. In this folder, it represents the medium-scale MILP result.
employee_summary.csv	Employee-level roster summary showing workdays, hours, night shifts, contract bounds, and standard role information. In this folder, it represents the medium-scale MILP result.
milp_roster_output.xlsx	Consolidated MILP output workbook combining roster assignments, coverage summary, employee summary, and model status. In this folder, it represents the medium-scale MILP result.
model_status.csv	Solver or algorithm status file containing objective value, feasibility status, coverage totals, and key run settings. In this folder, it represents the medium-scale MILP result.
roster_matrix.csv	Readable employee-by-day roster matrix showing assigned shift-role codes for presentation and review. In this folder, it represents the medium-scale MILP result.
sa_assignment_long.csv	Long-format SA roster output after post-MILP improvement, listing employee, day, shift, role, and assignment flag. In this folder, it represents the medium-scale SA result.



sa_coverage_summary.csv	SA coverage audit by day, shift, and role after the improvement process. In this folder, it represents the medium-scale SA result.
sa_demand_input.csv	Copy of the demand table used to evaluate the SA solution. In this folder, it represents the medium-scale SA result.
sa_employee_summary.csv	Employee-level summary after SA, including workdays, hours, night shifts, and shift/role distribution. In this folder, it represents the medium-scale SA result.
sa_iteration_history.csv	Iteration-level SA search history showing temperature, current score, best score, accepted/rejected moves, and violation measures. In this folder, it represents the medium-scale SA result.
sa_model_status.csv	SA run status and final score summary, including initial score, final score, improvement, and violation metrics. In this folder, it represents the medium-scale SA result.
sa_phase2_output.xlsx	Consolidated medium-scale Phase-2 SA output workbook.
sa_roster_matrix.csv	Readable employee-by-day roster matrix after SA improvement. In this folder, it represents the medium-scale SA result.

3.3. Data Input with Medium Scales Operation

Appendice 3-3: Data Input with Medium Scales Operation

Files	Purpose
employee_assignment_preferences.csv	Stores role-and-shift-specific assignment preferences with soft-penalty values. In this folder, it represents the medium-scale input set.
employee_availability.csv	Lists employee leave or availability restrictions by planning day. In this folder, it represents the medium-scale input set.
employee_day_requests.csv	Stores day-off or day-on requests and fixed-request flags used in the roster model. In this folder, it represents the medium-scale input set.
employee_fixed_assignments.csv	Stores any mandatory pre-assigned employee-day-shift-role assignments to be enforced by the model. In this folder, it represents the medium-scale input set.
employee_history.csv	Provides pre-horizon employee work history used for rest, consecutive-work, and night-shift continuity checks. In this folder, it represents the medium-scale input set.
employee_incompatibility.csv	Lists employee pairs that should not be assigned together, including priority and penalty level. In this folder, it represents the medium-scale input set.
employee_master.csv	Defines each employee's contract limits, target workdays/hours, days off, night-shift limits, and consecutive-work limits. In this folder, it represents the medium-scale input set.



employee_pairing.csv	Stores employee pair rules such as incompatible or preferred pairings for soft or hard schedule control. In this folder, it represents the medium-scale input set.
employee_preferences.csv	Stores shift-level employee preferences or avoid requests with associated soft-penalty values. In this folder, it represents the medium-scale input set.
employee_skills.csv	Maps each employee to eligible operational roles, forming the skill/qualification matrix for assignment feasibility. In this folder, it represents the medium-scale input set.
employee_standard_profile.csv	Defines each employee's standard role and standard shift for stability and roster-consistency scoring. In this folder, it represents the medium-scale input set.
holiday_calendar.csv	Provides calendar flags for weekday, weekend, and public-holiday logic in the planning horizon.
penalty_config.csv	Defines objective-function or score weights for coverage, preferences, fairness, stability, and other penalties.
scale_scenario_summary.csv	Summarizes the medium-scenario scaling parameters and resulting input totals.
scaled_inputs_medium.xlsx	Consolidated workbook of medium-scale MILP inputs after demand scaling.
shift_structure.csv	Defines shift IDs, shift times, start/end minutes, duration, night-shift flag, and minimum rest hours.
shift_transition_rest.csv	Defines whether next-day shift transitions satisfy minimum rest-hour rules.
staffing_coverage_bands.csv	Wide-format staffing demand table by day and shift with min/opt/max coverage for each role.
staffing_coverage_bands_long.csv	Long-format coverage-demand table by day, shift, and role; main demand input format for MILP and SA.
staffing_coverage_bands_long_scaled_trace.csv	Audit trace showing how long-format coverage demand was scaled for the medium scenario.
staffing_demand_daily.csv	Daily staffing demand summary by shift and role before or alongside coverage-band construction.
standard_role_shift_preferences.csv	Defines role-shift preference rules used to support stability and standard-assignment scoring.
weekend_policy.csv	Defines weekend scheduling rules such as maximum working weekends and Saturday-Sunday consistency settings.

4. [Validation Structure Dictionary](#)

The validation folder documents the customer-arrival scaling experiment. The purpose is to test whether the MILP model remains logical and feasible when customer arrivals are scaled upward or downward. The uploaded validation package contains seven customer scale scenarios: 0.90, 1.00, 1.10, 1.20, 1.30, 1.40, and 1.50. These validation scenarios provide a summary of how the generated

boarding times in the MILP model respond to varying levels of incoming customers and subsequently the associated staff plans will remain feasible, reasonable and operate successfully under each scenario where there is an increase in incoming customer demand.

Appendice 4-1: Root of Validation

Validation file/folder	What it contains	Purpose
milp_customer_arrival_scaling_validation.py	Runs the validation framework by scaling customer arrivals, regenerating scenario demand, checking feasibility, and solving MILP per scenario.	Main validation script.
customer_arrival_scaling_validation_summary.csv	Compact cross-scenario validation summary.	Used for quick comparison across scale factors.
customer_arrival_scaling_validation_summary.xlsx	Workbook version of the cross-scenario validation summary.	Used for report figures/tables and easier review.
scale_0p90/ to scale_1p50/	Each folder stores one complete validation scenario: scaled simulation, employee inputs, precheck, demand file, and MILP solution.	Provides traceability from scaled arrival demand to final MILP feasibility result.

4.1. Files Repeated inside Each Validation Scenario

Each scenario folder should be kept as a complete unit. The folder name identifies the customer-arrival scaling factor. For example, scale_1p20 represents 120% of the baseline customer arrival level.

Appendice 4-2: Files Repeated inside Each Validation Scenario

Item	Type	What it contains	Purpose
scenario_precheck.xlsx	Excel validation file	Pre-solve checks for the scenario, such as demand size, employee capacity, and feasibility indicators.	Used before MILP solving to identify possible infeasibility causes.



scenario_staffing_demand_daily.xlsx	Excel demand file	Scenario-specific daily staffing demand generated after customer-arrival scaling.	Direct scenario demand used to test MILP robustness.
employee_inputs/	Folder	Employee input CSVs used for that scenario.	Keeps scenario-specific employee and demand inputs together.
simulation/	Folder	Scaled arrival baseline, scenario DES workbook, and detailed DES event logs.	Documents how scaled customer demand affects operational workload.
milp_solution/	Folder	MILP output workbook, roster exports, coverage summary, employee summary, model status, and scenario status.	Stores the validation result for that demand scale.

4.2. Scenario Employee Input Files

Appendice 4-3: Scenario Employee Input Files

File name	Purpose
employee_master.csv	Employee contracts and limits for the scenario.
employee_skills.csv	Role qualifications used by the scenario MILP.
employee_availability.csv	Leave and hard unavailability restrictions.
employee_preferences.csv	Soft shift preferences and penalty values.
employee_history.csv	Previous-period rest and consecutive-work history.
employee_incompatibility.csv	Employee pair restrictions or penalties.
skill_pool_targets.csv	Qualified-pool sufficiency targets.



staffing_demand_daily.csv	Scenario-specific daily shift-role demand input.
standard_role_shift_preferences.csv	Role-shift pattern preferences.
generation_validation_report.csv	Scenario input validation checks.

4.3. Scenario Simulation Files

Appendix 4-4: Scenario Simulation Files

File pattern	What it contains	Purpose
arrival_baseline_scaled_<scale>.xlsx	Scaled one-day arrival baseline for the scenario.	Shows how customer arrivals were scaled before simulation.
Final_data_input_milp_customer_scale_<scale>.xlsx	Scenario DES output workbook.	Main workload and staffing recommendation workbook for the scaled scenario.
*_arrival_events_28d.csv	Event-level customer arrival log.	Provides traceability from scaled demand to generated arrivals.
*_customer_log.csv	Customer-level simulation log.	Validates customer flow, dwell time, admissions, departures, and table reset behavior.
*_service_events.csv	Role service-event log.	Supports role workload calculation and wait-time KPI validation.



*_dish_events.csv	Dirty dish and dishwashing batch log.	Supports DS workload validation.
*_stock_events.csv	Stock consumption and refill log.	Supports SLS workload validation.

4.4. Scenario MILP Solution Files

Appendice 4-5: Scenario MILP Solution Files

File name	What it contains	Purpose
milp_roster_output.xlsx	Main MILP workbook for that validation scenario.	Complete review of the solved scenario roster.
assignment_long.csv	Long-format employee assignment output.	Checks who was assigned to each day, shift, and role.
roster_matrix.csv	Employee-by-day roster matrix.	Readable scenario roster.
coverage_summary.csv	Required versus actual staffing by day-shift-role.	Checks whether scaled demand was covered.
employee_summary.csv	Employee workload, hours, night shifts, and contract indicators.	Checks employee-level feasibility under the scenario.
model_status.csv	Solver status, objective value, shortage/overstaff, and violation counts.	Main scenario feasibility result.
scenario_status.csv	Scenario-level run status and summary.	Allows quick review of whether each scaling scenario completed successfully.

5. Coding Video Performance

The code execution process and result demonstration are provided in the following video link:

<https://youtu.be/iVsLps6FvY8>